

Seismo 101 & Noise

Know your noise - Get the most out of it

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CODES for practicals!

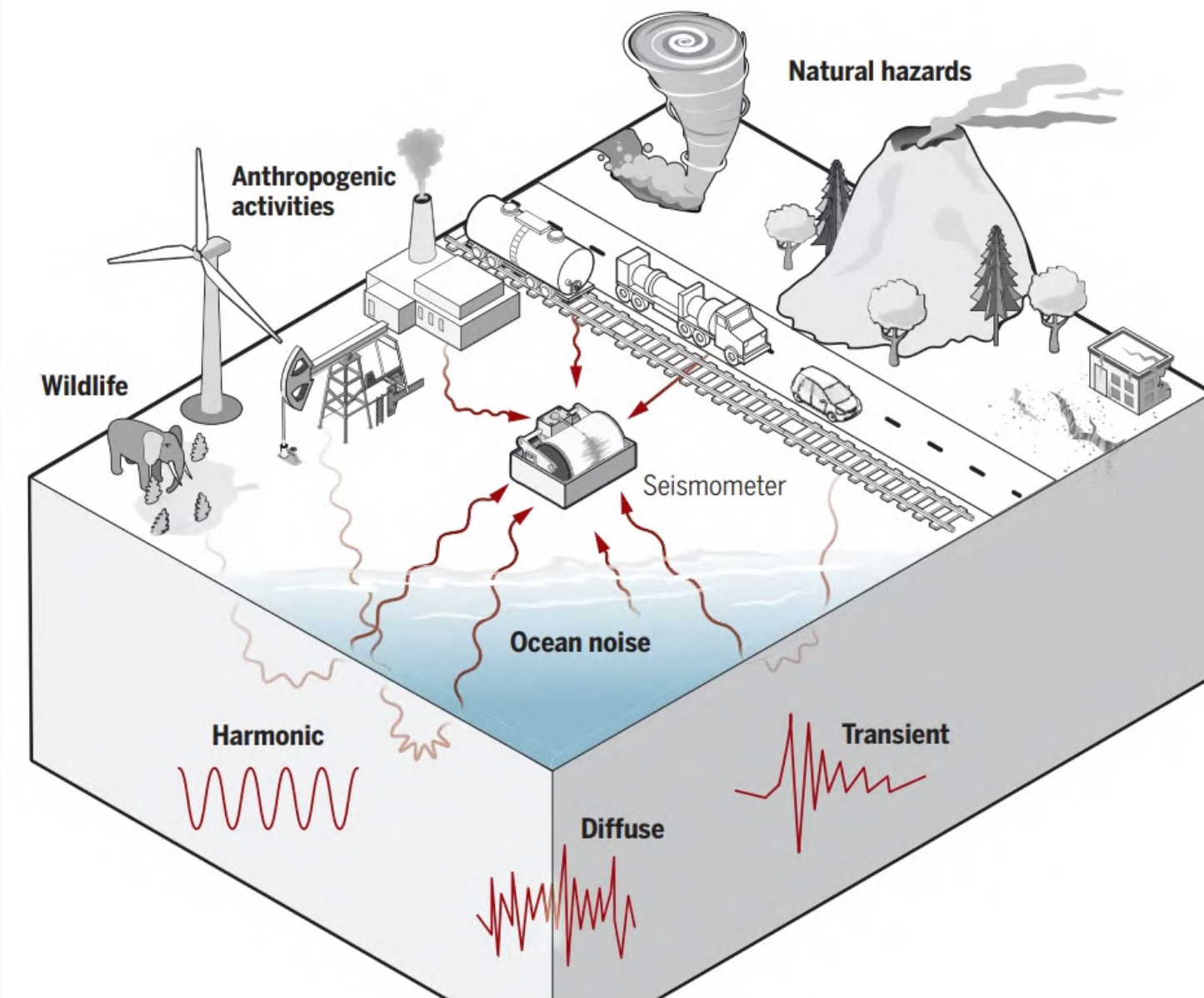
- open a console (bash, console, cmd, etc) - and run the command
- *git clone <https://github.com/heinerigel/skience2026.git>*
- it'll create a skience26 folder structure with the codes for the practicals
- if you already did it before *now*, remove the skience26 folder and redo it (or you might need to run complicated git commands...)

DATA for practicals!

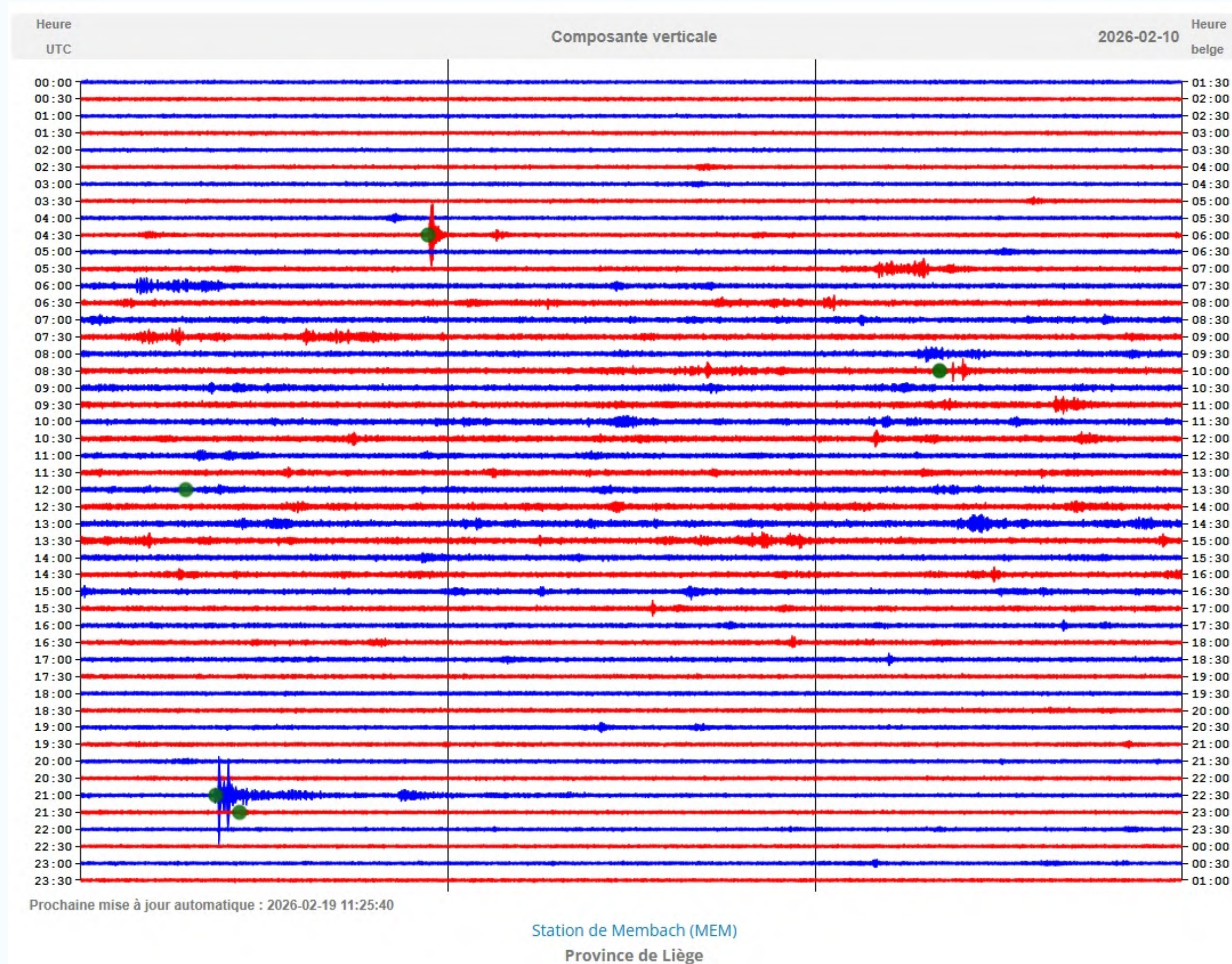
- Two USB disk are going around
- copy SKIENCE26_DATA to your computer in any local folder: **NOT IN THE skience26 git FOLDER!**

Part 1

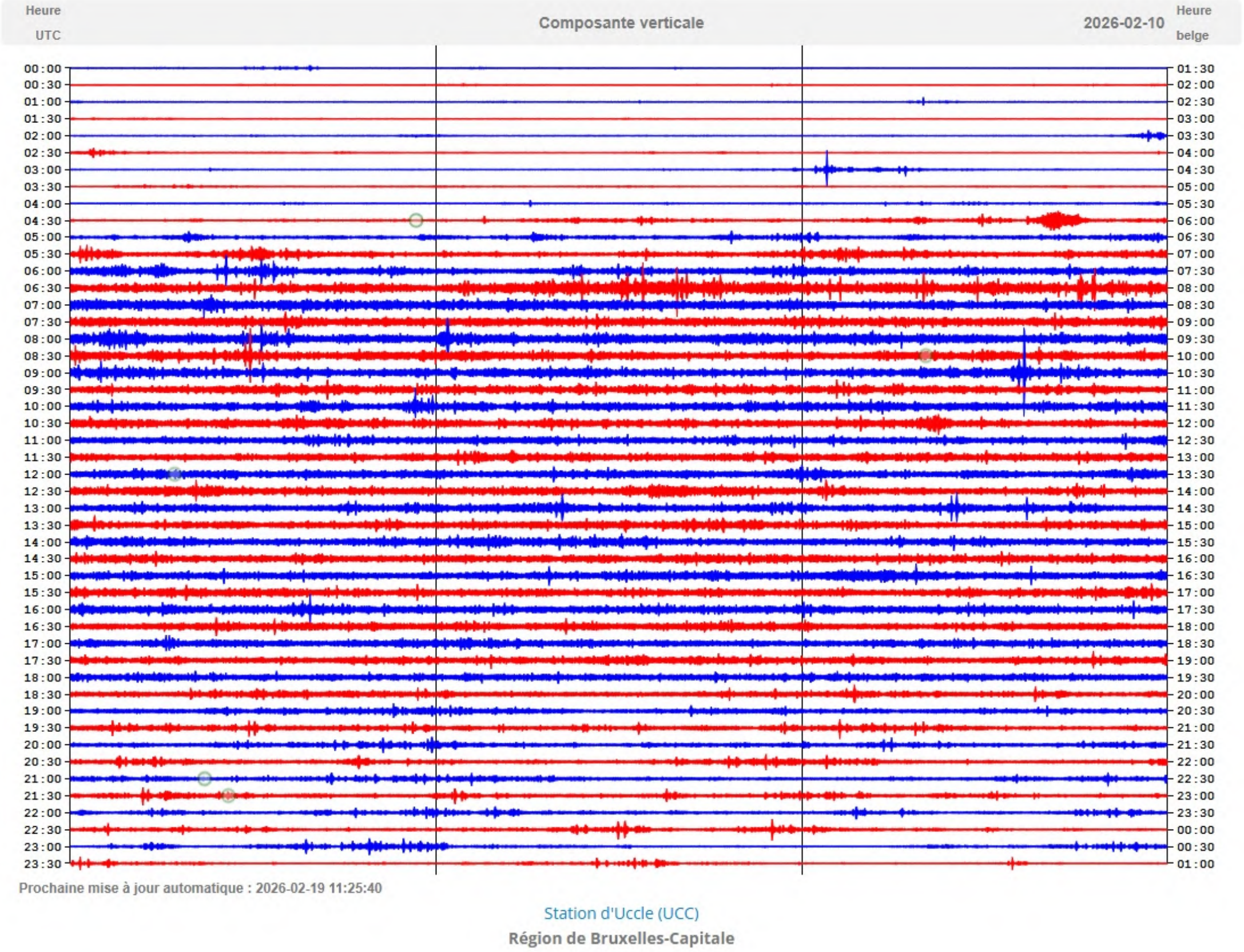
The Earth is NEVER silent



What are we looking at?



What are we looking at?



What is “ground motion”?

- Ground motion = displacement of the Earth's surface as a function of time
- Amplitudes range from **nanometres** (quiet night, far from cities) to **metres** (large earthquakes)
- Frequencies range from **~0.00001 Hz** (Earth's free oscillations) to **~100 Hz** (near-surface exploration)
- We measure it typically in three components: vertical + two horizontal (N-S, E-W)

The seismologist's view of the world

Signal vs noise: a false dichotomy

To a seismologist, **everything recorded is a signal** — the label "noise" depends on your question:

- Studying **earthquakes & faulting**? → microseisms are noise, tectonic tremor is signal
- Studying **crustal & mantle structure**? → microseisms are your data (ANT), earthquakes are your source
- Studying **the ocean**? → microseisms are a proxy for wave height and storm tracks
- Studying **glaciers & ice sheets**? → icequakes, calving events, and glacial tremor are signal; ocean microseisms are noise (maybe)
- Studying **permafrost & Arctic change**? → freeze-thaw cracking, frost quakes are signal
- Studying **volcanoes**? → harmonic tremor, VT, LP events are signal; Non-transient wavefield contains information (dv/v)
- Studying **cities & human activity**? → cultural noise *is* the signal (urban seismology, lockdown studies)
- Studying **atmospheric & climate processes**? → infrasound coupling, pressure loading are signal

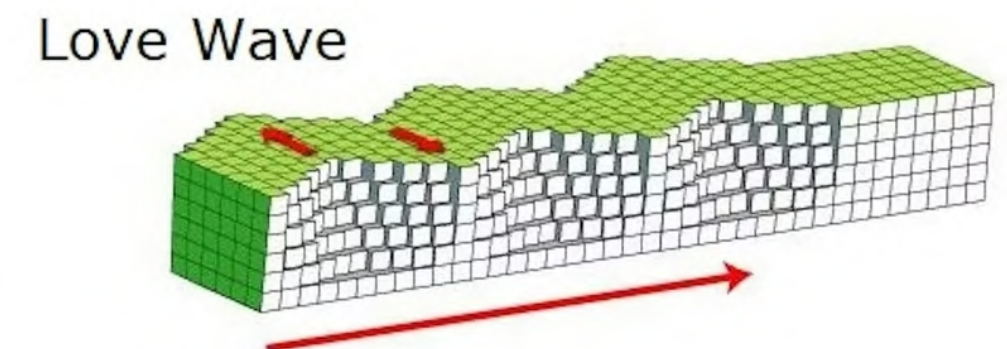
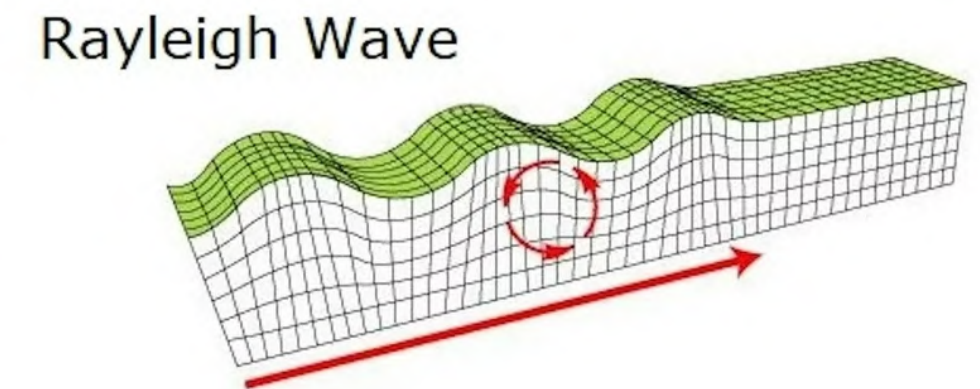
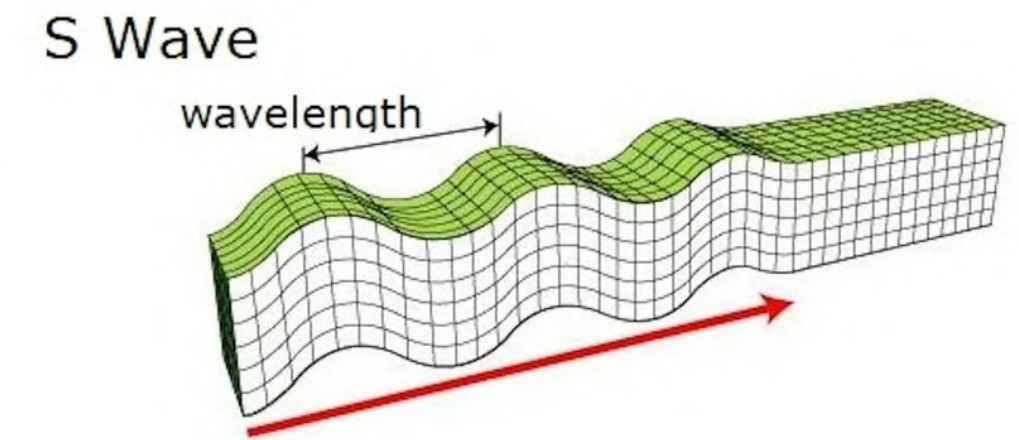
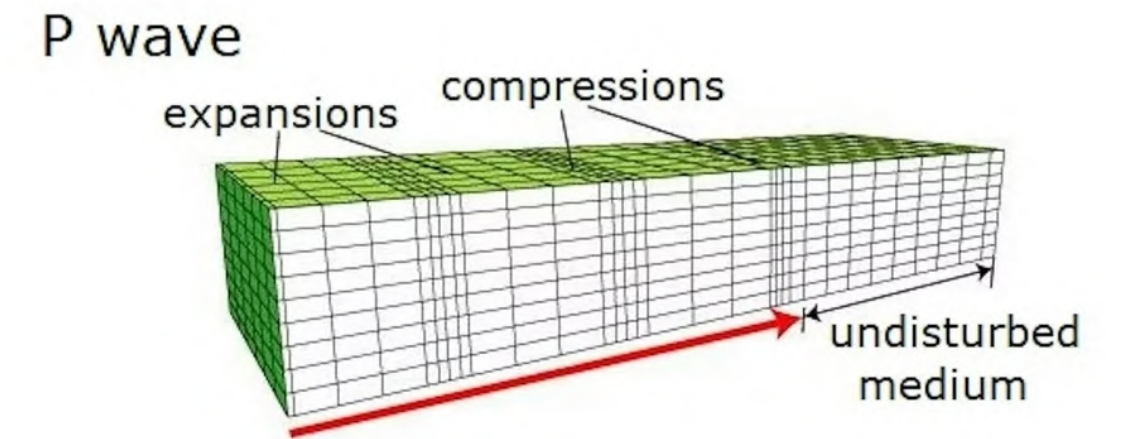
Same waveform, many sciences — **context defines signal !**

Wave types 101

P, S and surface waves - how energy travels through the Earth

Seismic waves = elastic waves propagating through the Earth

- **Body waves** travel through the interior:
 - *P-waves* (compressional) — push-pull motion, fastest, travel through solid & fluid
 - *S-waves* (shear) — side-to-side motion, slower, **cannot travel through liquids**
- **Surface waves** travel along the surface, slower but larger amplitude at distance:
 - *Rayleigh waves* — elliptical particle motion in the vertical plane
 - *Love waves* — horizontal shear, no vertical component
- Surface waves **disperse** — different frequencies travel at different speeds
- Surface waves dominate the ambient noise field



Part 2

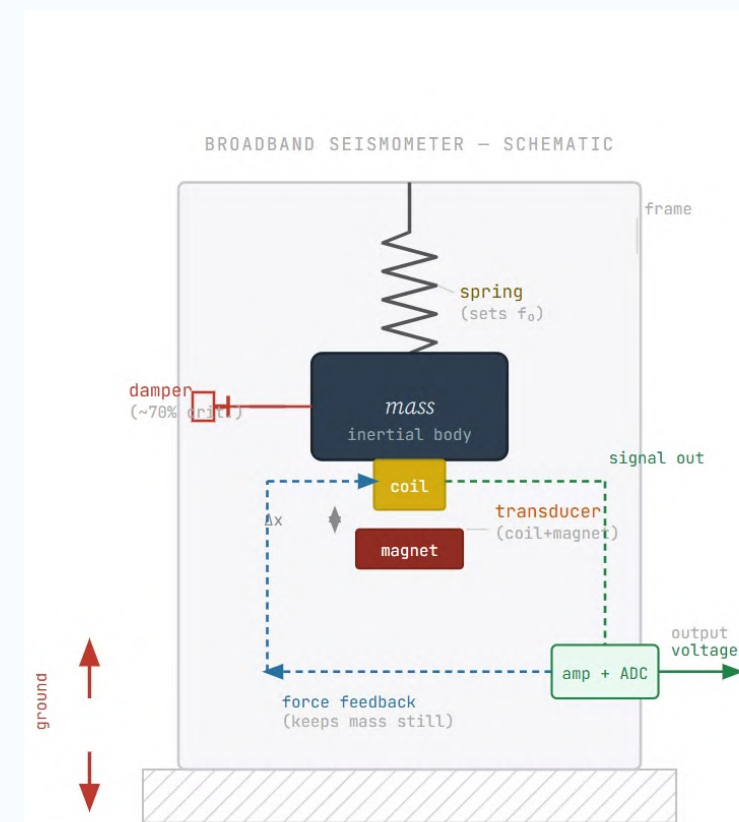
Measuring Ground Motion



The Seismometer

A mass that "stays still" while the ground moves around it

- Core principle: **inertia** — a suspended mass tends to remain stationary when the ground moves
- The relative motion between mass and frame = ground motion (to first order)
- Three key components:
 - *Suspension system* — spring or pendulum, sets the natural frequency f_0
 - *Damping* — prevents the mass from oscillating forever (usually ~70% critical damping)
 - *Transducer* — converts relative motion to voltage (electromagnetic coil, capacitive, etc.)
- Modern broadband seismometers use **force-feedback** (electrostatic or electromagnetic): the mass is kept stationary by an active feedback loop → the feedback current *is* the measurement
- Output: velocity (most common), displacement, or acceleration



- **Seismic mass**
Heavy inertial body. Tends to remain stationary due to inertia while the frame — attached to the ground — moves around it.
- **Spring suspension**
Sets the natural frequency f_0 of the pendulum. Broadband sensors have very low f_0 (seconds to tens of seconds).
- **Damper**
Prevents the mass from ringing at f_0 . Typically set to ~70% of critical damping for a flat response near the corner.
- **Transducer (coil + magnet)**
Converts relative motion Δx between mass and frame into a voltage. The coil moves through the magnetic field of the fixed magnet.
- **Force-feedback loop**
Actively keeps the mass stationary by injecting a restoring force. The feedback current is proportional to ground velocity — this is the output signal.

Broadband vs Short-Period instruments

Sensitivity ranges and trade-offs

- **Short-period (SP)** sensors: $f_0 \sim 1\text{--}2\text{--}4.5\text{--}10$ Hz, cheap, robust, small — used in local/regional networks and dense arrays
- **Broadband (BB)** sensors: $f_0 \sim 0.01\text{--}0.1$ Hz, flat response from ~ 100 s to 50 Hz — the workhorse of global seismology
- **Very broadband (VBB)**: f_0 down to ~ 360 s (e.g. STS-1) — needed for normal modes and Earth tides
- **Strong motion (accelerometers)**: not velocity but acceleration, no clipping during large earthquakes — essential for engineering seismology
- Key trade-offs:
 - Bandwidth $\leftrightarrow$ cost & size
 - Sensitivity $\leftrightarrow$ dynamic range (clipping)
 - Low $f_0 \leftrightarrow$ susceptibility to tilt and atmospheric noise



The Geophone

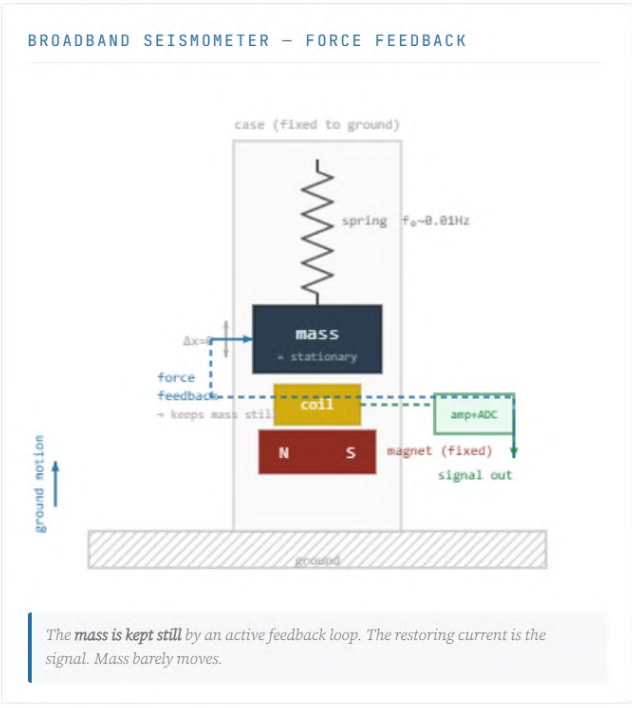
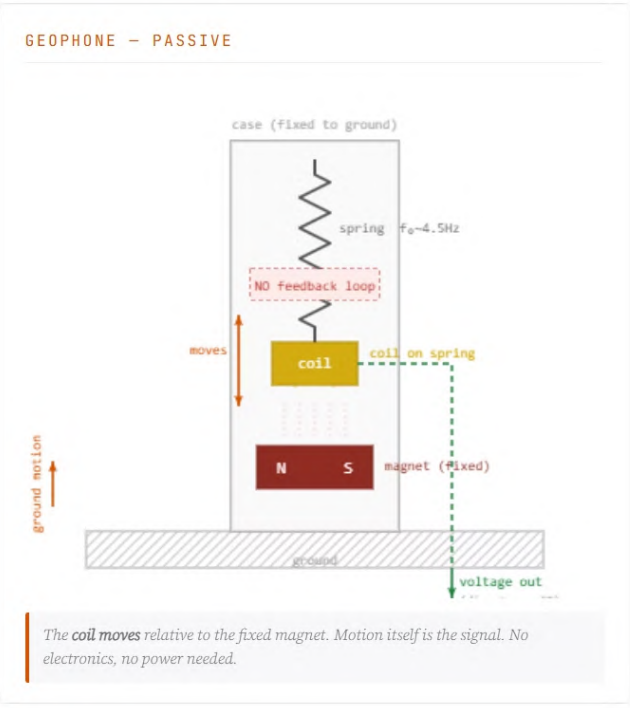
Velocity sensor — cheap, robust, used in exploration and dense arrays

- A geophone is essentially a **short-period seismometer** with no force feedback
- A coil suspended on a spring moves relative to a fixed magnet → voltage proportional to velocity
- Natural frequency f_0 typically **1, 2, 4.5 or 10 Hz** — low response below f_0
- Key advantages: **rugged, cheap (~10–100 €), small, passive** (no power needed for the sensor itself)
- Used in: oil & gas exploration, dense nodal arrays, microseismic monitoring, DAS companions
- Modern "nodal" systems pack geophone + digitizer + battery in one ~1 kg unit → 1000s deployable
- Limitation: no broadband response — could be blind to teleseisms, surface waves at long periods



Geophone vs Broadband seismometer

	Broadband Seismometer (force-feedback)	Geophone
Moving part	Mass fixed (force-feedback keeps it still)	Coil moves, magnet fixed to case
Transducer	Feedback current proportional to ground velocity	Coil in magnetic field → velocity voltage
Feedback	Active force-feedback loop	None — pure passive
Natural freq f_0	Low: 0.008–0.1 Hz	High: 1, 4.5 to 10 Hz
Bandwidth	Far below f_0 thanks to feedback	Above f_0 only, very low below f_0
Sensitivity	Very high	Lower
Size / weight	kg-scale, temperature-controlled/isolated(?)	~50–200g, palm-sized
Cost	5,000–50,000 €	10–100 €
Power	Needs power for electronics	Passive (no power)



MEMS & other modern sensors

From lab to smartphones

- **MEMS** (Micro-Electro-Mechanical Systems): silicon microfabricated proof mass on a capacitive suspension
 - Measures acceleration (accelerometer)
 - Mass, spring, and damper etched onto a chip — mm scale
 - Self-noise $\sim 10\text{--}100\times$ worse than broadband, but good enough for strong motion and dense networks
 - Found in every smartphone — the basis of citizen seismology (e.g. MyShake, LastQuake)
- **Optical sensors**: laser interferometry for ultra-low noise — next frontier for gravimetry and geodesy

Bundles

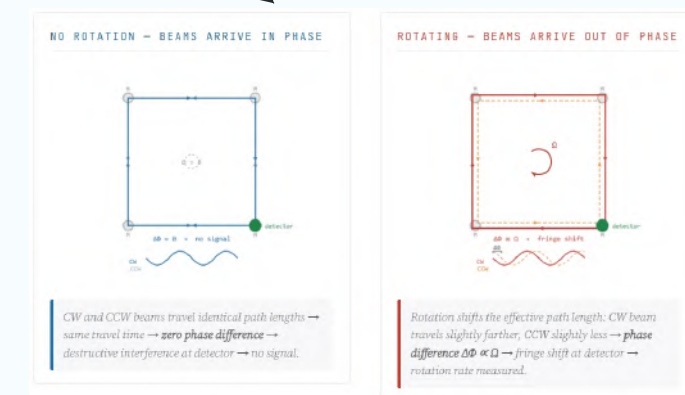
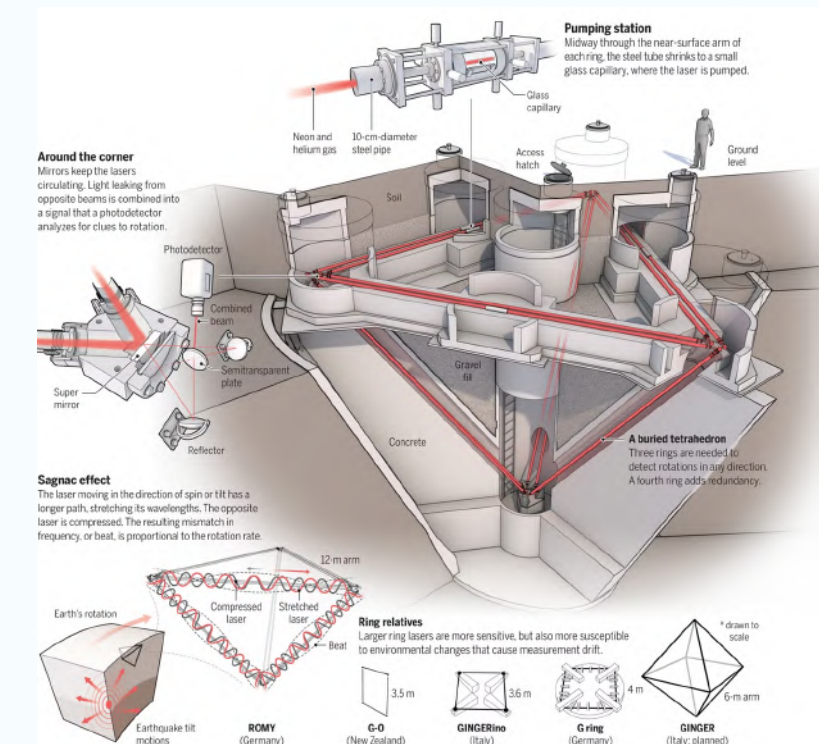
- **Raspberry Shake**: MEMS or classical geophone + Raspberry Pi — $\sim \$400$, plug-and-play seismic station - Essentially Lennartz patent
- **Nodal systems** (e.g. Fairfield ZLand, SmartSolo): self-contained geophone + digitizer + battery, $\sim 1\text{ kg}$
- Trade-off: cost & accessibility $\leftrightarrow$ sensitivity & low-frequency performance



Rotational sensors

Measuring rotation rate, not just translation

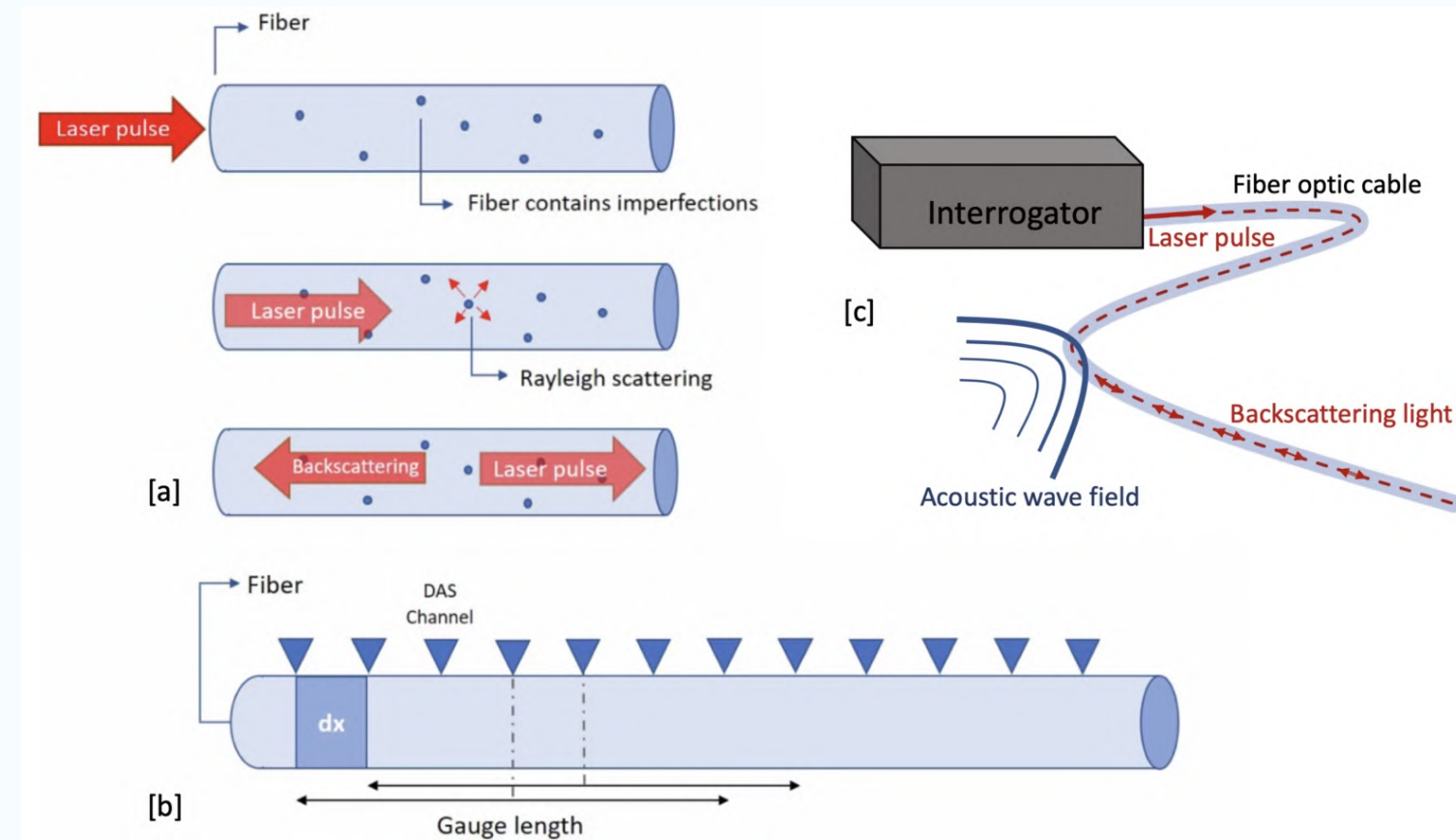
- Classical seismometers measure **translational** ground motion (displacement, velocity, acceleration)
- The ground also **rotates** — tilt, twist, and rocking — invisible to translation sensors alone
- Rotation rate measured in **rad/s**, typically nrad/s sensitivity needed for seismology
- Two main technologies:
 - *Ring laser gyroscope (RLG)* — laser beams counter-propagating in a closed cavity; rotation causes a phase shift (Sagnac effect) → e.g. ROMY (Munich), G-ring (Wettzell)
 - *Fiber optic gyroscope (FOG)* — same Sagnac principle but in coiled optical fiber → portable
- Why it matters:
 - Separates **tilt from horizontal acceleration** (a classical ambiguity in broadband sensors)
 - Wavefield **decomposition**: independent retrieval of P, S, surface wave components
 - Input for **6-component seismology** (3 translation + 3 rotation)
 - Still an emerging field — few permanent deployments worldwide (but here it's the good place to learn about it 🤝)



Distributed Strain Sensors

The new buzz

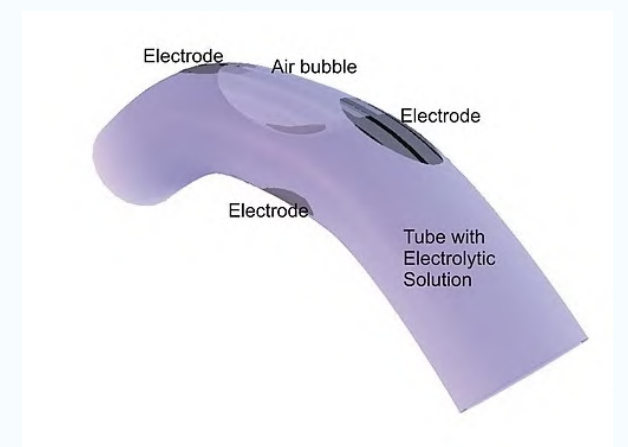
Check Tom's presentation on Tuesday morning :-)



Tiltmeters

Long-period ground tilt: between seismology and geodesy

- Tiltmeters measure the **angle of the ground surface** relative to the local gravity vector
- Units: **nrad to μ rad** — equivalent to mm/km of surface deformation
- Why it matters for seismology:
 - Long-period tilt contaminates broadband horizontal seismograms (tilt $\leftrightarrow$ horizontal acceleration ambiguity)
 - Essential for **volcanic unrest** monitoring (magma intrusion tilts the surface), landslide monitoring, etc
 - Records **solid Earth tides** and **atmospheric loading**
- Two main types:
 - **Electrolytic** / *bubble tiltmeter* — fluid seeks level, capacitive readout; cheap, moderate sensitivity
 - **Long-baseline** *tiltmeter* — water tube tens to hundreds of metres long; extremely sensitive, used in observatories (e.g. Strasbourg, BFO)
 - *can be transformed to a* **Borehole** *tiltmeter* — installed at depth to reduce surface noise
- Bandwidth: DC to ~ 0.01 Hz — complementary to seismometers at the low-frequency end
- Key limitation: **very sensitive to temperature and atmospheric pressure** — requires careful installation



Gravimeters

Measuring gravity changes: superconducting, spring-based, and atom interferometry

- Gravimeters measure **(variations in) gravitational acceleration g** — not ground motion, but mass redistribution
- Units: **nm/s²** (nanogals) — Earth's $g \approx 9.81 \text{ m/s}^2$, we measure variations at the 10^{-9} level
- Why it matters for seismology & geophysics:
 - **Solid Earth tides** — elastic deformation of the Earth by Moon & Sun ($\sim 200 \text{ nm/s}^2$)
 - **Atmospheric & hydrological loading** — mass above and around the station
 - **Co-seismic & post-seismic deformation** — permanent gravity changes after large earthquakes
 - **Normal modes** — gravimeters are superb low-frequency seismometers below $\sim 1 \text{ mHz}$
- Four main types:
 - **Spring gravimeter** — mass on a spring, measures spring extension; portable, $\sim 10 \text{ nm/s}^2$ sensitivity
 - **Superconducting gravimeter (SG)** — magnetically levitated Nb sphere in a superconducting coil; $\sim 0.1 \text{ nm/s}^2$ sensitivity; best long-period instrument available
 - **Absolute gravimeters (AG)** - free-falling corner and interferometric laser
 - **Atom interferometry (AIG)** — free-falling laser-cooled atoms; absolute, drift-free; emerging technology
- SGs are the **best low-frequency seismometers** on Earth — record normal modes for weeks after M8+



Instrument self-noise

Internal noise sources: thermal (Brownian), electronic, digitizer quantization

- Every sensor has a **noise floor** — a level below which ground motion cannot be distinguished from internal noise
- Three main sources:
 - ***Thermal (Brownian) noise*** — random motion of the suspension mass due to thermal energy kT ; fundamental physical limit; sets the floor for mechanical sensors
 - ***Electronic noise*** — amplifier noise, Johnson noise in resistors, ADC input noise
 - ***Digitizer quantization noise*** — finite bit resolution of the ADC; flat white noise floor set by $\text{LSB}/\sqrt{12}$
- Self-noise is **instrument-specific** and frequency-dependent — typically worst at low frequencies
- Thermal noise floor scales as: $S_x(f) \propto kT / (m \cdot Q \cdot \omega_0)$ — heavier mass, higher Q , lower noise
- This is why broadband sensors are large and heavy, and why cooling helps (SGs, future quantum sensors)

Noise floor curves — how to measure self-noise

Sleeman method, two/three-station method

- You cannot measure instrument self-noise with a single sensor — ground motion and self-noise are **indistinguishable**
- Solution: use **multiple co-located sensors** and exploit the fact that self-noise is **uncorrelated** between instruments
- **Two-sensor method** (coherence method):
 - Cross-correlate two co-located sensors → coherent part = ground signal, incoherent part = noise
 - Requires sensors with similar self-noise levels; gives a noise **upper bound**
- **Three-sensor method** (Sleeman et al., 2006):
 - With 3 sensors A, B, C: self-noise of A = $P_{AA} - (P_{AB} \cdot P_{AC} / P_{BC})$
 - No assumption needed about which sensor is quieter
 - Gives individual self-noise estimate for each sensor
- Output: self-noise PSD in dB rel. $(\text{m/s}^2)^2/\text{Hz}$ — directly comparable to Peterson curves
- Practical requirement: sensors must be on the **same pier**, thermally isolated, over a sufficiently long time window

What is an instrument response?

The sensor doesn't measure ground motion directly

- Every seismometer records a **voltage** (or counts after digitisation) — not ground displacement or velocity directly
- The relationship between ground motion input and recorded output is the **instrument response**
- It is a function of **frequency** — the sensor amplifies some frequencies more than others
- Mathematically: **Output(f) = Input(f) × Response(f)**
- Or equivalently in the time domain: output = input convolved with the impulse response
- The response encodes:
 - **Sensitivity** — overall gain (V per m/s, or counts per m/s)
 - **Shape** — how gain varies with frequency (flat band, roll-offs, notches)
 - **Phase** — time delay introduced at each frequency
- To recover true ground motion: **deconvolve** the response → "remove the instrument response"
- Stored in standard formats: **SEED/StationXML** — poles, zeros, gain stages

Poles & Zeros, gain & sensitivity

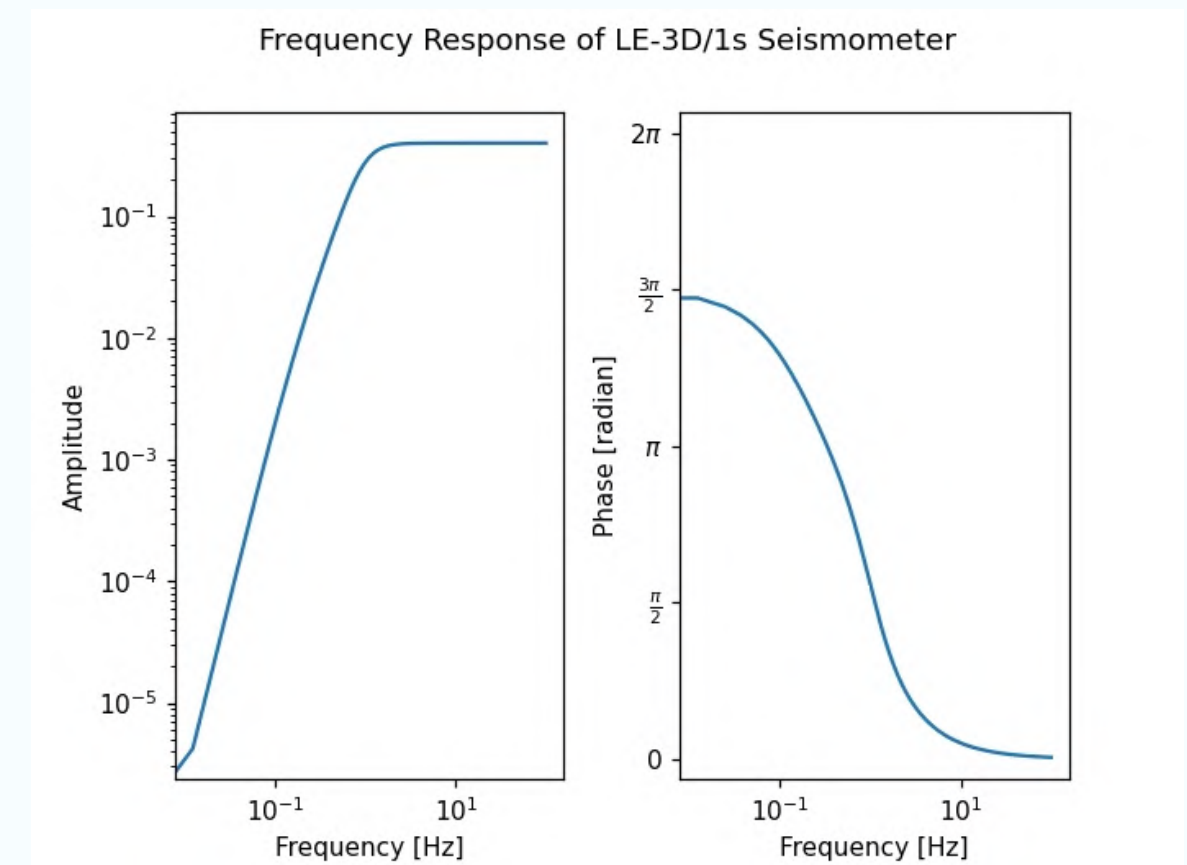
sparing you the full math

- The instrument response $H(f)$ is most compactly described in the **Laplace/z domain** as a rational function
- **Zeros** — frequencies where the response goes to zero (numerator roots)
- **Poles** — frequencies where the response would diverge (denominator roots) — physically prevented by damping
- For a broadband velocity seismometer:
 - 2 zeros at the origin ($s=0$) $\rightarrow$ response $\propto f^2$ at very low $f \rightarrow$ high-pass behaviour
 - 2 complex conjugate poles near $f_0 \rightarrow$ the resonance corner
- **Sensitivity** — the overall scalar gain at a reference frequency (e.g. 1 Hz): in $V/(m/s)$ or $counts/(m/s)$
- **Gain stages** — the full response is a chain: ground motion $\rightarrow$ sensor $\rightarrow$ amplifier $\rightarrow$ digitiser (ADC) $\rightarrow$ each stage has its own gain and response
- Total sensitivity = product of all stage gains
- Stored in **StationXML** as: poles, zeros, normalization factor A_0 , sensitivity

The response in frequency domain

Why a flat response is the goal

- In the frequency domain the instrument response $H(f)$ has two components:
 - **Amplitude response $|H(f)|$** — how much each frequency is amplified or attenuated
 - **Phase response $\angle H(f)$** — time delay introduced at each frequency
- The ideal seismometer has a **flat amplitude response** across its target band → all frequencies recorded with equal fidelity
- And a **linear phase response** → constant time delay, no waveform distortion
- Real sensors deviate from this ideal near the corner frequencies f_0 and f_c :
 - Amplitude rolls off → sensitivity lost
 - Phase rotates → waveform shape distorted
- **Why it matters practically:**
 - Comparing amplitudes across stations requires knowing $|H(f)|$
 - Measuring arrival times requires knowing $\angle H(f)$
 - Computing magnitude, moment tensor, source spectra all require response removal
- Modern FIR digital filters (in the digitiser) are designed to be **linear phase** — a key advantage over analogue filters



Minimum vs Linear phase?

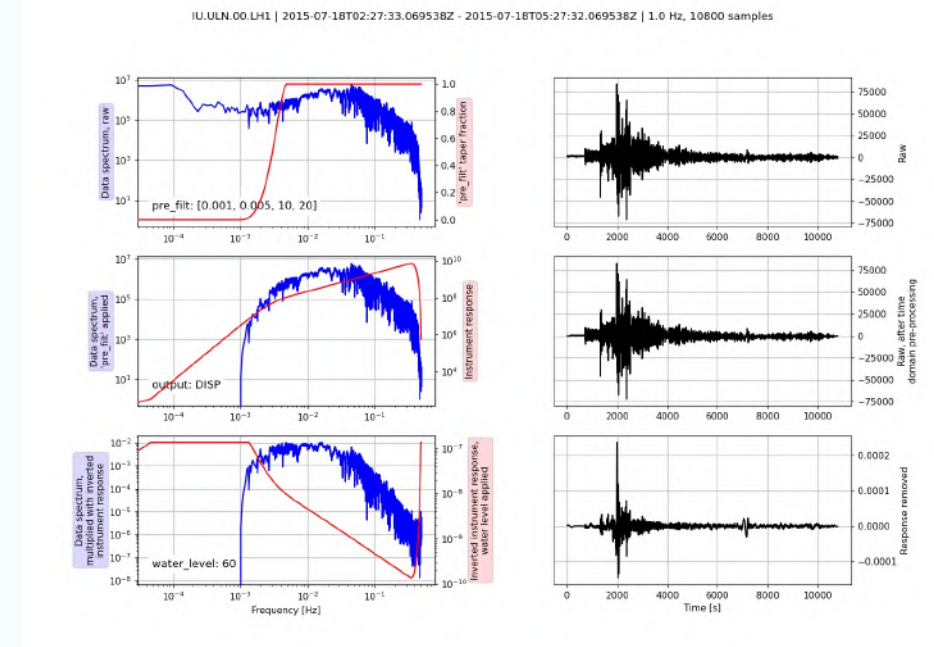
A practical choice when configuring your datalogger

	Linear Phase	Minimum Phase
Phase response	Linear – constant delay at all frequencies	Non-linear – frequency-dependent delay
Waveform shape	Preserved	Distorted near corners
Impulse response	Symmetric, has precursor (fixed delay)	One-sided, causal, no precursor
Group delay	Fixed latency (seconds to tens of seconds)	Near-zero at onset
Real-time use	✗ Too much delay	✓ Low latency → early warning, triggering
Research use	✓ Clean waveforms, no correction needed	✗ Needs phase correction
Ambient noise / ANT	Preferred	Acceptable with care
Waveform inversion	Preferred	Needs correction
StationXML	Must document FIR coefficients + delay	Must document FIR coefficients

Removing the instrument response

Deconvolution — getting back to physical units (m/s, m/s²)

- Recorded signal: $\mathbf{V(f)} = \mathbf{U(f)} \times \mathbf{H(f)}$
- To recover ground motion: $\mathbf{U(f)} = \mathbf{V(f)} / \mathbf{H(f)}$ → division in frequency domain = deconvolution ⚠ ⚠
- In practice:
 - FFT the recorded trace
 - Divide by $H(f)$ (complex division — corrects both amplitude and phase simultaneously)
 - Inverse FFT → ground motion in physical units
- ⚠ ⚠ **Critical caveat:** where $|H(f)| \rightarrow 0$ (outside the passband), division amplifies noise catastrophically
- Solution: **pre-filtering** (water-level or cosine taper) before deconvolution — limit the frequency band to where the sensor is sensitive
- Output can be **displacement, velocity, or acceleration** depending on how many times you integrate or differentiate
- In ObsPy: `trace.remove_response(output='VEL', pre_filt=(f1,f2,f3,f4))`
- The `pre_filt` corners define the safe frequency band — choosing them correctly is critical



Part 3

From Ground Motion to Numbers

H

H

Z

Orientation (Z = vertical)

Instrument (H = high-gain seismometer)

Band (H = 80-250 sps, broadband)

Try Pitch

F $\geq$ 1000 sps broadband	H High-gain seismometer	Z Vertical
G $\geq$ 1000 sps short period	L Low-gain seismometer	N North-South
D 250-1000 sps short period	N Accelerometer (strong motion)	E East-West
C 250-1000 sps broadband	G Gravimeter	1 Horizontal, arbitrary azimuth
E 80-250 sps short period	M Mass position	2 Horizontal, arbitrary azimuth
H 80-250 sps broadband	D Barometer / pressure	3 Non-standard (~vertical)
B 10-80 sps broadband	F Hydrophone	T Transverse (rotated)
S 10-80 sps short period	I Infrasound	R Radial (rotated)
M 1-10 sps	J Rotational	U/V/W Triaxial native frame
L ~1 sps long period	S Strain meter	A/B/C Triaxial non-standard
V ~0.1 sps very long period	A Tiltmeter	
U ~0.01 sps ultra long period	P Geophone (short period)	

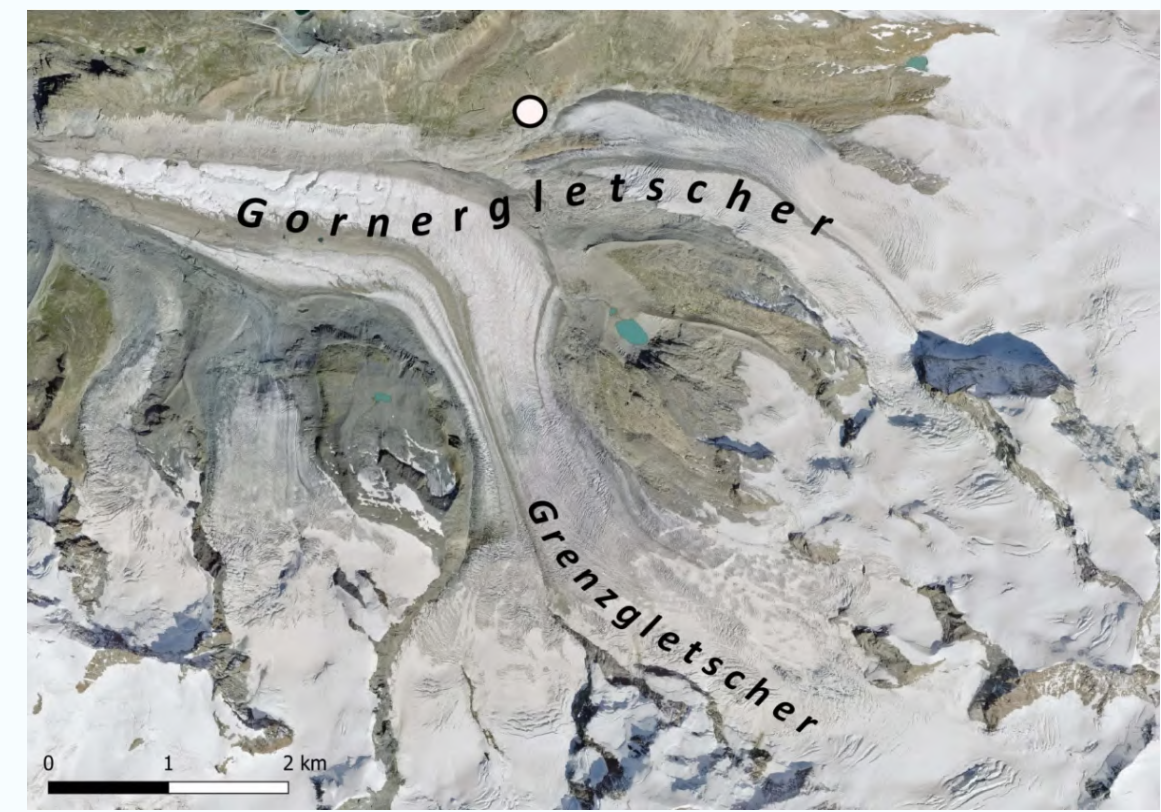
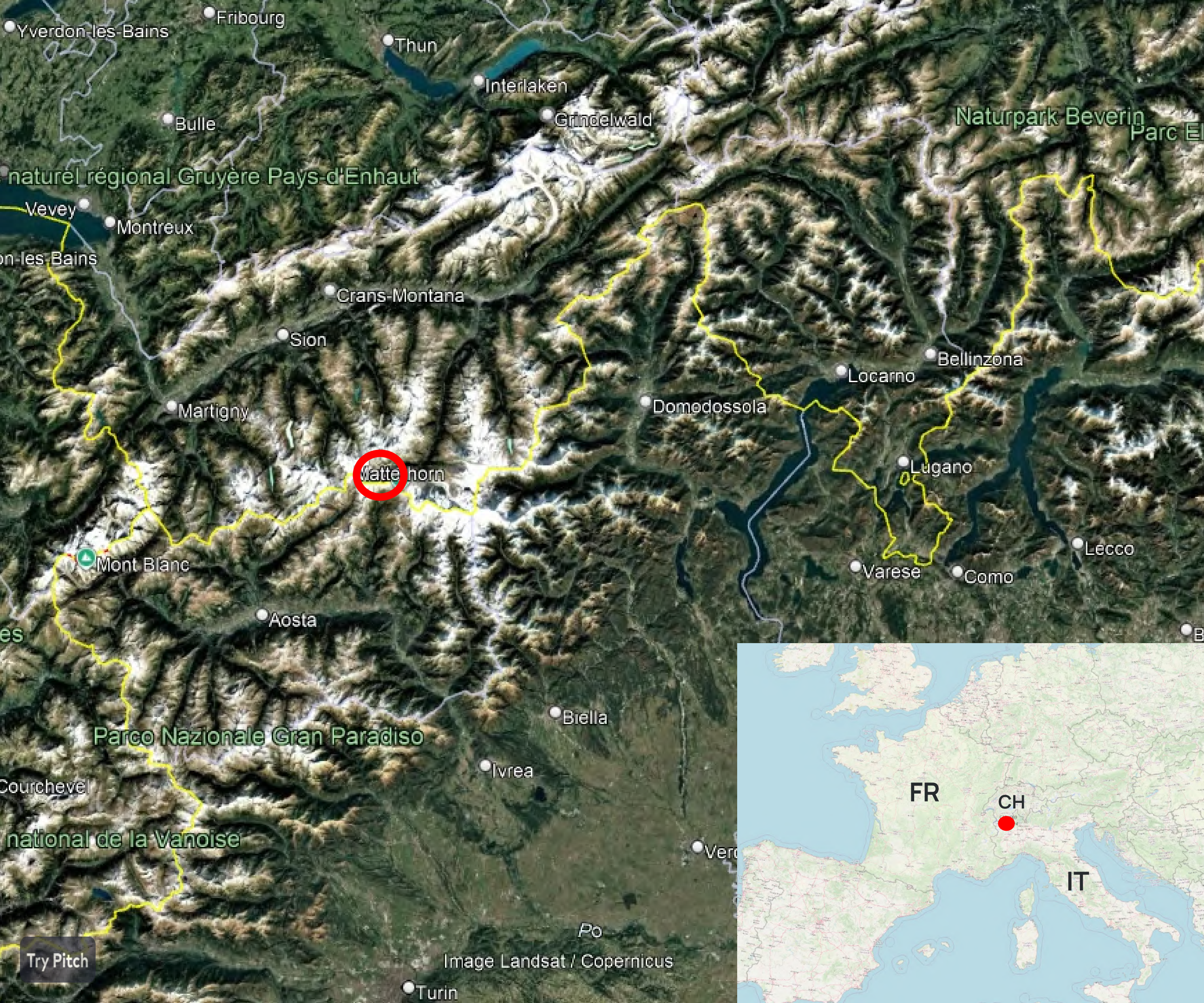
practical

The Grenzgletscher

The data - the glacier → see introduction

Why Monitor with Seismological Tools?

- **from Icequakes** into glacier dynamics: seismic signals reveal crevasse formation, basal slip, subglacial water pressure changes, and ice flow speed
- **Glacial lake outburst floods:** Seismic monitoring captures precursor signals before sudden drainage events — thousands of icequakes/day recorded during Gornersee drainage episodes (Walter et al., Cambridge Journal of Glaciology)
- **Basal icequakes** reveal subglacial water pressure variations — critical for understanding how meltwater injection drives flow acceleration
- **Early warning potential:** Seismic clustering patterns can signal structural failure zones before they become visible on the surface
- **Because it's fun!**

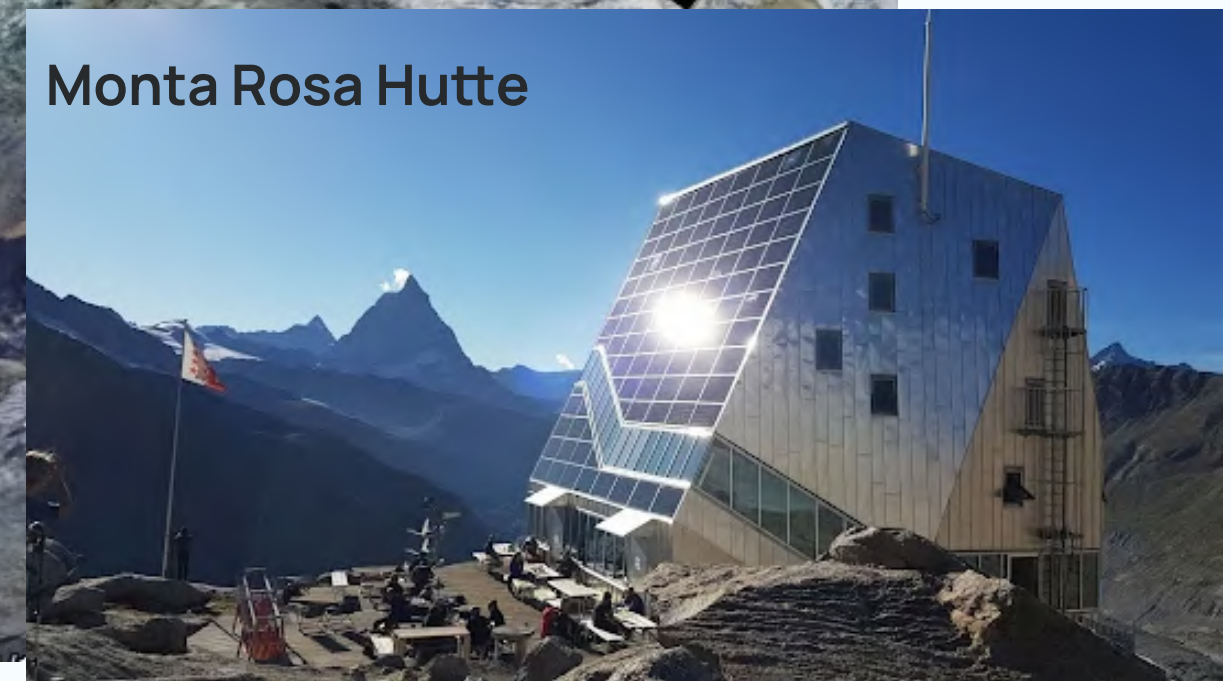
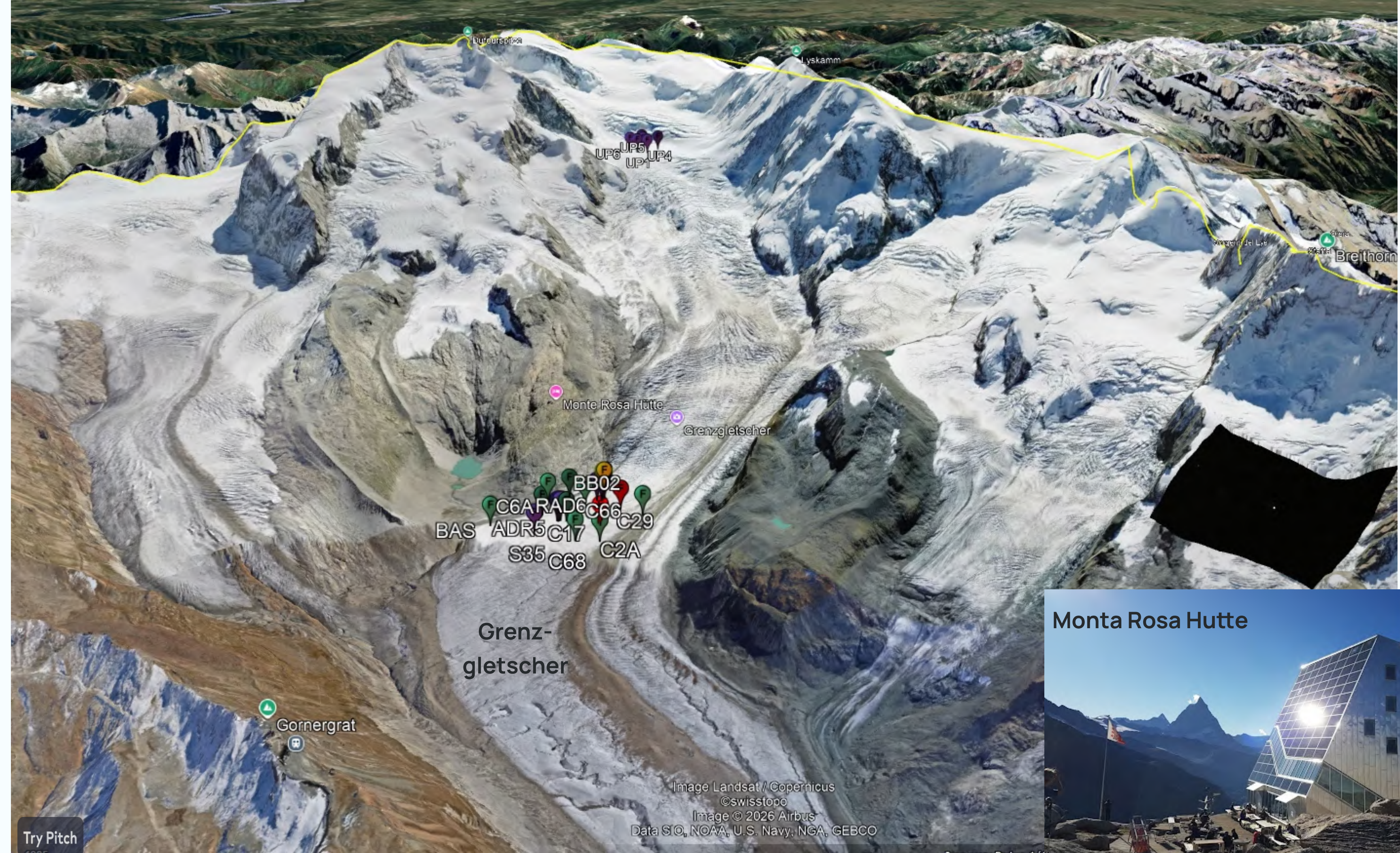


GRENZGLETSCHER — Glacier Info

- **Location:** Monte Rosa massif, Swiss-Italian border, Canton Valais, near Zermatt
- **Length:** ~14–15 km (as the main flow tributary of the Gorner system)
- **Width:** ~0.5–1.5 km
- **Elevation range:** ~2,200 m to 4,634 m

Melting — Historical Info

- Retreating **~50 m per year** in length; losing **several meters of thickness** annually



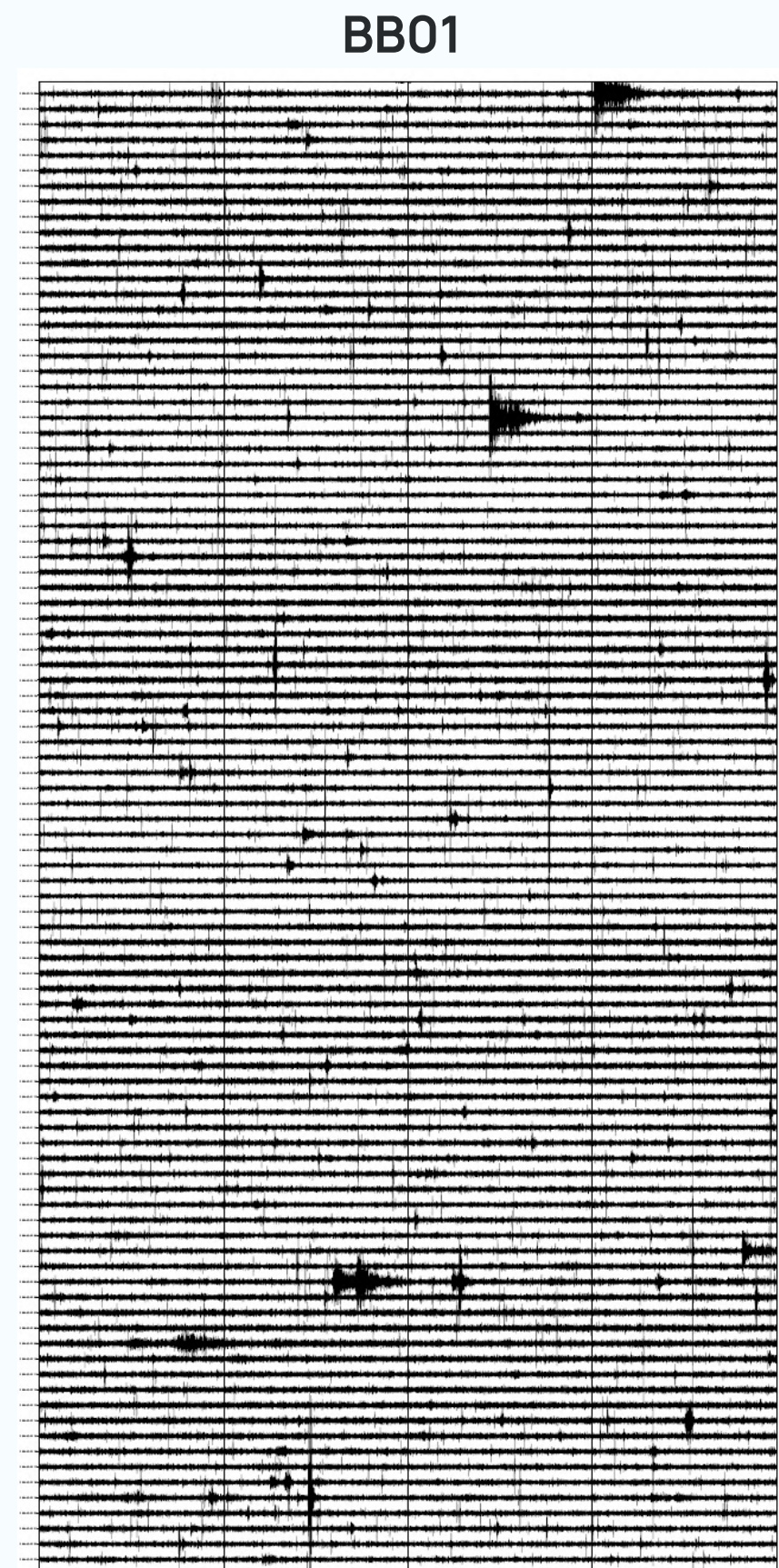
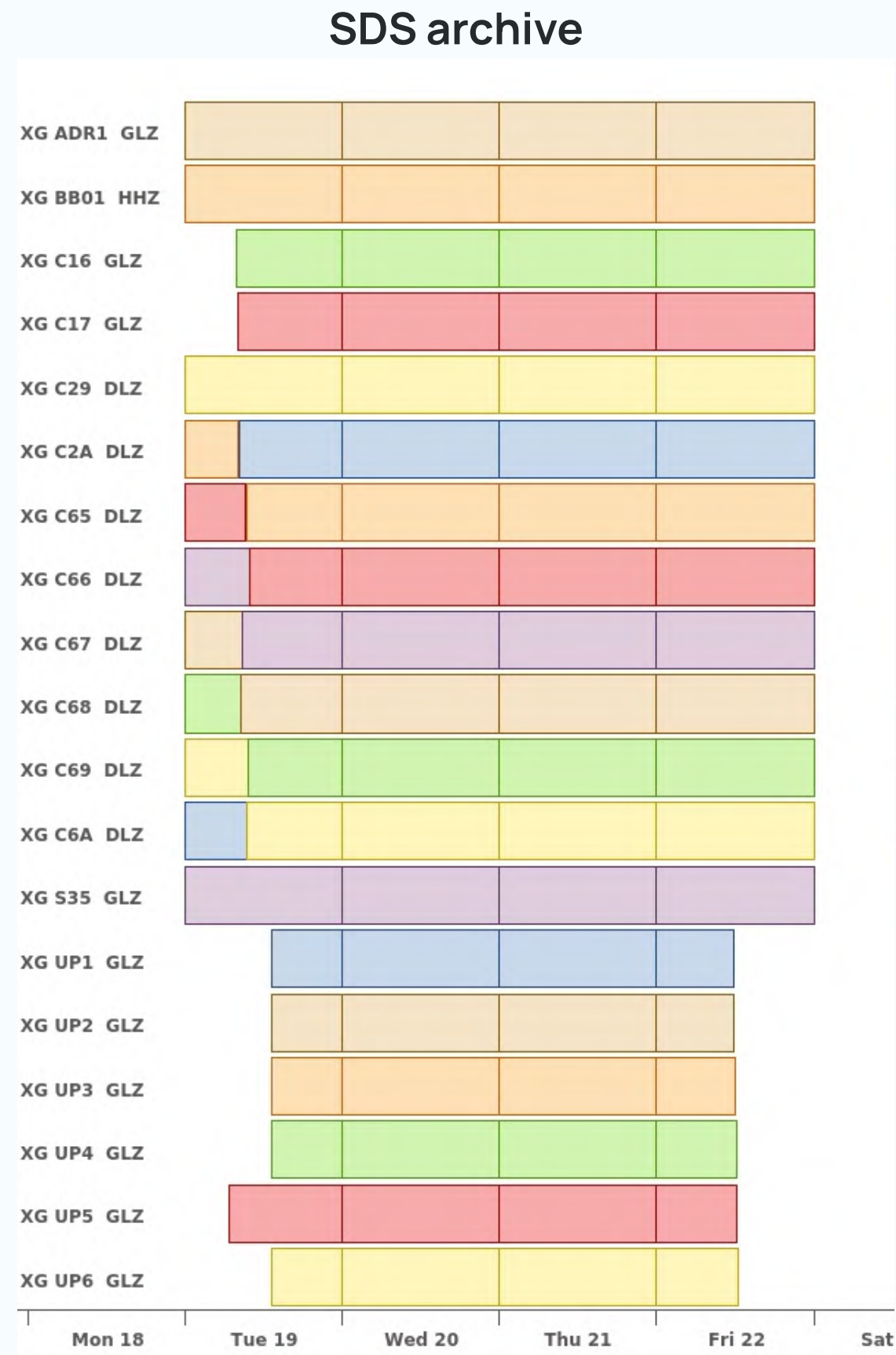


Gornersee (ice-dammed lake at the Gorner/Grenz confluence) – formed annually since the 1950s, draining in summer outburst floods – has now ceased to exist as the glaciers separated



The data - the glacier

- 4 days of data
- 19 stations
- Lower part
 - 1 ADR array with SmartSolo **ADR1**
 - 1 ADR array with 4 Broadband Nanometrics Trillium compact **BB01**
 - **C16** → **C6A** SM6-3D geophones
- Upper part: 6 SmartSolo 5Hz IGU 16HR **UP**



practical

The Grenzgletscher

Los geht's!



Open a console

cd to the folder where the notebooks are downloaded

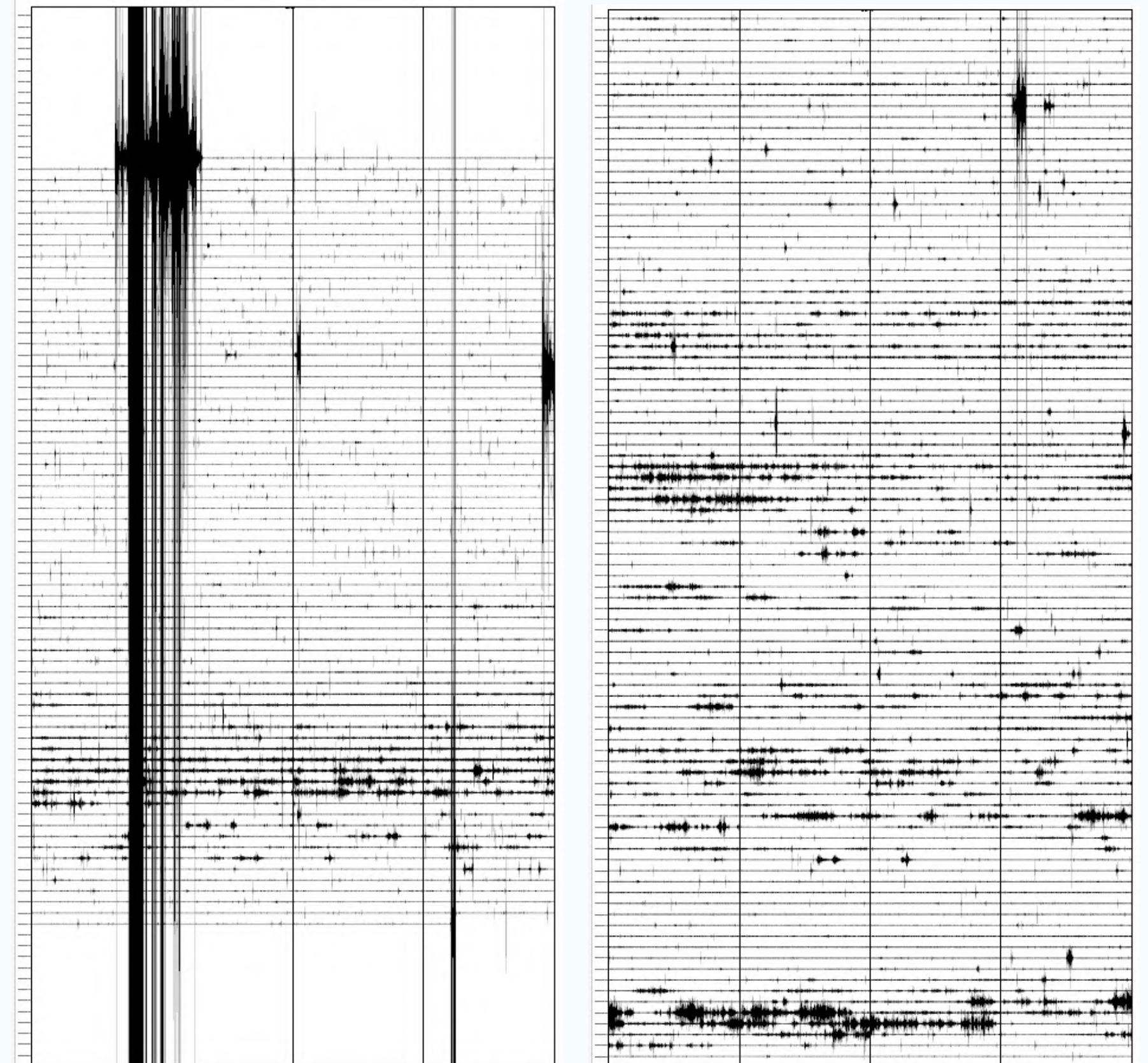
start "jupyter notebook"

Try Pitch ⚡⚡⚡⚡ Set computer to "maximum performance" (~~"battery saving"~~) 🔋 🔋 🔋 🔋

TIME - NOTEBOOK 00: EXPLORE YR DATA

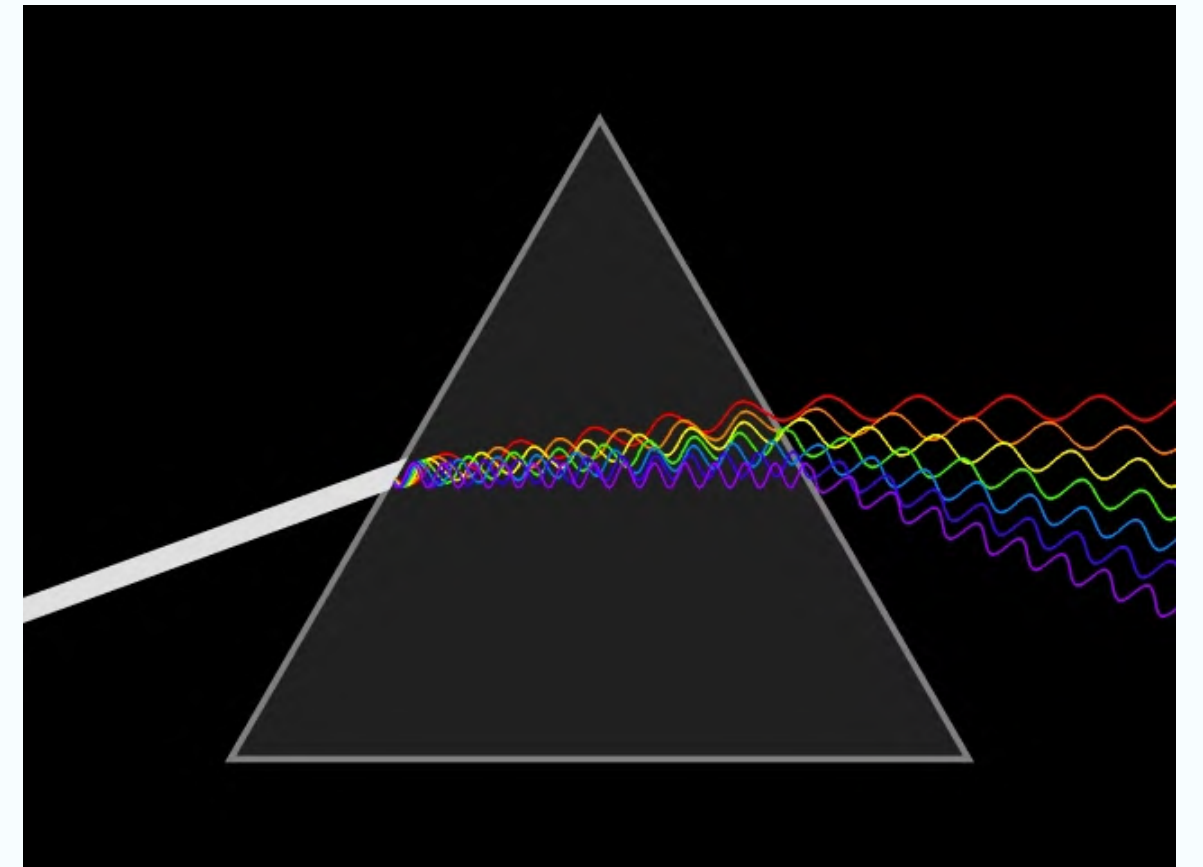
Using noise to probe local site structure

- Give SDS path
- Load yr data from the SDS archive
- Choose station
- Obspy: merge all days
- plot hourly time series
- trim if needed (installation noise)
- do it all-in-one



Part 4

Seismic Noise Signal & Background



What is “Seismic Noise”

Make some noise

- **Seismic noise** = all recorded ground motion that is not the target signal
- But "noise" is observer-dependent — the same vibration is noise to one scientist, signal to another (→ echo from Part 1)
- Physically: the Earth's surface is **never at rest** — continuous broadband ground motion exists at all times and all locations
- Sources span an enormous range:
 - Natural: ocean waves, wind, atmospheric pressure, tides, rivers, volcanism
 - Anthropogenic: traffic, industry, trains, cities
 - Instrumental: sensor self-noise, thermal drift, tilt
- Noise is **always present** — even during the quietest night at the most remote station
- The noise field is **not random white noise** — it has a rich, structured spectrum with distinct peaks and slopes
- Understanding and characterising noise is a prerequisite for any serious seismological analysis

The noise spectrum – Peterson's NLNM/NHNM

The "noise floor" of the Earth

- Peterson (1993) analysed recordings from **75 globally distributed quiet stations** to define the envelope of Earth's natural noise
- **NLNM** (New Low Noise Model) – the quietest ground ever observed; a physical lower bound on seismic noise
- **NHNM** (New High Noise Model) – the noisiest "normal" conditions; above this something is likely wrong with the installation
- The corridor between NLNM and NHNM defines the **dynamic range** any useful seismic network must cover
- Key features of the spectrum:
 - **~0.003–0.01 Hz** – Earth's free oscillations, tidal forcing; very quiet at good sites
 - **~0.05–0.1 Hz** – primary microseismic peak (ocean swell, ~20 s period)
 - **~0.1–0.3 Hz** – secondary microseismic peak (dominant noise source globally, ~7 s period)
 - **~1–10 Hz** – cultural/anthropogenic noise window; highly site-dependent
 - **>10 Hz** – local noise, wind, sensor self-noise

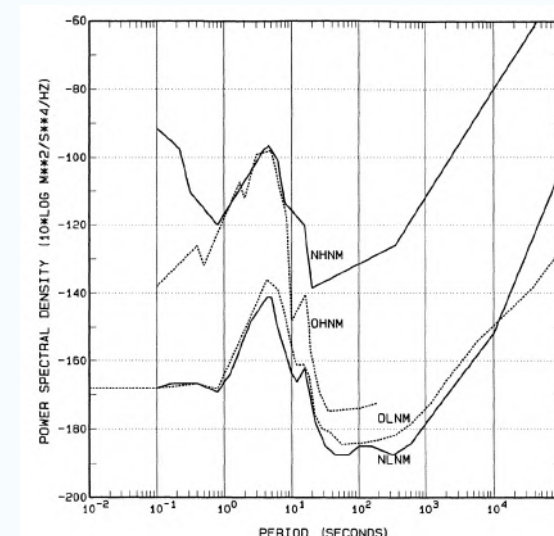


Figure 15.--Comparison of new and original noise models. The OHNM was based on SRO data, which accounts for the large difference between the OHNM and NHNM at long periods.

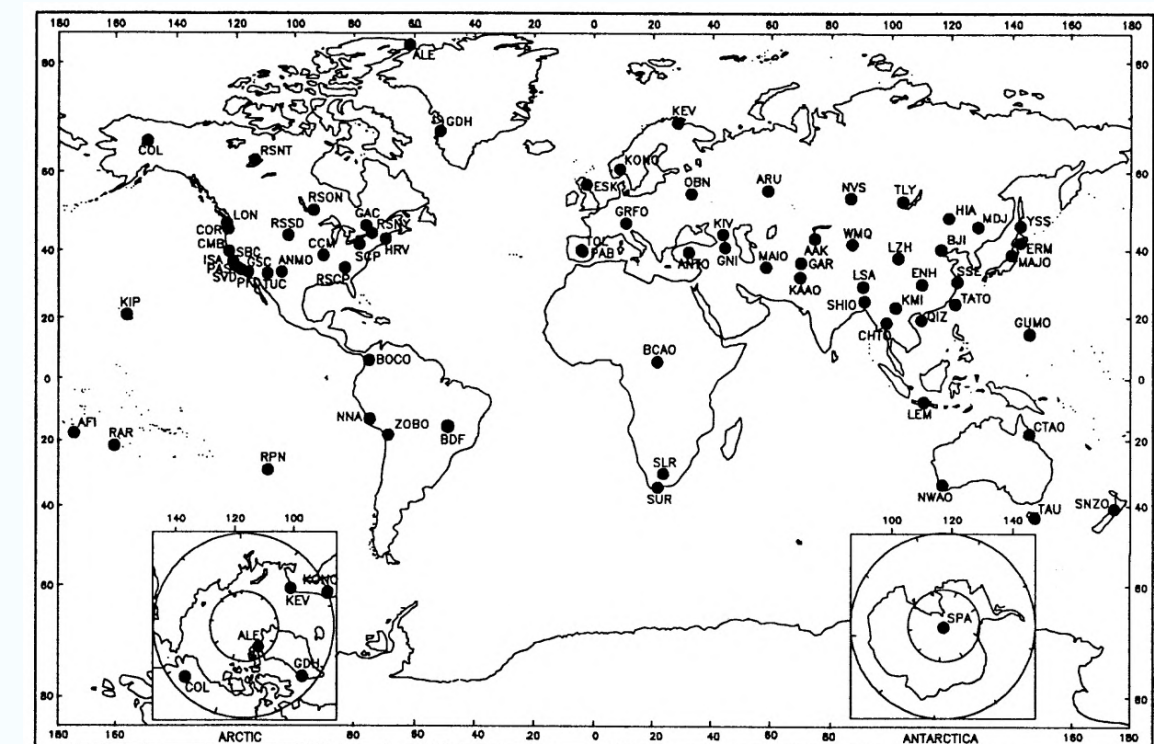


Figure 1.--Map showing the approximate locations of the stations used in this report.

Self-noise vs. Peterson NLNM

Can your sensor "see" the Earth's quiet? - The ultimate benchmark for any seismic sensor

- The **Peterson NLNM** (1993) is the quietest ground motion ever observed — it defines the physical noise floor of the Earth itself
- A sensor is **scientifically useful** only where its self-noise lies below the NLNM
- This single plot tells you everything about an instrument's worth:
 - **Frequency range** — where does the sensor respond at all?
 - **Dynamic range** — gap between self-noise and NLNM
 - **Blind spots** — where the sensor cannot hear the Earth even at its quietest
- Key benchmarks:
 - STS-1: below NLNM from ~0.003–10 Hz
 - STS-2: below NLNM from ~0.01–10 Hz
 - Geophone (4.5 Hz): below NLNM only above ~2–3 Hz
 - MEMS: rarely below NLNM — useful for strong motion only
 - Superconducting gravimeter: below NLNM from DC to ~1 mHz — unrivalled at ultra-long periods

Tidal, atmospheric and other long-period noise

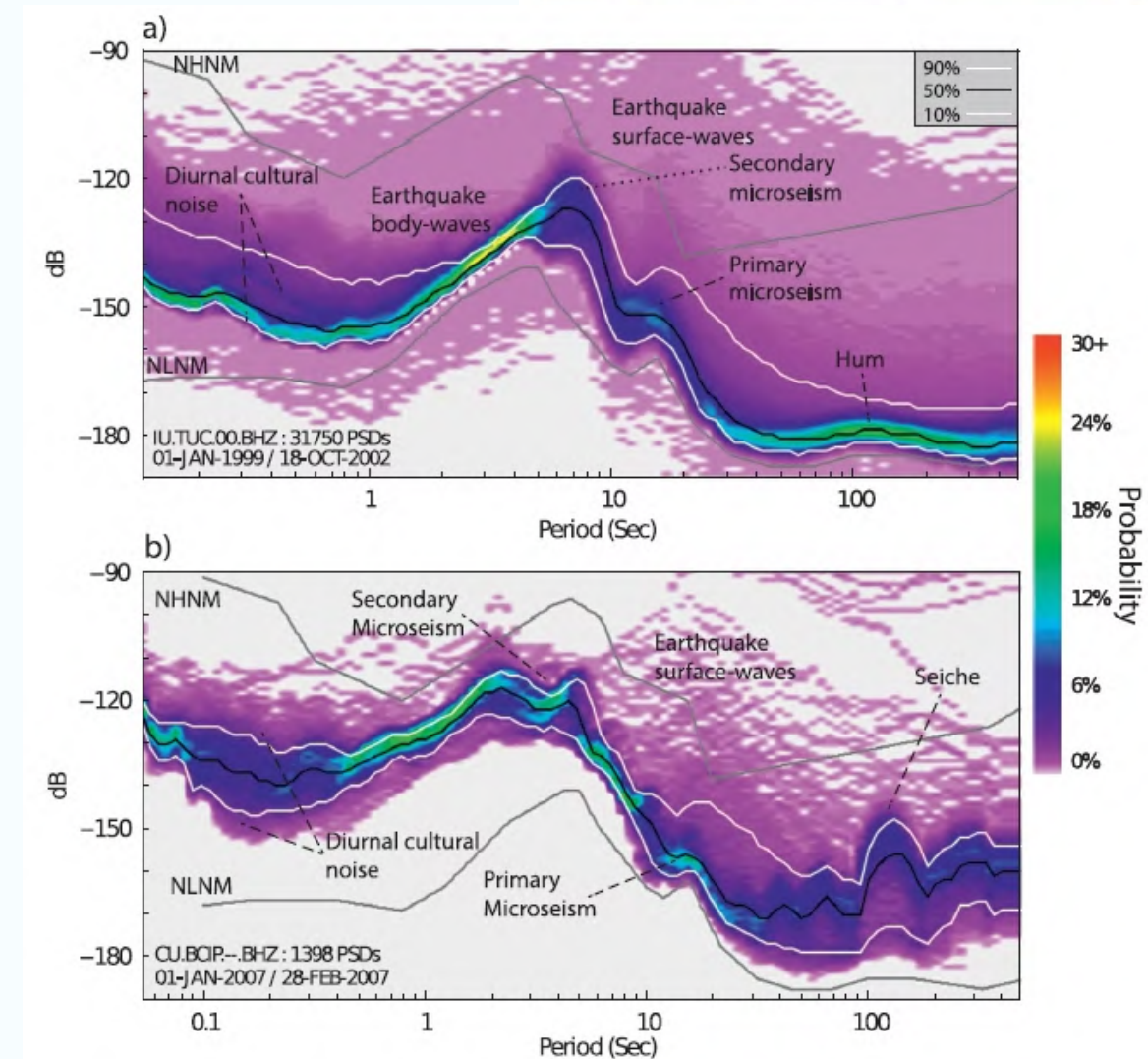
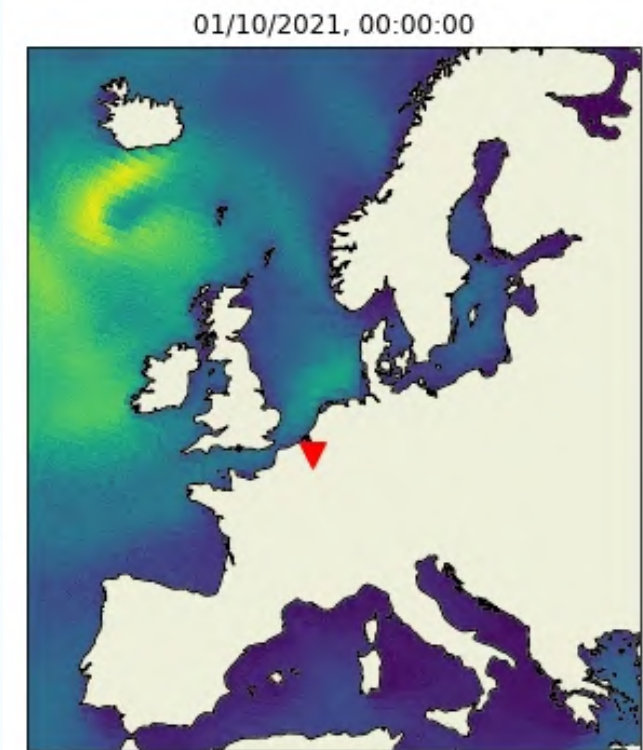
The low-frequency world

- Below ~ 0.01 Hz (periods > 100 s) the noise field is dominated by **deterministic or quasi-deterministic** signals rather than random noise
- **Solid Earth tides:**
 - Elastic deformation of the Earth by gravitational forcing of Moon & Sun
 - Dominant periods: 12.4 h (M2), 24 h (S1), 25.8 h (K1), ~ 14 days (Mf)
 - Amplitude: $\sim 200\text{--}400$ nm/s² in gravity; mm-scale deformation
 - Fully predictable from ephemeris $\rightarrow$ used to **calibrate and validate** gravimeters and tiltmeters
- **Atmospheric loading:**
 - Surface pressure variations flex the crust $\rightarrow$ ground deformation
 - $\sim 1\text{--}10$ nm/s² per hPa; correlated with weather systems
 - Must be corrected for in precise geodetic and gravimetric work
- **Hydrological loading:**
 - Groundwater, soil moisture, snow load $\rightarrow$ seasonal crustal deformation
 - $\sim$ mm-scale vertical displacement; $\sim$ weeks to months period
- **Infrasound coupling:**
 - Atmospheric acoustic waves couple into the ground at low frequencies
 - Relevant for volcano monitoring and nuclear test detection
- **Earth's hum:**
 - Continuous free oscillations excited by ocean-atmosphere coupling, even without earthquakes
 - $\sim 2\text{--}7$ mHz; discovered in the 1990s, still not fully understood

Primary and secondary microseismic peaks

Ocean waves and their interaction with the solid Earth

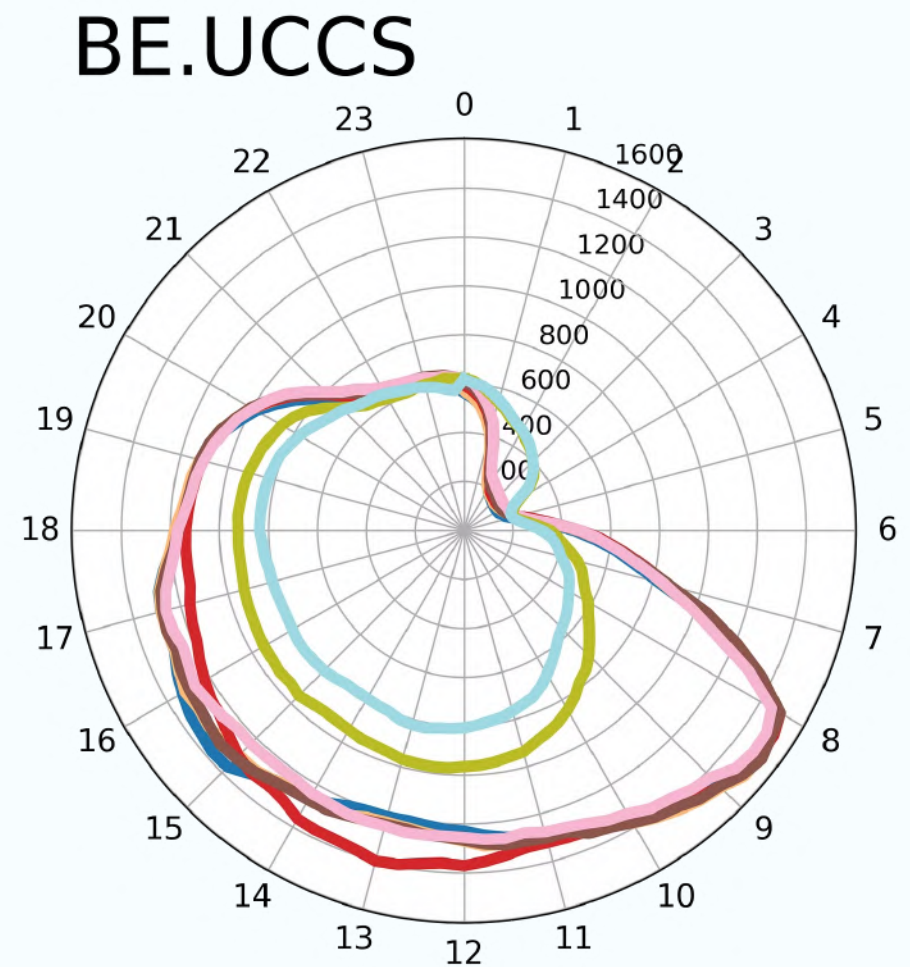
- The two most prominent features of the global noise spectrum – visible at **every** seismic station on Earth
- **Primary microseismic peak (0.05–0.1 Hz, T10–20 s):**
 - Generated by **direct ocean wave loading** on shallow seafloor near coastlines
 - Frequency = ocean wave frequency
 - Dominates at coastal stations; weaker inland
 - Rayleigh waves propagating from the coast
- **Secondary microseismic peak (0.1–0.3 Hz, T5–8 s):**
 - Generated by **non-linear interaction of opposing ocean wave trains** (Longuet-Higgins)
 - Frequency = **twice** the ocean wave frequency → hence "**secondary**"
 - Generated in the open ocean – can be thousands of km from any coast
 - Dominant noise source globally; visible even at inland stations
 - Both Rayleigh and Love waves
 - Amplitude varies seasonally – stronger in winter (stormier oceans), weaker in summer
 - Used as a **proxy for ocean wave height** and storm tracking



Cultural noise

Trains, traffic, industry — the human footprint on the ground

- Anthropogenic noise dominates the **1–50 Hz** band at most stations worldwide
- Sources:
 - **Road traffic** — continuous broadband hum, peaks at ~2–10 Hz; strongest during rush hours
 - **Trains** — impulsive + harmonic, strong ~1–15 Hz; detectable tens of km away
 - **Industry** — machinery harmonics, rotating equipment; often monochromatic lines in spectrogram
 - **Wind turbines** — ~1–2 Hz fundamental + harmonics; blades' RPM
 - **Construction** — impulsive, broadband spikes
 - **Pumping / hydraulic activity** — very low frequency, 0.1–1 Hz
- Strong **diurnal pattern** — quieter at night and weekends; this is exploited in noise studies
- The **COVID-19 lockdowns (2020)** provided a global natural experiment:
 - 50% reduction in high-frequency noise at urban stations worldwide
 - Equivalent to gaining ~1 week of extra quiet per year at some stations
- Cultural noise is **not just a nuisance** — its spatial and temporal patterns map human activity (urban seismology)



Wind, rain & environmental coupling

When the atmosphere talks to the ground

- **Wind:**

- Direct pressure on the vault/surface → tilt and deformation of the ground → long-period noise (0.001–0.1 Hz)
- Wind on trees, buildings, vegetation → transmitted to shallow soil → broadband noise
- Bernoulli effect on sensor housing → pressure fluctuations directly on the mass
- One of the dominant noise sources at surface broadband stations above ~50 mHz
- Mitigation: wind shields, burial, vault design

- **Rain:**

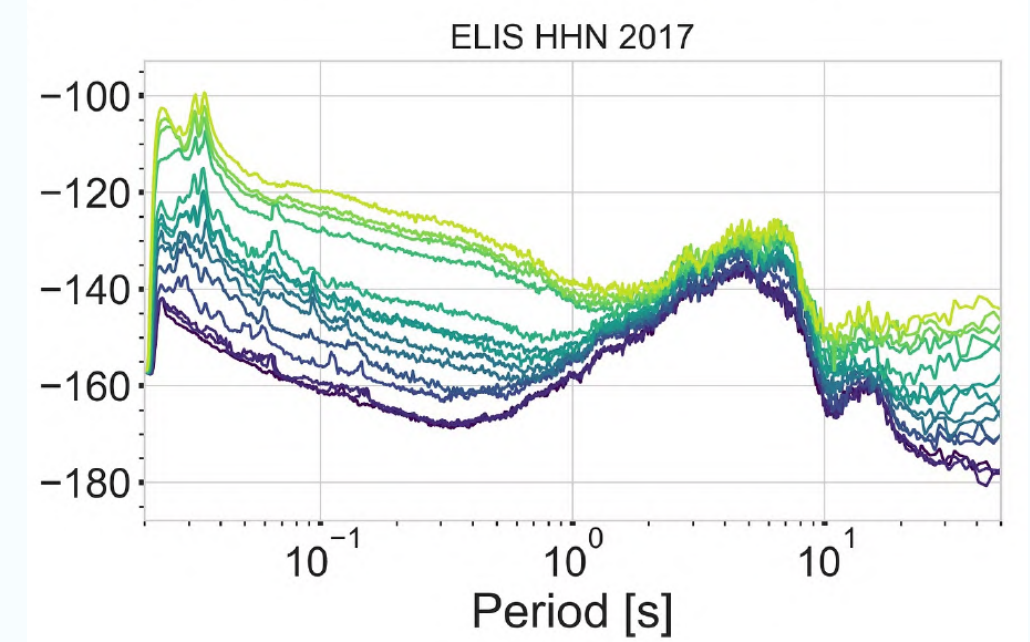
- Impulsive broadband noise from raindrop impacts on the ground and vault (1–50 Hz)
- Saturates shallow soil → changes V_s → apparent velocity shifts in noise monitoring
- Heavy rain can mask small local earthquakes

- **Temperature:**

- Thermal expansion of the vault, pier, and instrument → slow drift in tilt and output
- Daily thermal cycle visible as a ~24h signal — can mimic tidal periods
- Reason broadband sensors are thermally insulated and buried

- **Barometric pressure:**

- Direct elastic loading of the crust ($\sim 1 \text{ nm/s}^2$ per hPa)
- Couples into tiltmeter and gravimeter records strongly
- Correctable if a co-located barometer is available — standard practice at geodetic observatories



practical

The Grenzgletscher

Los geht's!



Open a console

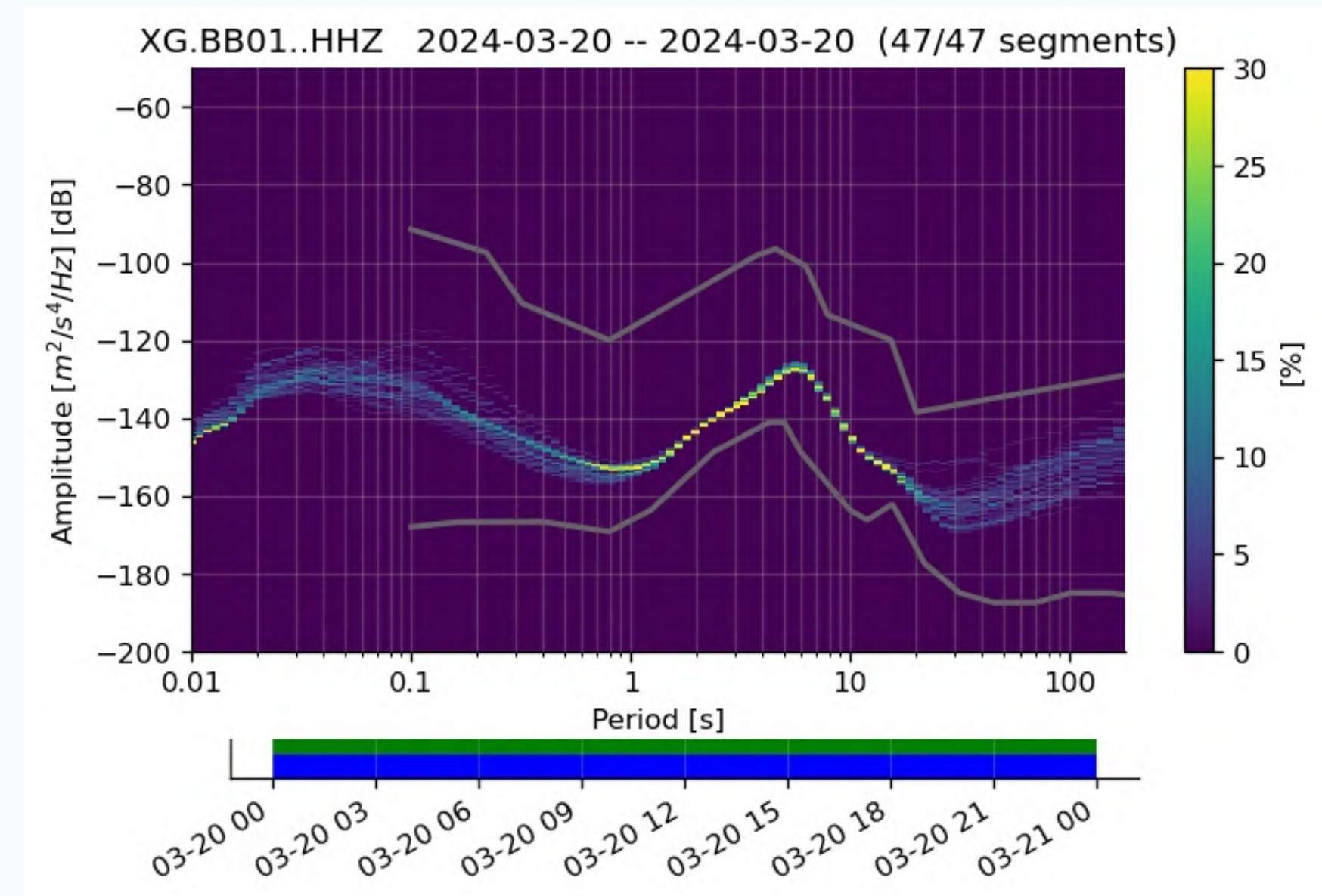
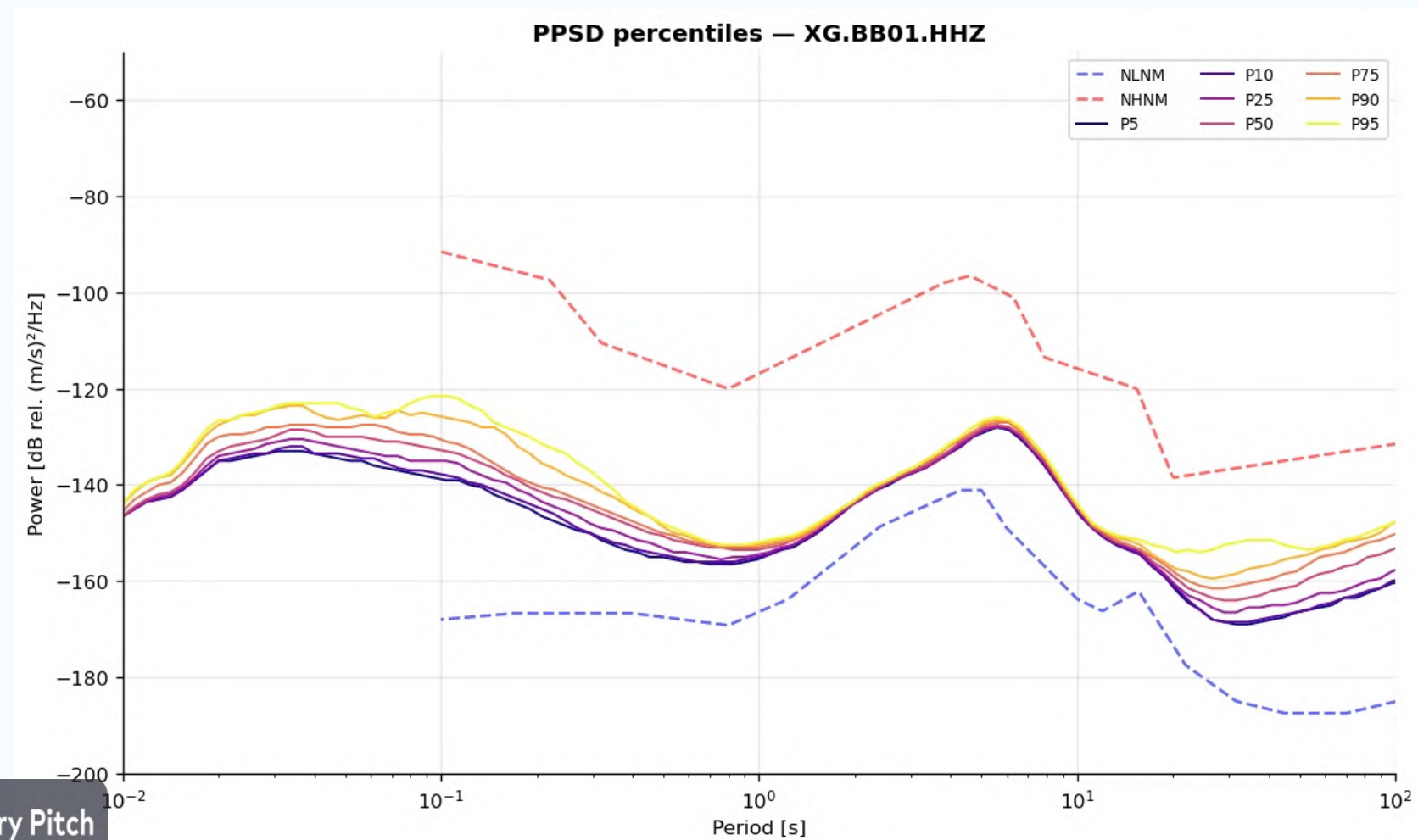
cd to the folder where the notebooks are downloaded

start "jupyter notebook"

Try Pitch ⚡⚡⚡⚡ Set computer to "maximum performance" (~~"battery saving"~~) 🔋 🔋 🔋 🔋

TIME - NOTEBOOK 10: (P)PSD

Using noise to probe local site quiteness



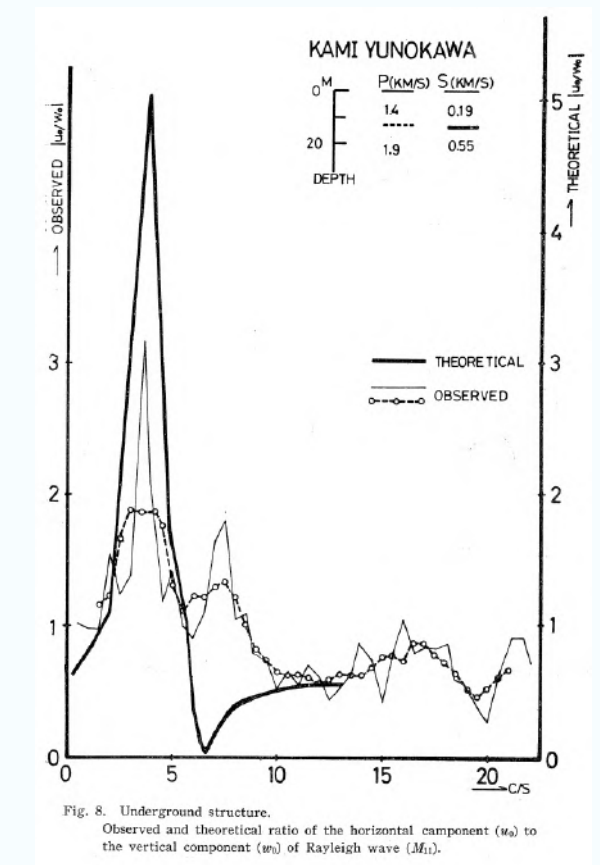
Part 5

Noise *as a* Tool

From nuisance to data – the ambient noise revolution

How seismic noise became one of the most powerful tools in seismology

- For most of the 20th century, seismic noise was treated as a **problem to minimise**
- But exploitation of noise for site characterisation started **much earlier** than the cross-correlation era:
 - **Nogoshi & Igarashi (1971)** – horizontal-to-vertical spectral ratio (HVSr) of noise to probe site resonance
 - **Nakamura (1989)** – popularised HVSr as a practical engineering tool worldwide
 - **Lachet & Bard (1994), SESAME project (2004)** – systematic validation and guidelines for HVSr
- These methods used noise **empirically** – without a full theoretical framework – but they worked
- **Paradigm shift ~2000–2005:** theoretical foundation established for noise as a wavefield:
 - **Weaver & Lobkis (2001)** – Green's function retrieval from diffuse noise (laboratory)
 - **Campillo & Paul (2003)** – first seismological application
 - **Shapiro et al. (2005)** – ambient noise tomography of California
- Today the full arc spans **site characterisation** → **crustal imaging** → **global tomography** → **monitoring**
- The noise that frustrated 20th century seismologists now drives some of the most cited papers in geophysics



What waves does noise carry?

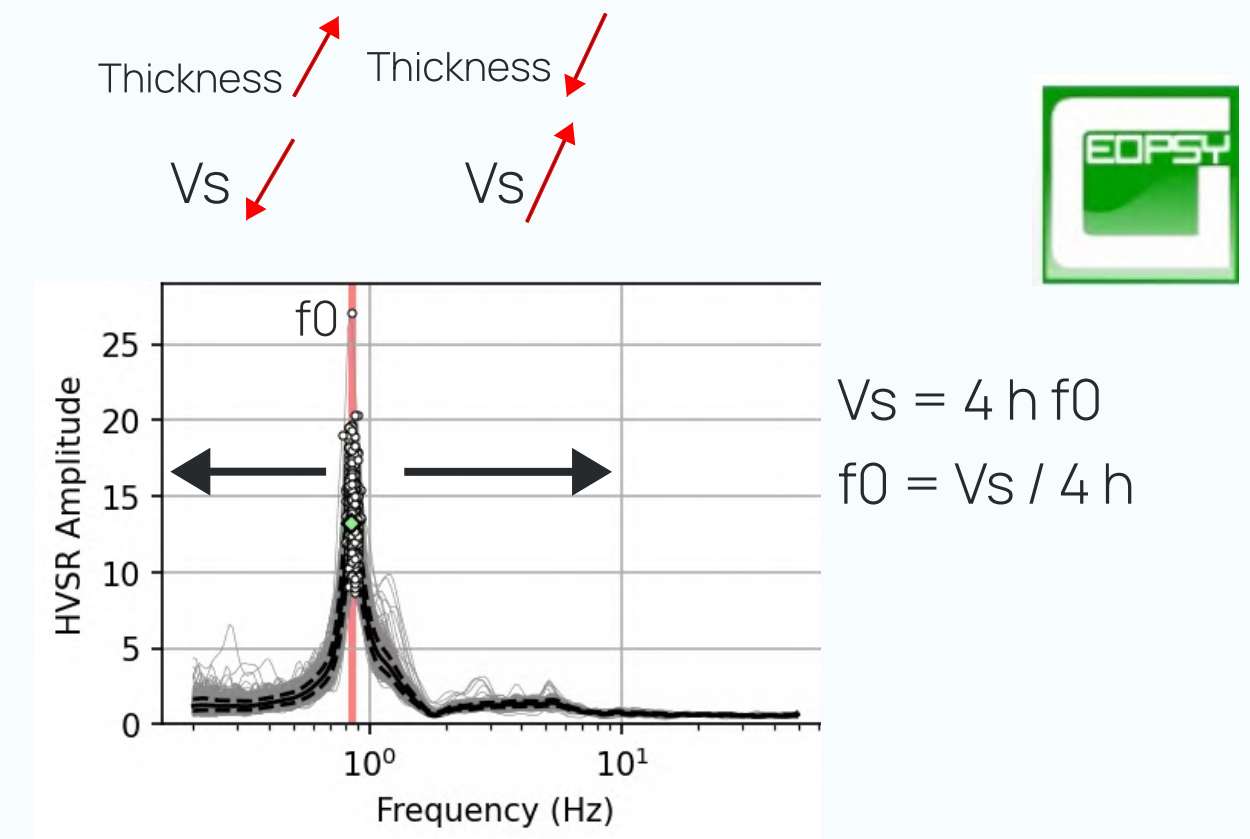
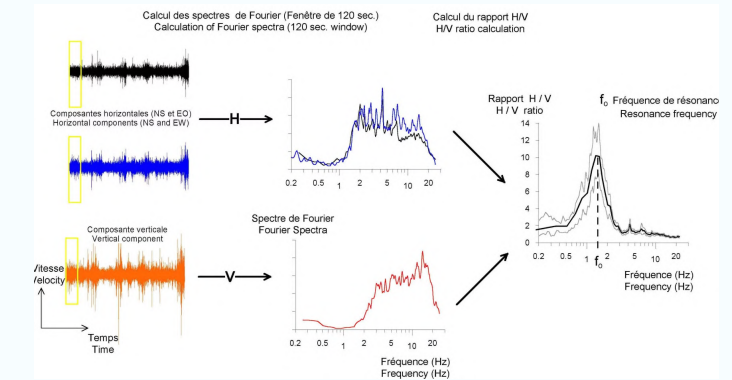
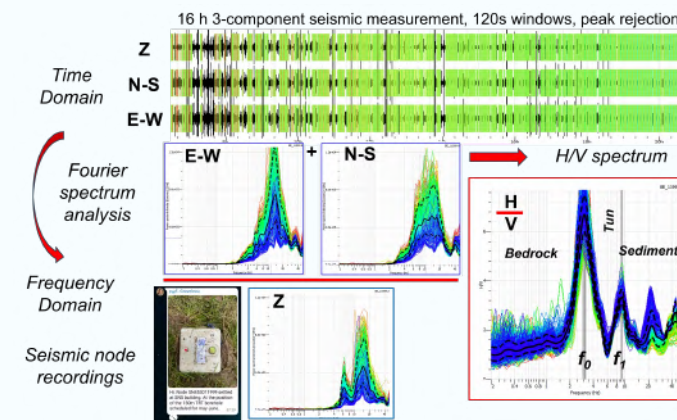
Surface waves dominate — dispersion, frequency content, and why that matters

- The ambient noise wavefield is **not a single wave type** — it is a superposition of all wave types
- But it is **not equal**: surface waves dominate the noise field in the microseismic band , for example
 - Rayleigh waves: ~60–80% of energy in vertical component noise
 - Love waves: significant on horizontal components
 - Body waves (P, S): present but weaker in the noise field — more important at higher frequencies
- Why surface waves dominate:
 - Generated efficiently by ocean-atmosphere forcing at the surface
 - Geometrical spreading in 2D ($\propto 1/\sqrt{r}$) vs. body waves in 3D ($\propto 1/r$) → surface waves decay slower
 - Trapped in the crust → energy guided along the surface
- **Dispersion** — surface wave velocity depends on frequency:
 - Long periods sample deeper structure (faster)
 - Short periods sample shallow structure (slower)
 - This frequency-depth relationship is what makes ANT work as an imaging tool
- **Ellipticity** of Rayleigh waves also carries depth information → exploited by HVSR

HVSR

Using noise to probe local site structure

- **HVSR** (Horizontal-to-Vertical Spectral Ratio) — ratio of the amplitude spectrum of horizontal components to vertical component of ambient noise
- Originally empirical (1971 & 1989): the peak frequency of HVSR $\approx$ **fundamental resonance frequency f_0 of the soil column**
- Physical interpretation (still debated):
 - Nakamura: body wave interpretation — S-wave amplification
 - More accepted today: **ellipticity of Rayleigh waves** — the H/V ratio of a Rayleigh wave changes with frequency, with a peak near the resonance of the shallow structure
- What HVSR gives you:
 - **f_0** — site fundamental frequency (inversely related to soil thickness and V_s)
 - **Amplitude of peak** — proxy for impedance contrast
 - **Shape of curve** — constrains V_s profile when inverted
- Practical advantages: **single station, passive, cheap, fast** — no source needed
- Limitations:
 - amplitude not directly = amplification factor
 - sensitive to instrument orientation, wind, traffic, noise! and H/V trough interpretation is tricky

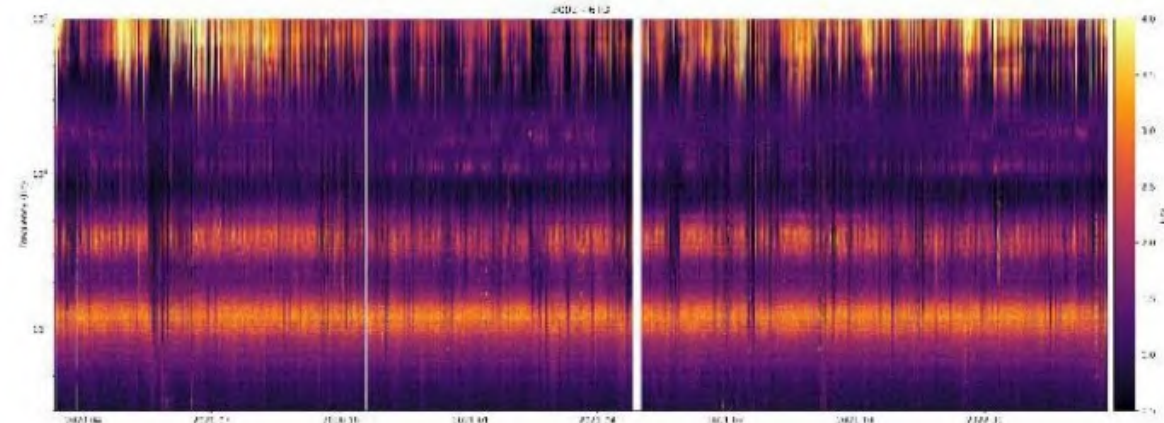
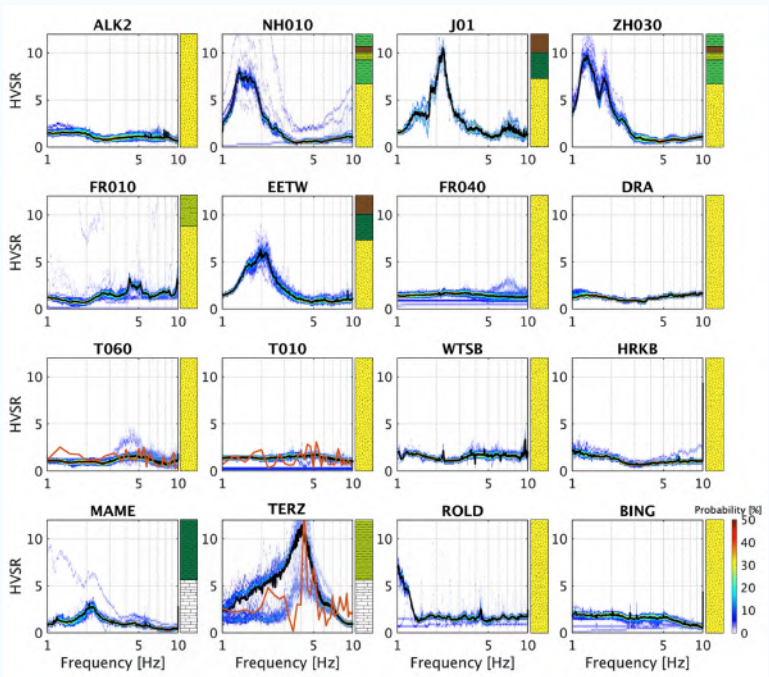
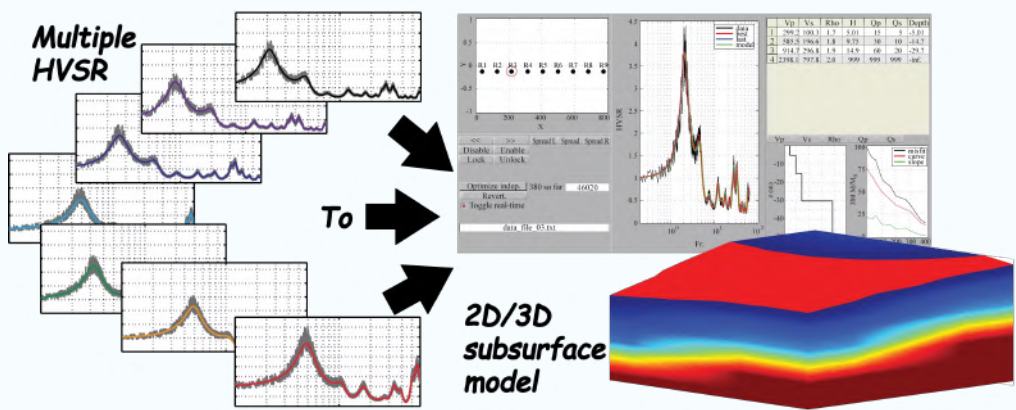
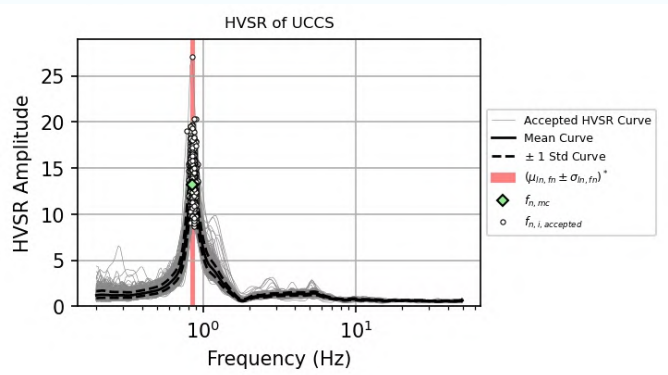
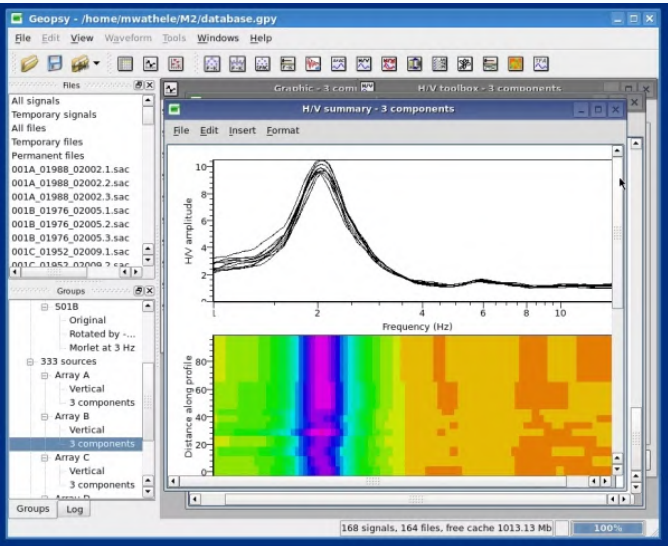


HVSR

Using noise to probe local site structure

- **Programs**

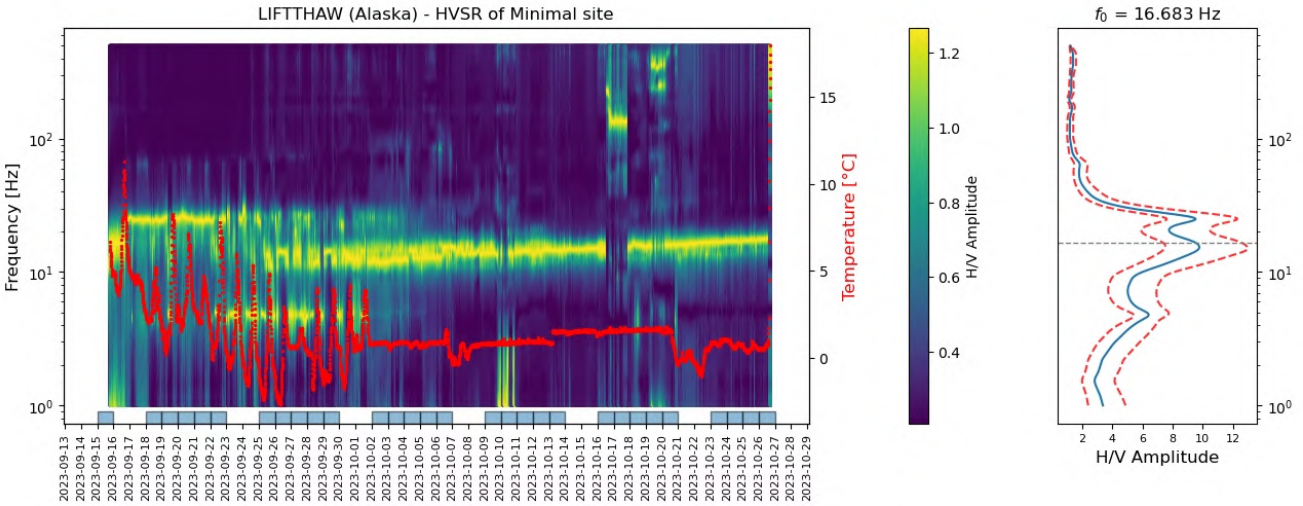
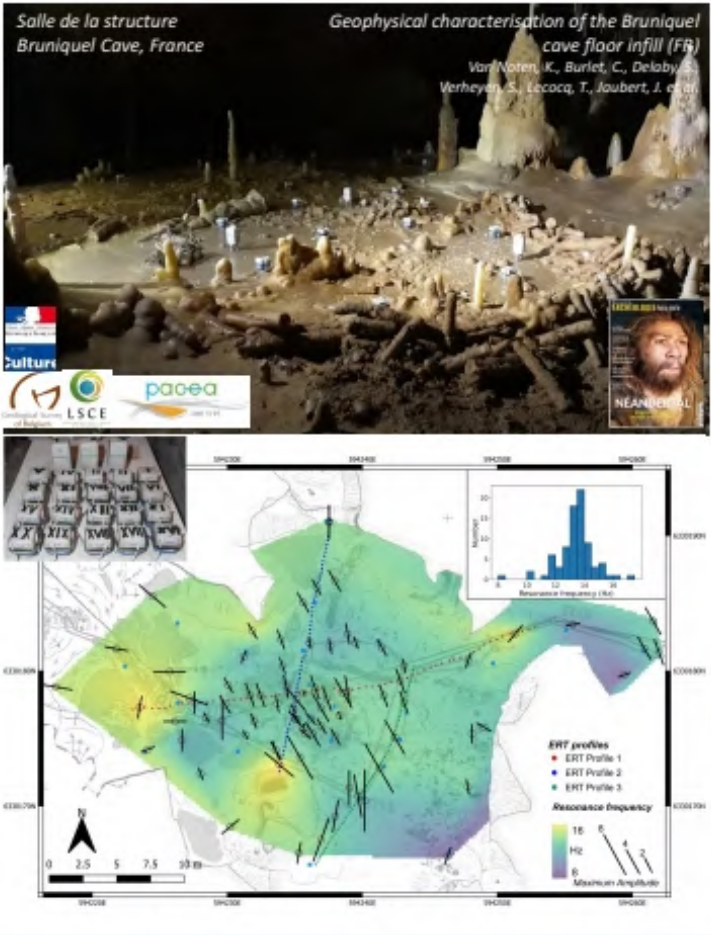
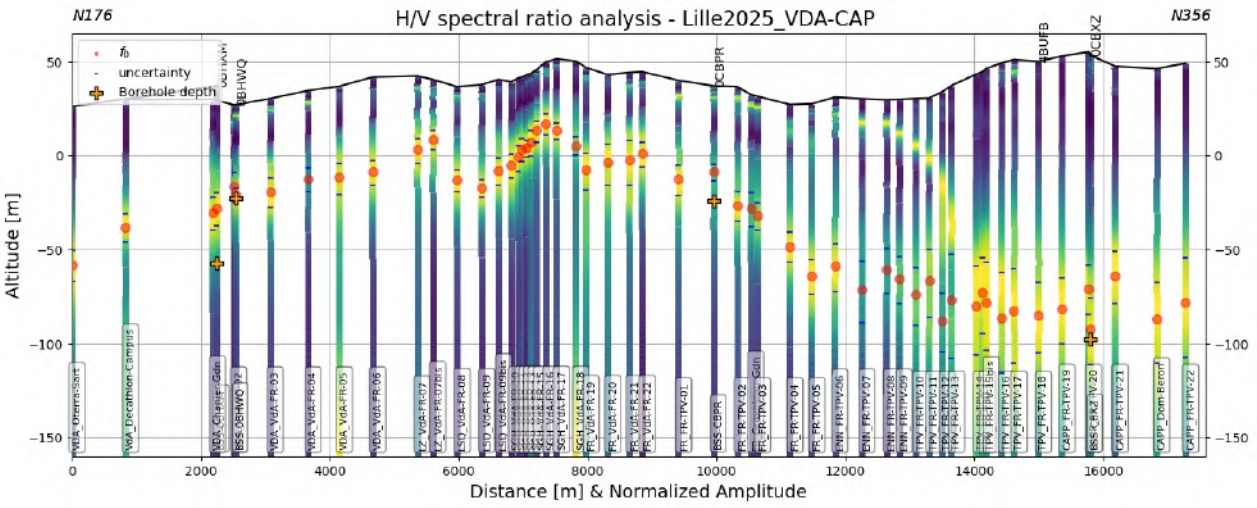
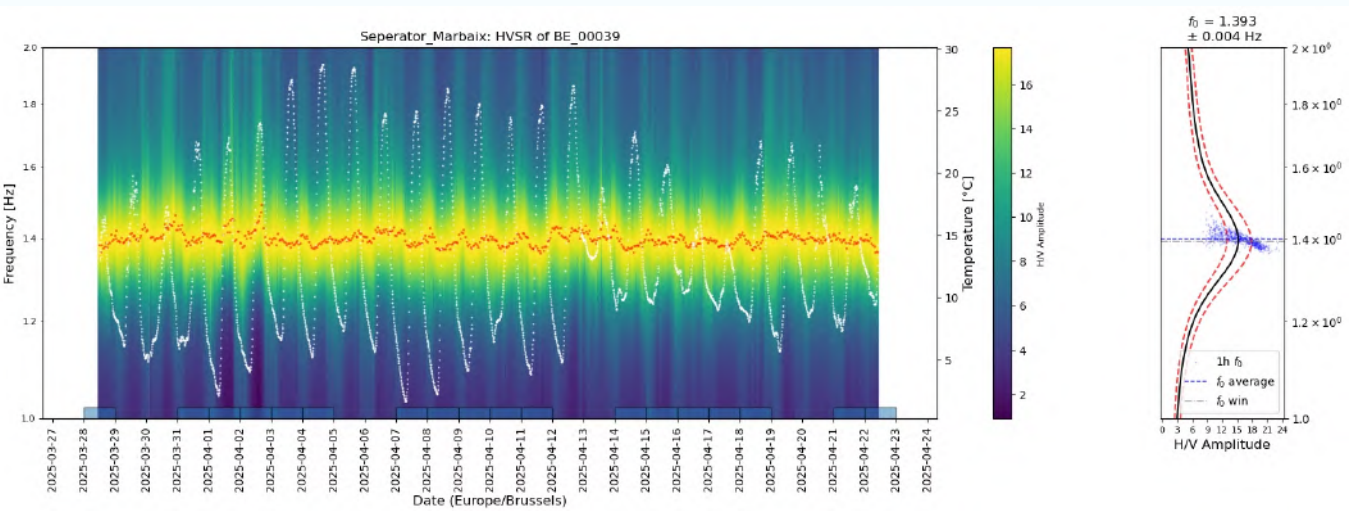
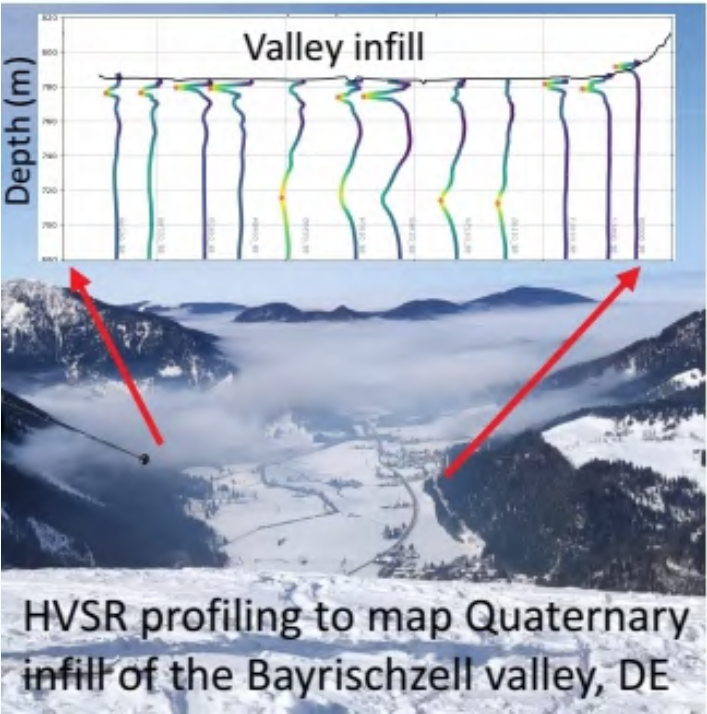
- GEOPSY - Whatelet et al. 2020
 - + manual data treatment
 - - coding to automate
- HVSRPY - Van Tassel 2025 - **TODAY**
 - + python
 - - docs / processing-steps
- OPENHVSR - Bignardi 2016 (Matlab - never tested)
 - - matlab matlab matlab
- PSD -ratios - MSNoise - Lecocq et al. 2014
 - own codes (Van Ginkel et al 2024)
- + automated
- + link with psd etc
- - advanced



HVSR

Using noise to probe local site structure

- Applications
 - Earthquake microzonation and amplification
 - Station site characterisation
 - Geological cross-sections & basement mapping
- **Glacier thickness mapping**
 - Earthquake Engineering
 - Structural mapping in caves
 - Permafrost monitoring



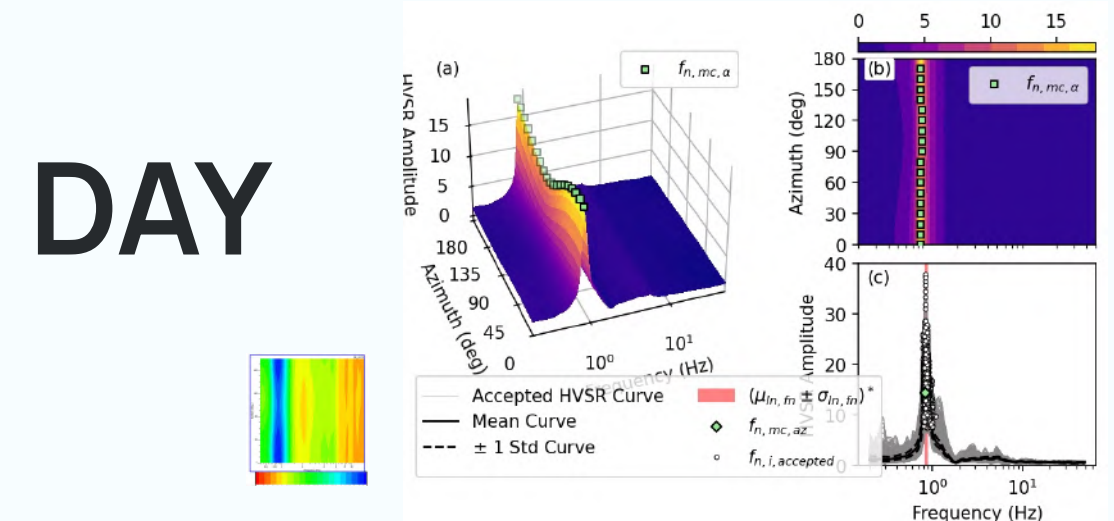
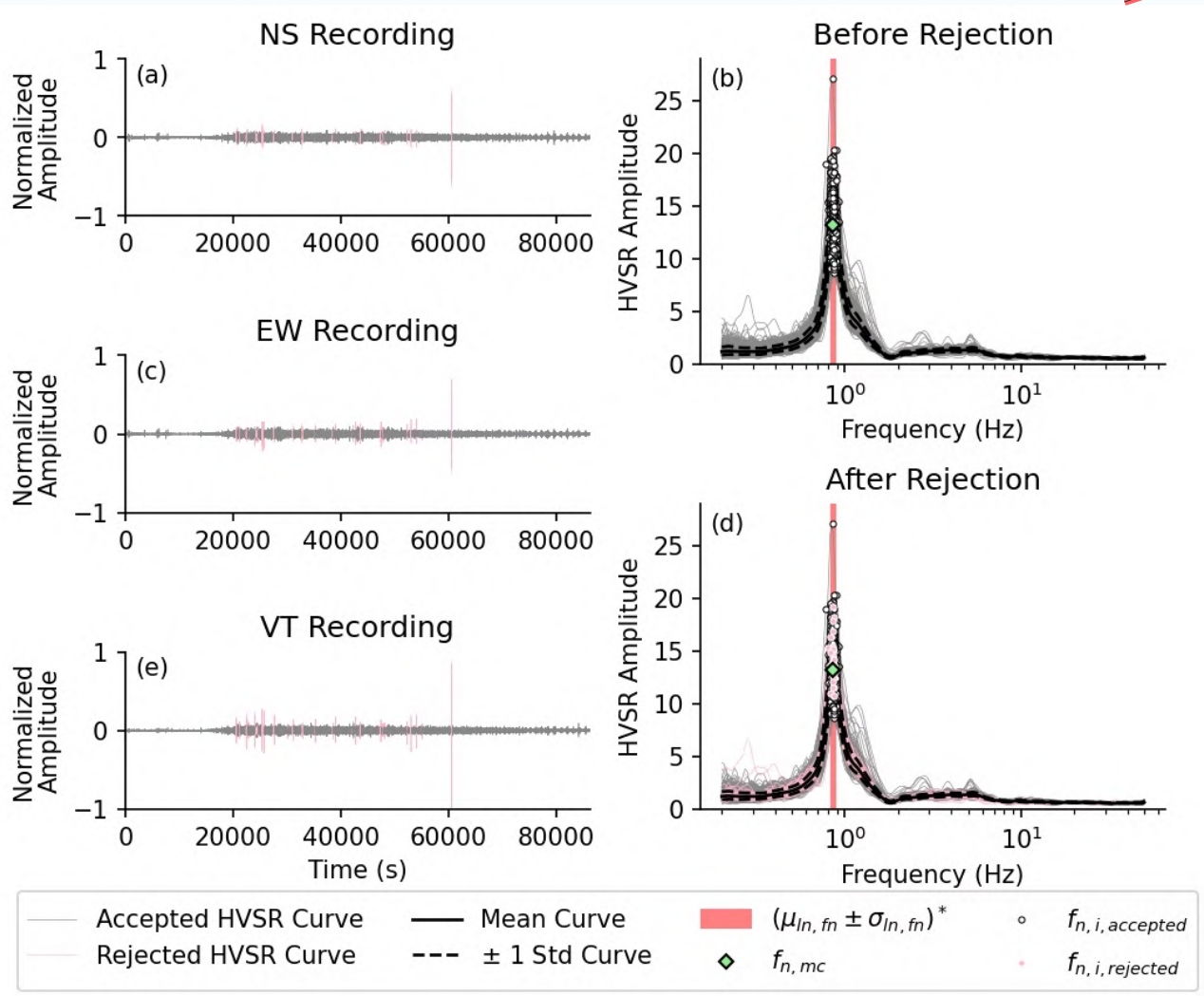
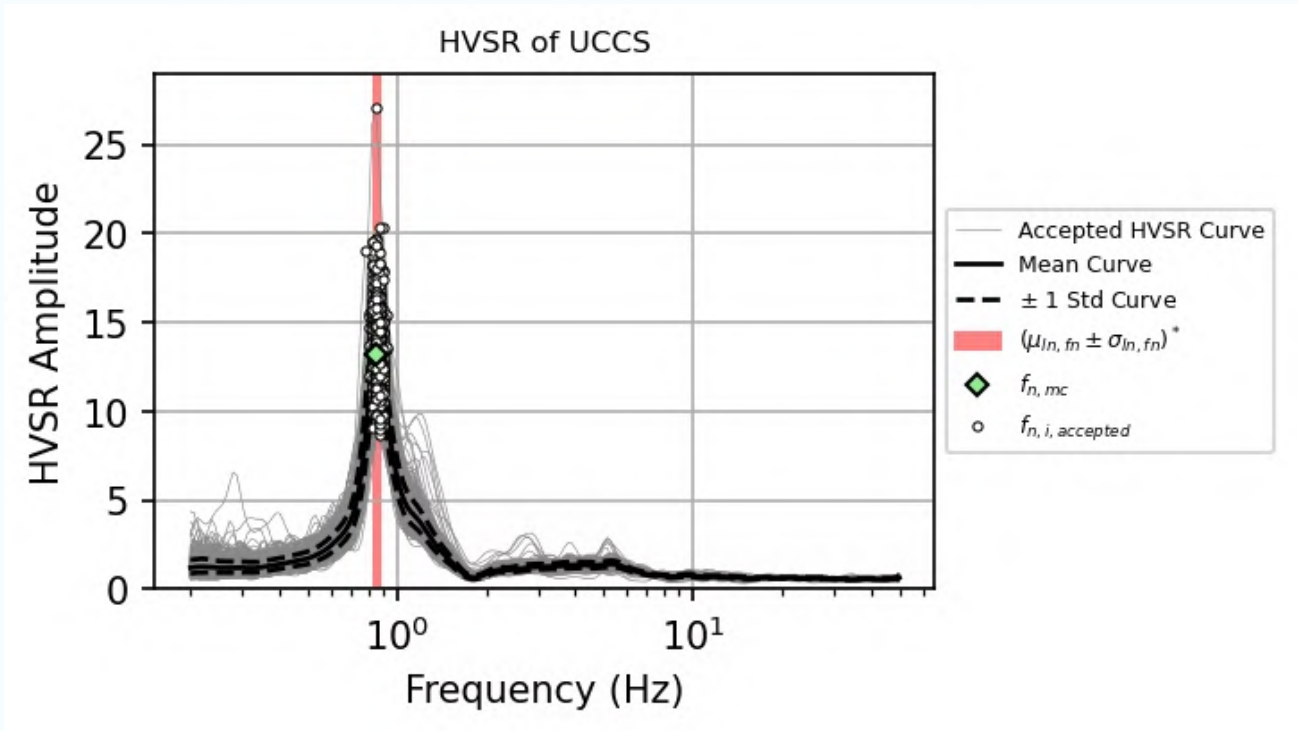
HVSR - NOTEBOOK 20: ONE STATION / ONE DAY

Using noise to probe local site structure

- Computes the HVSR of one station for one day in the SDS archive
- Automatic window rejection
- HVSR to virtual borehole

- **City seismometer case: UCCS**
- Grenzgletscher case

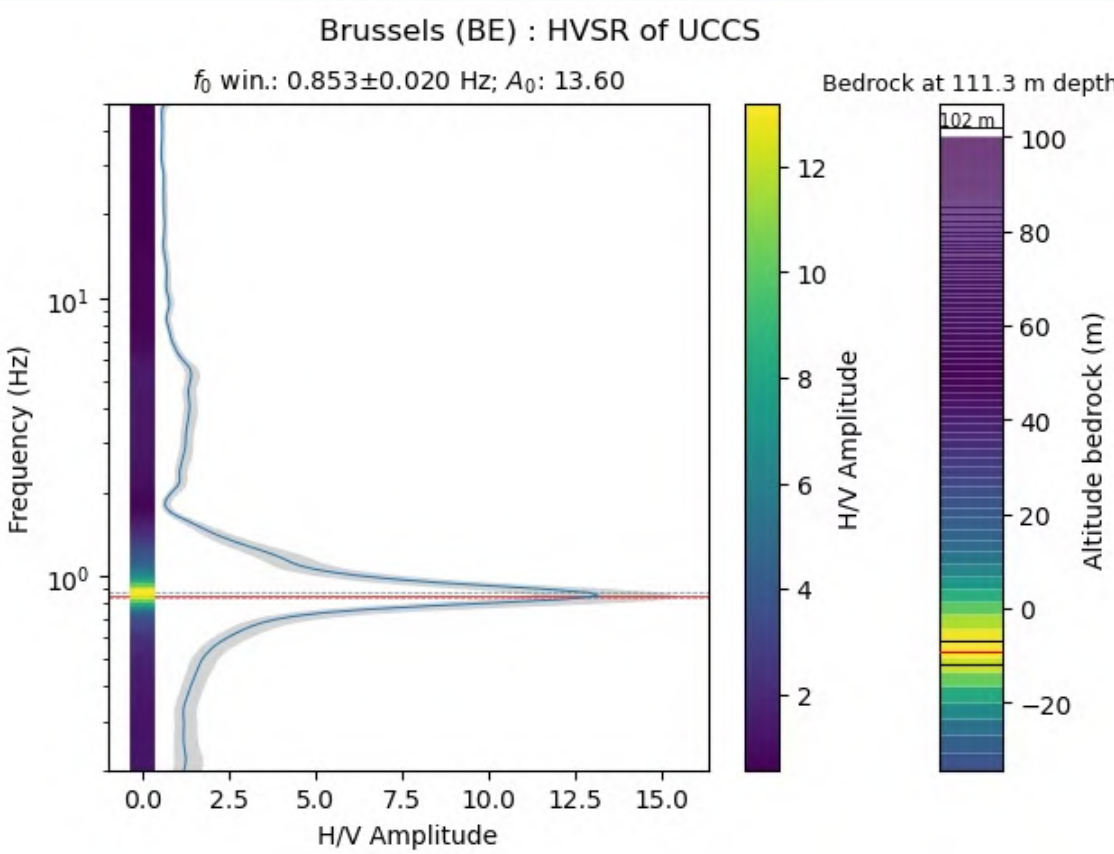
window rejection



Max f₀: 0.8464 Hz @ azimuth 10° with H/V amplitude of 17.508
Min f₀: 0.8632 Hz @ azimuth 100° with H/V amplitude of 12.431

Azimuth

Virtual Borehole
Requires Vs information



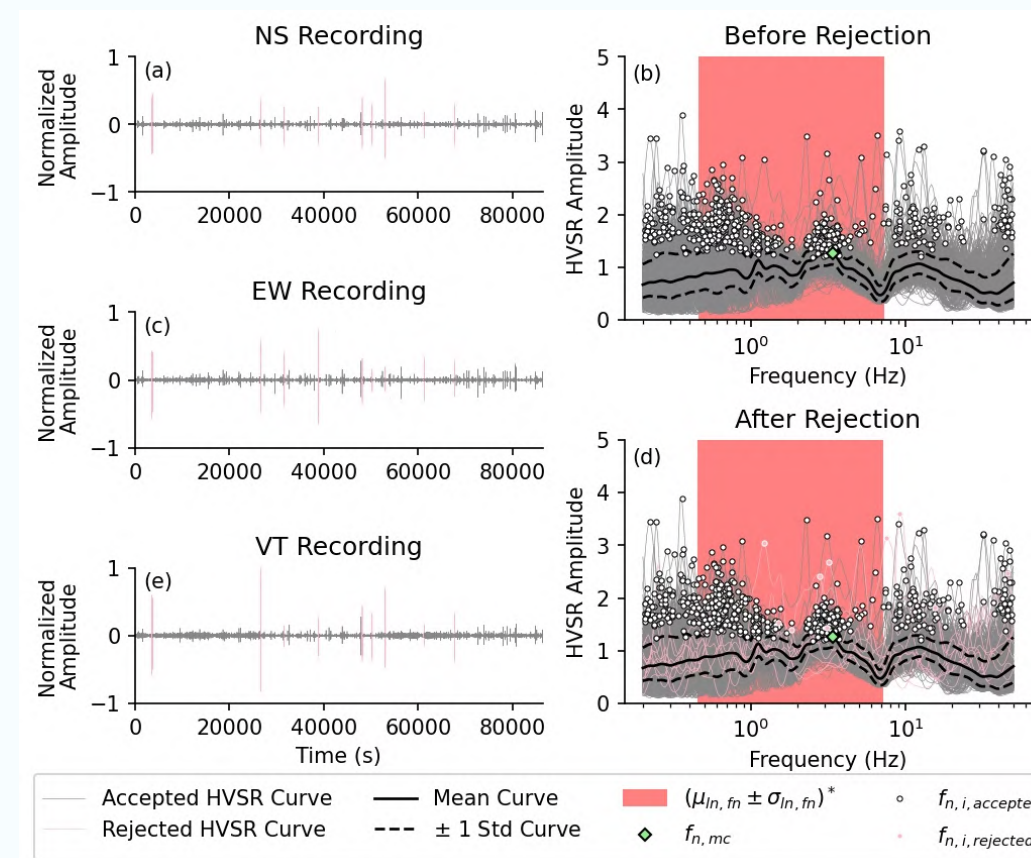
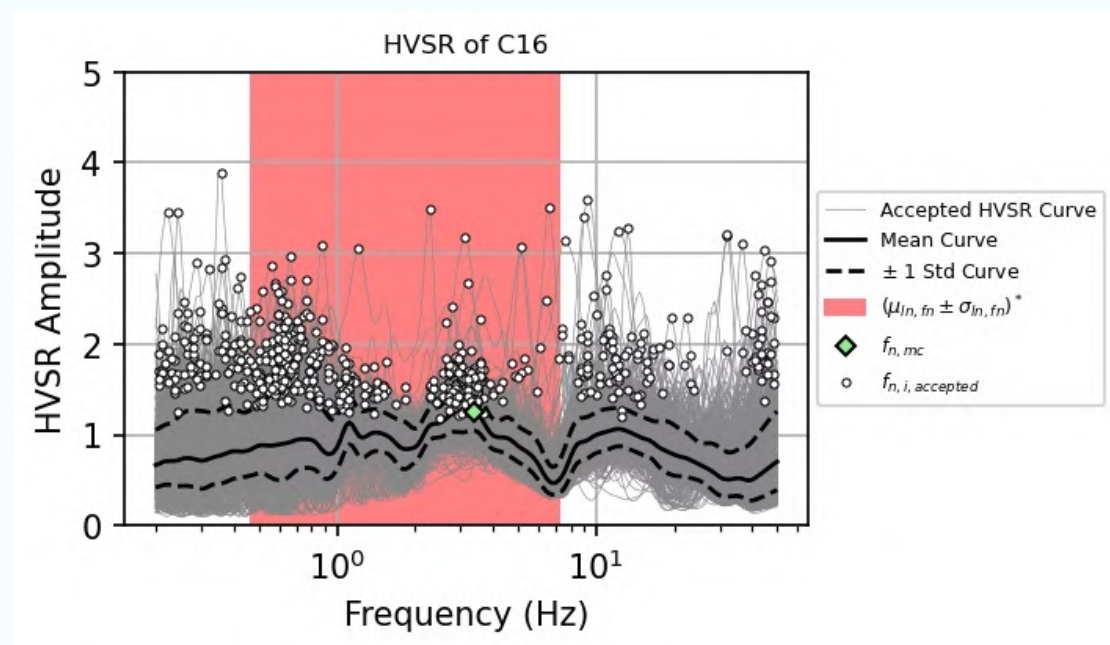
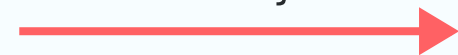
HVSR - NOTEBOOK 20: ONE STATION / ONE DAY

Using noise to probe local site structure

- Computes the HVSR of one station for one day in the SDS archive
- Automatic window rejection
- HVSR to virtual borehole

- City seismometer case: UCCS
- **Grenzgletscher case**

window rejection



Virtual Borehole

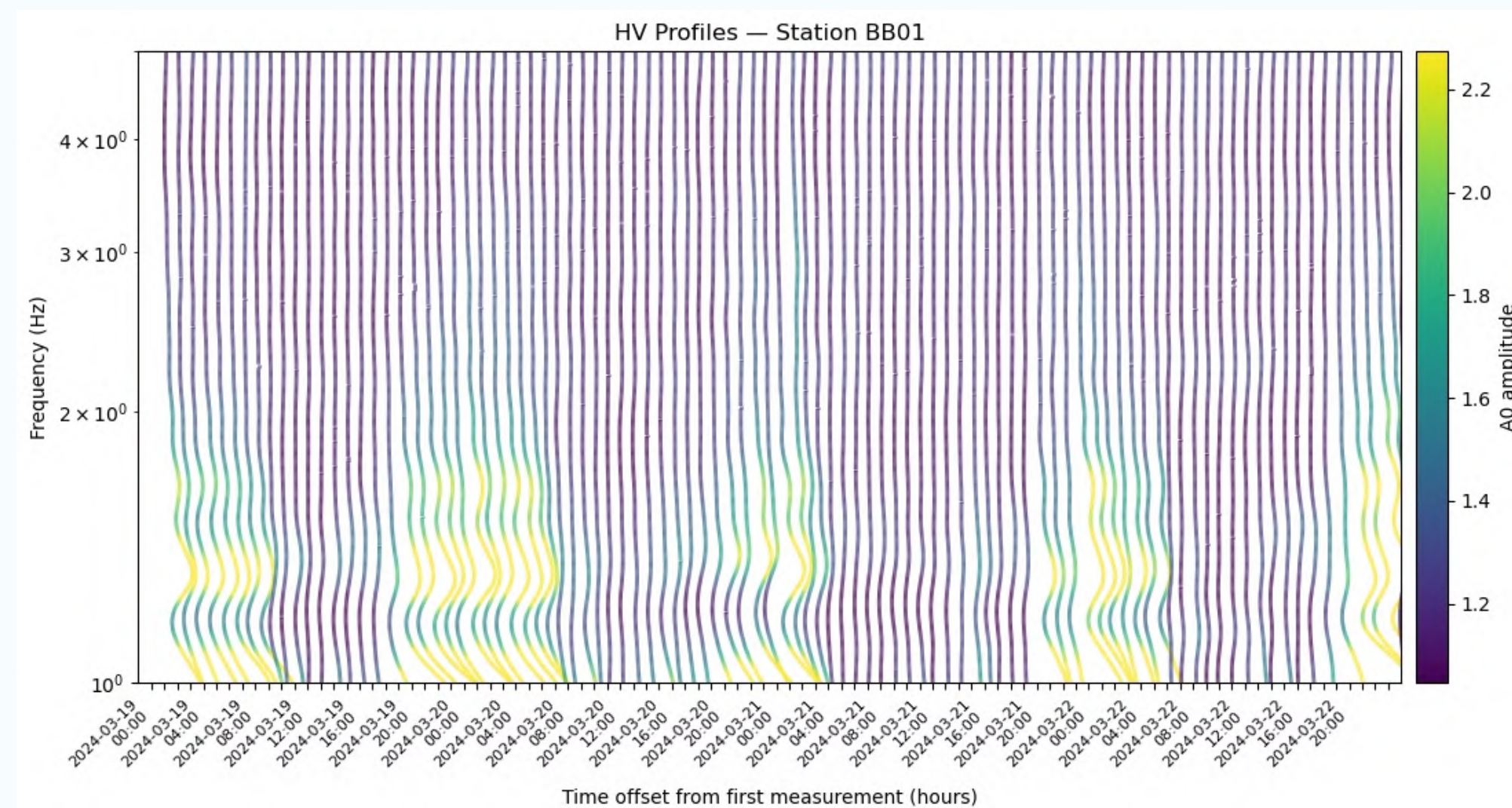
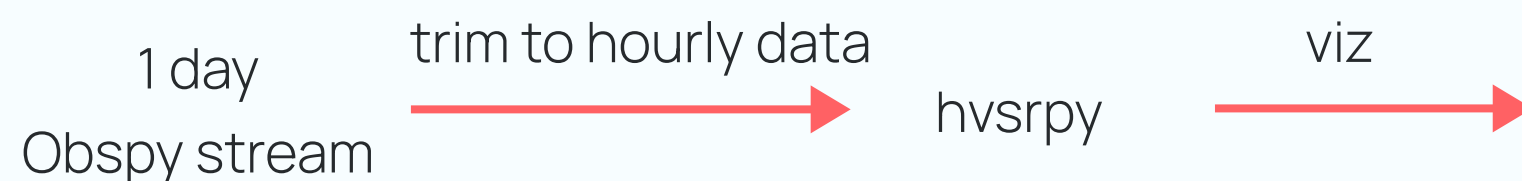
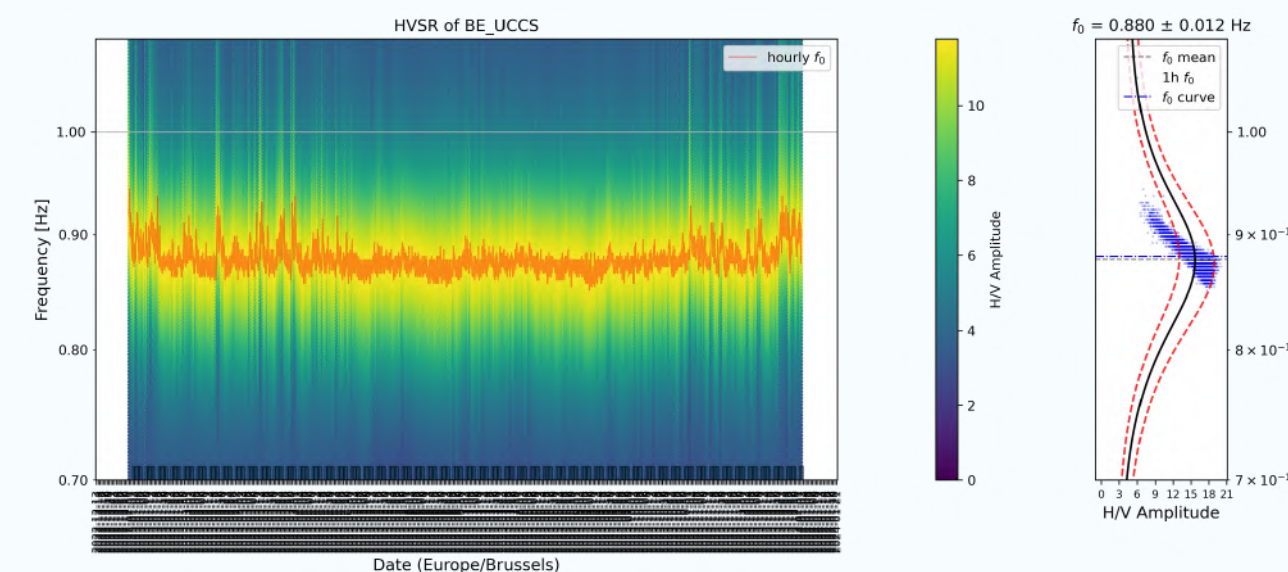
Requires Vs information



HVSR - NOTEBOOK 21: HOURLY HVSR

Is the average HVSR of one day sufficient to understand what's happening on a glacier?

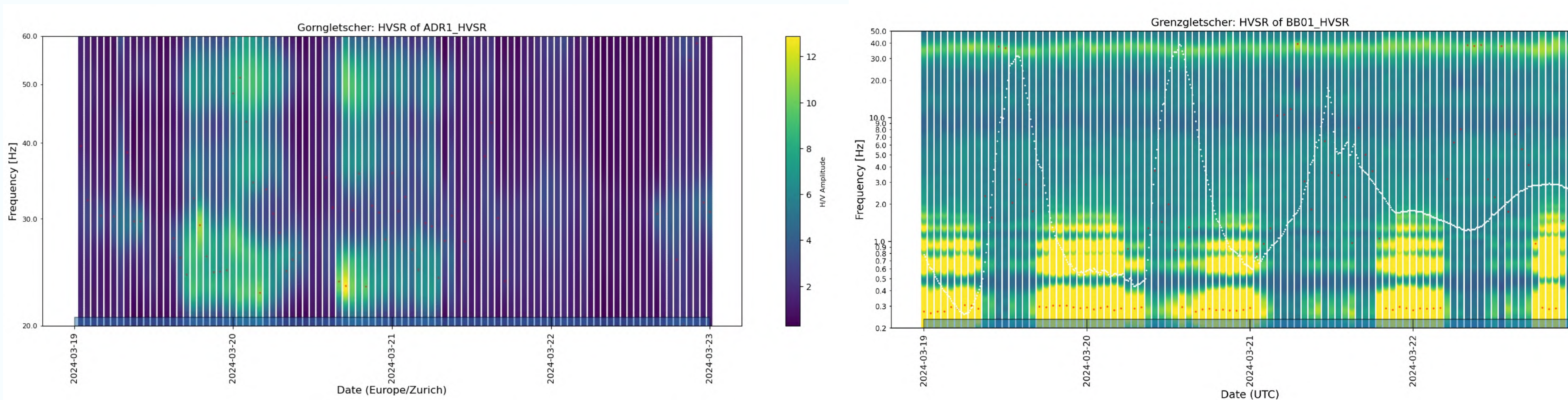
- Splits the SDS archive in e.g. hourly data (3600s) using **obspy**
- Computes hourly HVSR
- Automatic window rejection if needed
- Plots hourly HVSR curves in time



HVSR - NOTEBOOK 22: PLOTTING HOURLY HVSR

Is the average HVSR of one day sufficient to understand what's happening on a glacier?

- Play with frequency range for interactive plotting
- Zoom into frequency bands
- Play with max H/V amplitude values
- Discuss !



HVSR - conclusion

Is the average HVSR of one day sufficient to understand what's happening on a glacier?

- Work first with data that you know to understand the program / filter / window selection
- Then move to complex datasets
- Compute wide frequency band
- Zoom into frequency band of interest
- Compute HV_one day $\leftrightarrow$ HV_continuous
- HVSR is not a single tool, you need
 - velocity data
 - knowledge of wavefield
 - ...

CHALLENGE

Prize : 1 beer or soft

What are the two Teleseisms visible on the BB01 station

Compulsory: use ObsPy's FDSN clients in a notebook to show 1. search 2. traveltimes computation and 3. plot of theoretical arrival vs seismic data !

You can work in teams - winning team (whatever size) gets the prize/person !

bonus: What is the regional event ?

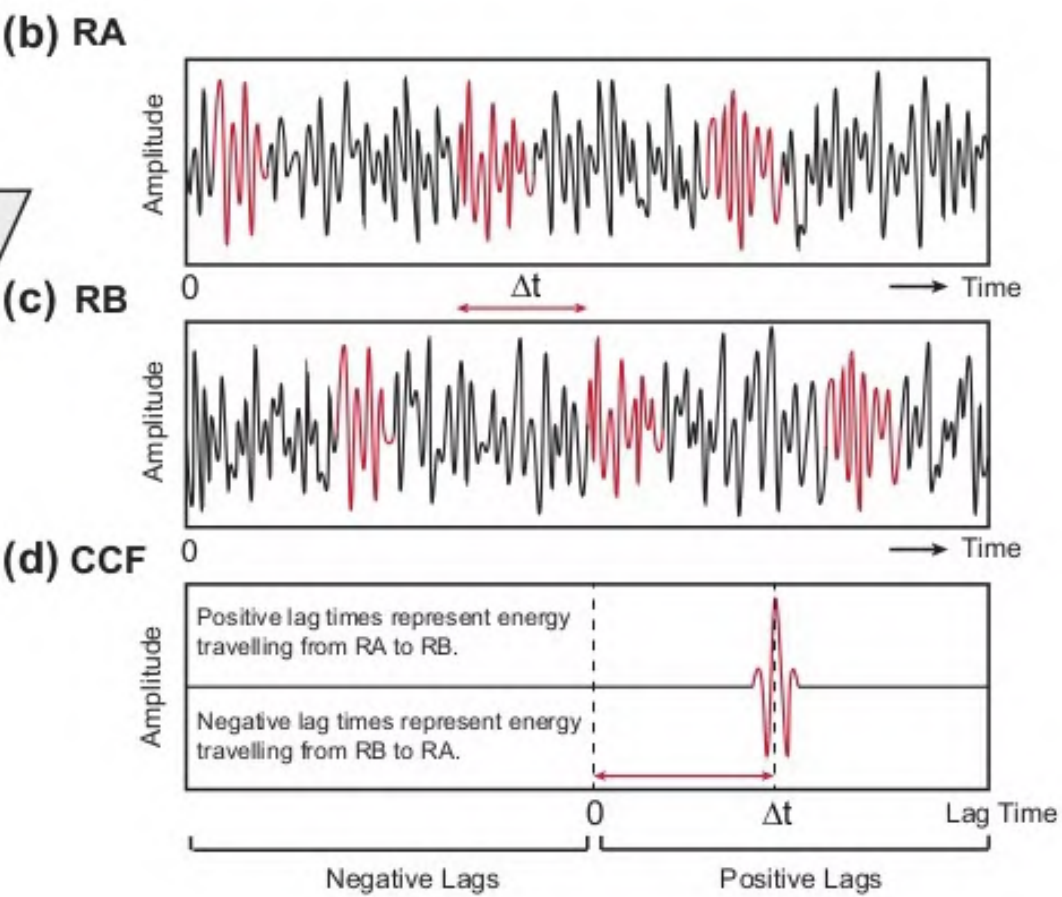
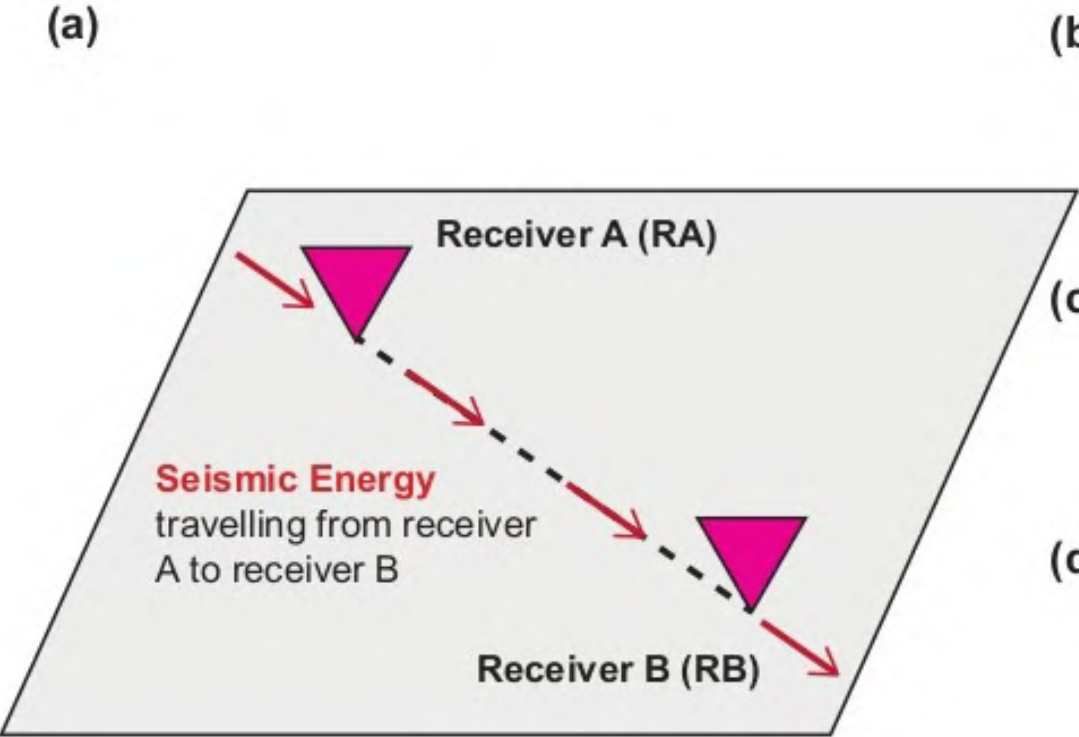
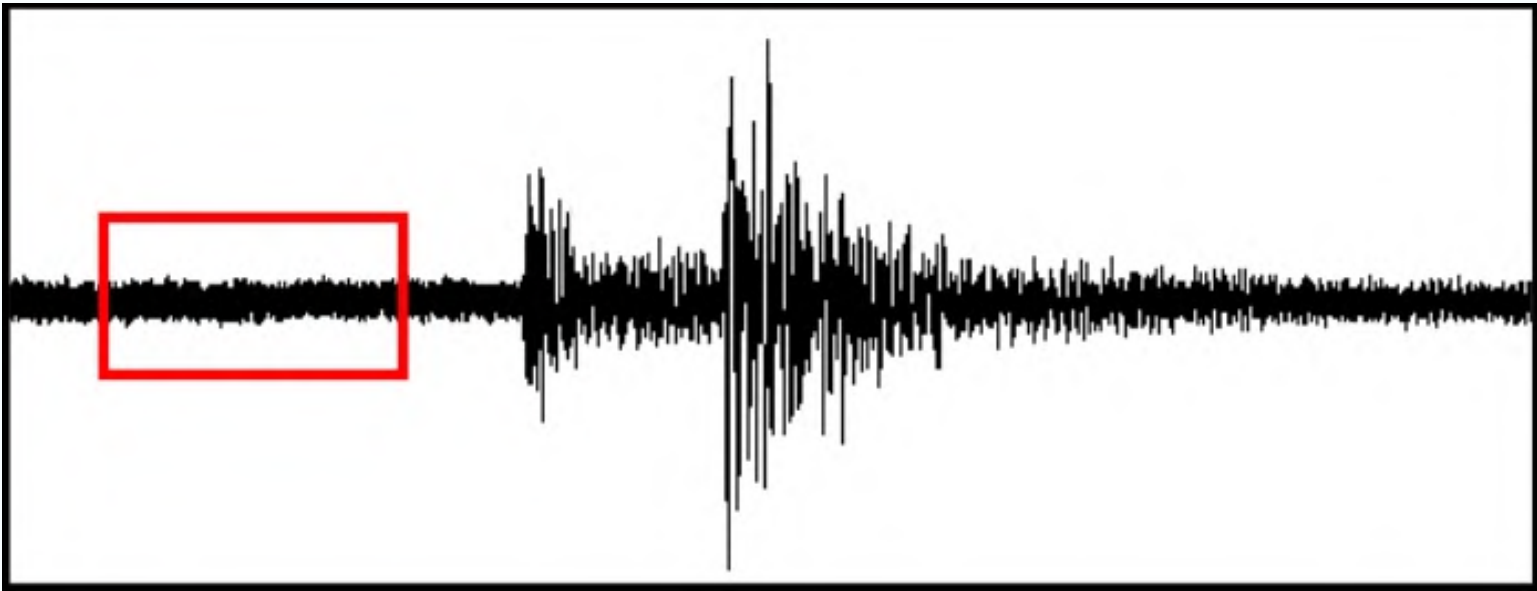
Cross-correlation and the Green's function

"Listening" between two stations

- **Key idea:** cross-correlate the continuous noise recorded at stations A and B → the result converges (after averaging many time windows) to the **Green's function** between A and B
- The Green's function = the seismic response at B if a point source fired at A — a **virtual seismogram**
- Mathematically: $C_{AB}(\tau) = \int u_A(t) \cdot u_B(t+\tau) dt \rightarrow G_{AB}(\tau) - G_{AB}(-\tau)$ (causal + acausal)
- Why it works: if noise sources are **diffuse and isotropic**, the cross-correlation stationary phase contributions sum to the Green's function
- In practice the noise field is **not perfectly isotropic** — dominant sources (oceans) bias the result → requires careful processing (spectral whitening, 1-bit normalisation, stacking)
- Processing workflow:
 - Long continuous records (months to years)
 - Cut into windows (e.g. 1h), pre-process each window
 - Cross-correlate all station pairs
 - Stack (linear or phase-weighted) → empirical Green's function (EGF)
- The EGF is dominated by **surface waves** → looks like a dispersive wavepacket → measure phase/group velocity

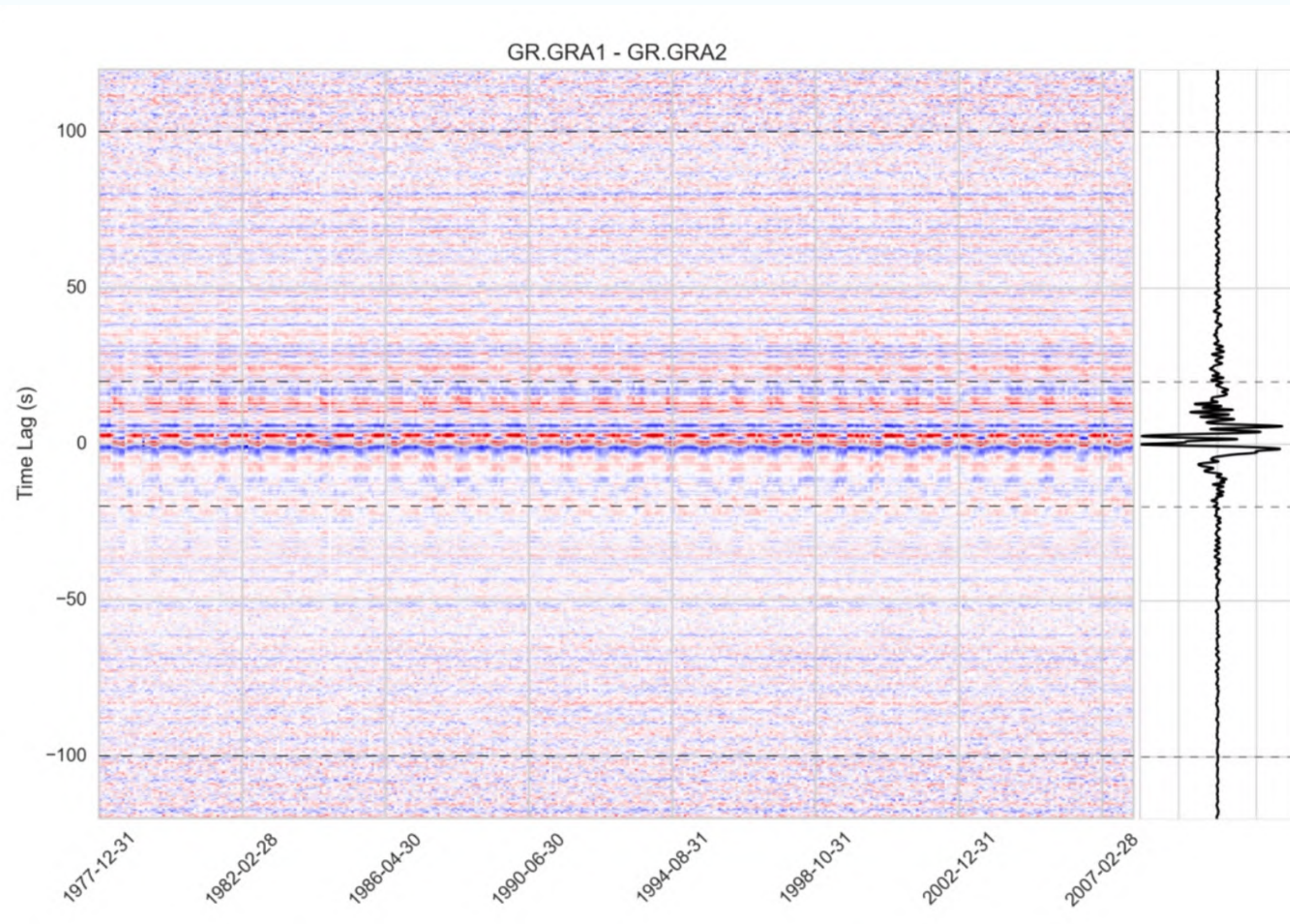
Cross-correlation and the Green's function

"Listening" between two stations



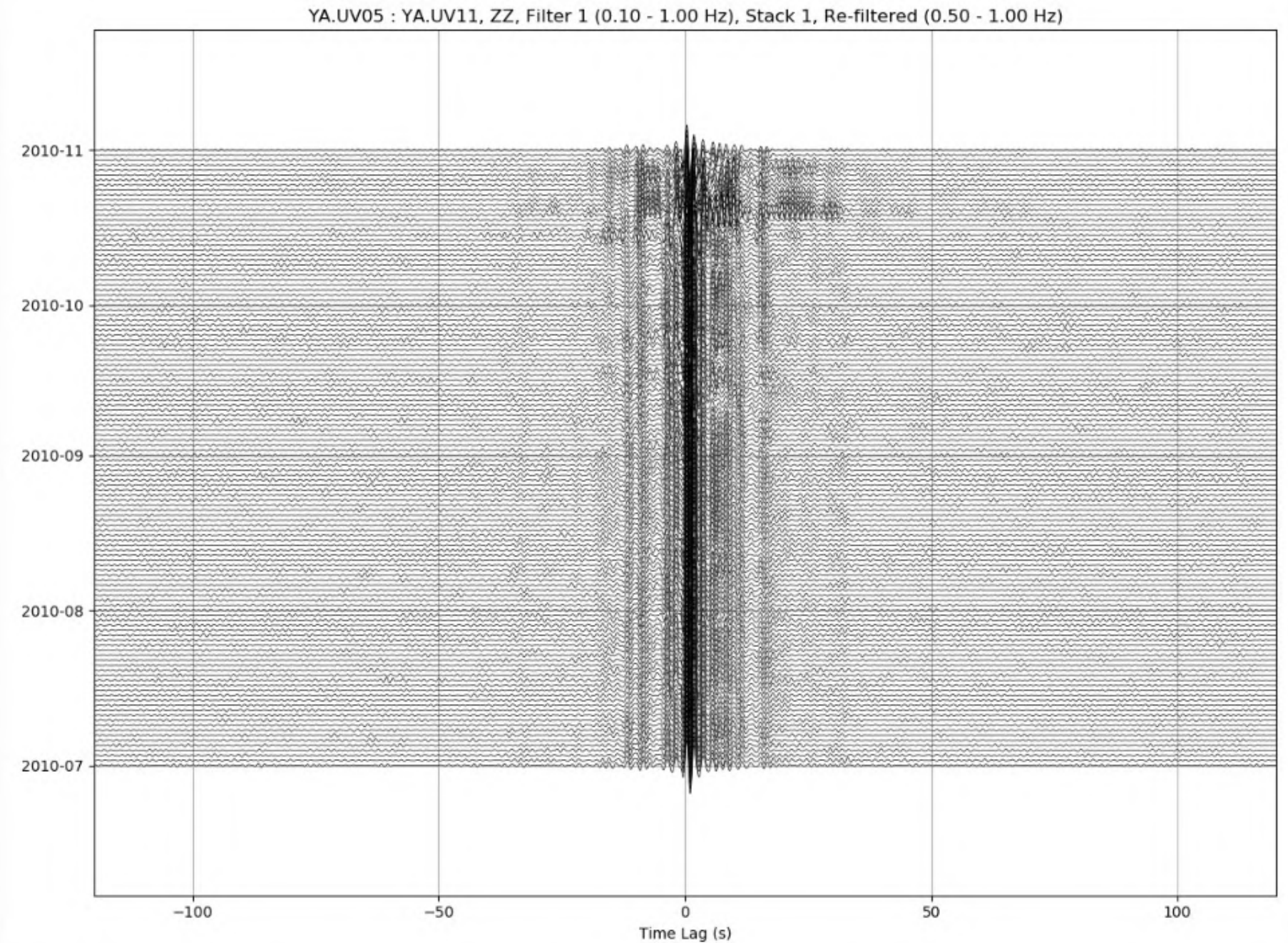
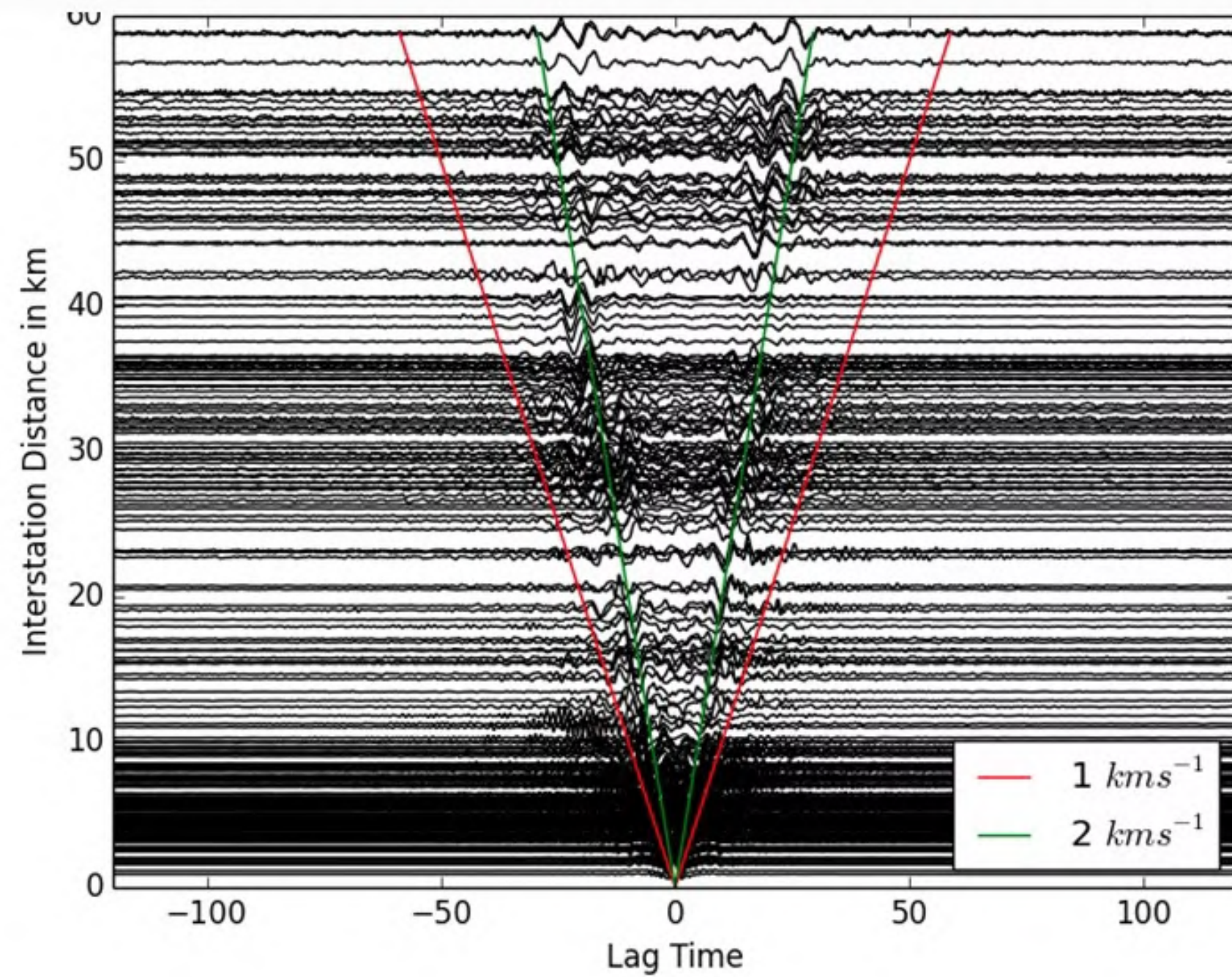
Cross-correlation and the Green's function

"Listening" between two stations



Cross-correlation and the Green's function

"Listening" between two stations



Ambient Noise Tomography

Mapping the crust without earthquakes

- **ANT workflow:**
 - Compute EGFs for all station pairs
 - Measure **phase or group velocity** of surface waves at each period from each EGF
 - Collect many inter-station paths → build a **dispersion dataset**
 - Invert the dataset for a 2D velocity map at each period (→ period-dependent maps)
 - Invert period-dependent maps for **3D Vs structure** (depth inversion)
- Resolution determined by **path coverage** — dense networks → high resolution
- Period range sets depth sensitivity:
 - 1–5 s → upper crust (0–15 km)
 - 5–20 s → mid-lower crust (15–40 km)
 - 20–100 s → upper mantle
- Key advantages over earthquake tomography:
 - No waiting for earthquakes — continuous data acquisition
 - Dense path coverage between all station pairs
 - Sensitive to **slow anomalies** (sedimentary basins, geothermal zones) not well sampled by teleseismic waves
- Applications: basin characterisation, geothermal exploration, volcanic imaging, lithospheric structure

Monitoring with noise — dv/v

Detecting velocity changes over time

- Extension to ambient noise (Sens-Schönfelder & Wegler 2006; Brenguier et al. 2008): replace repeating earthquakes with the **repeatable noise-derived EGF** → continuous monitoring without any earthquakes
- Method: compute EGFs for the same station pair at different times → **compare waveforms in the coda**
- A small velocity change $\delta v/v$ produces a time shift $\delta t = -(\delta v/v) \cdot t$ in the coda
- Measured by: **stretching** or **MWCS** (Moving Window Cross-Spectral) analysis
- Typical amplitudes: $\delta v/v \sim 10^{-4}$ to 10^{-2} (0.01% to 1%)
- Applications:
 - **Volcanic monitoring** — velocity drops before (most) eruptions (of ideal volcanoes)
 - **Earthquake damage** — co-seismic velocity drops, post-seismic recovery
 - **Hydrological cycle** — seasonal V_s changes
 - **Induced seismicity** — fluid injection monitoring
- Key tool: **MSNoise** — open-source package for large-scale dv/v monitoring

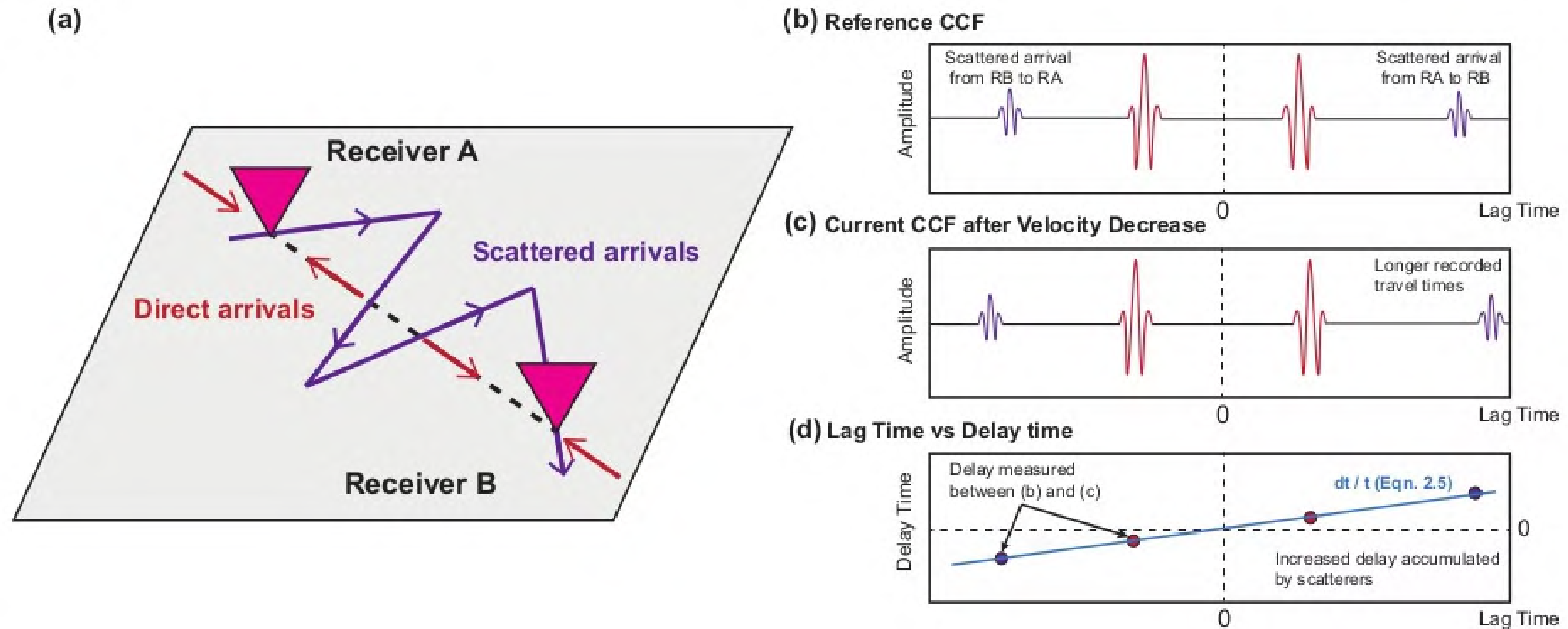
Monitoring with noise — dv/v

Detecting velocity changes over time

- If the Earth's structure **changes over time** (stress, fluid content, damage), seismic velocities change
- Ambient noise provides a **continuous, repeatable virtual source** → ideal for monitoring
- Conceptual roots: **Coda Wave Interferometry (CWI)**
 - Poupinet et al. (1984) — first observation of velocity changes between repeating earthquakes using coda cross-correlation (*doublets*)
 - Snieder et al. (2002) — formalised CWI theoretically: coda is sensitive to tiny perturbations in the medium because multiply-scattered waves sample the medium many times
 - Key insight: **coda waves are more sensitive to velocity changes than direct waves** — longer travel paths → larger accumulated time shift

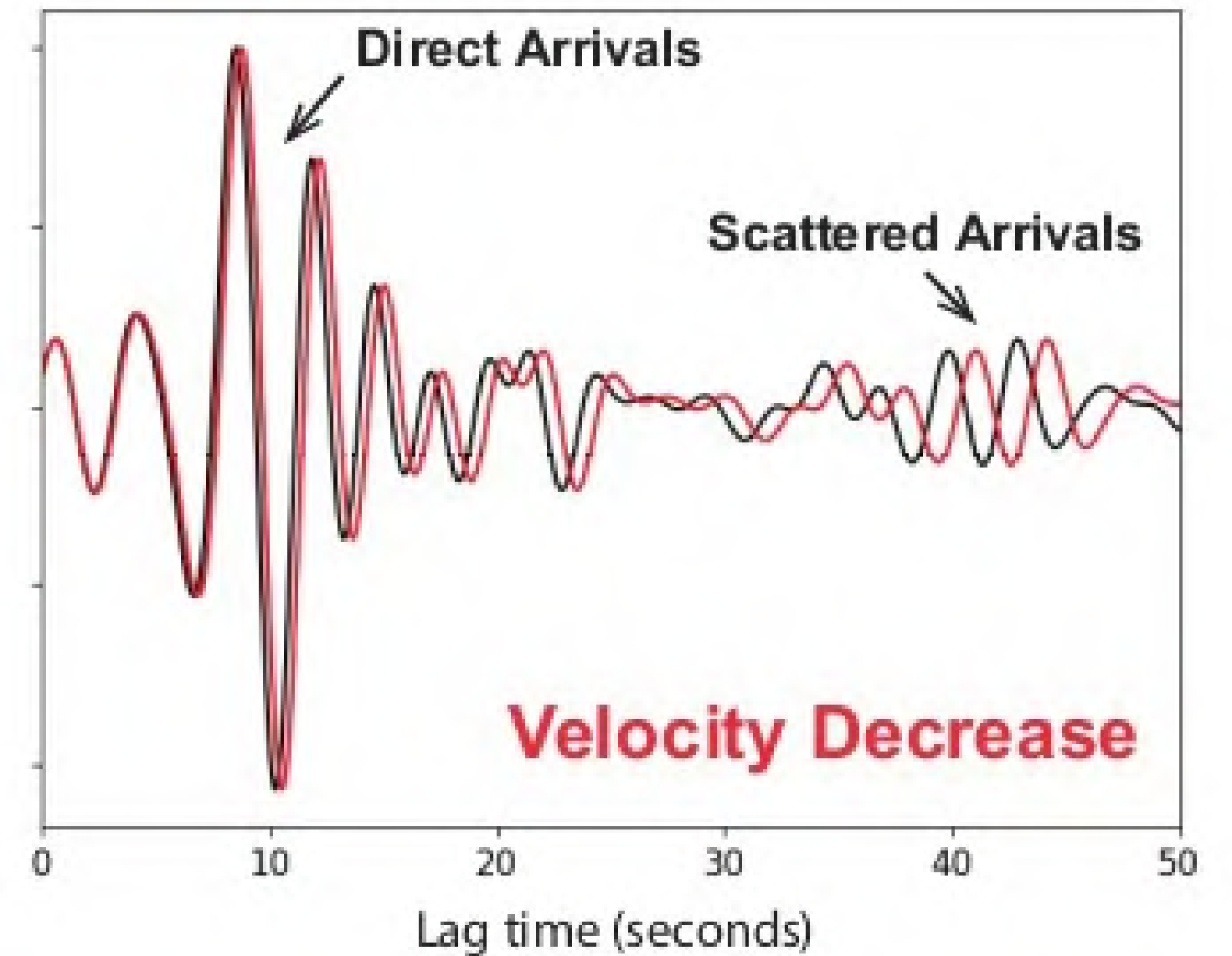
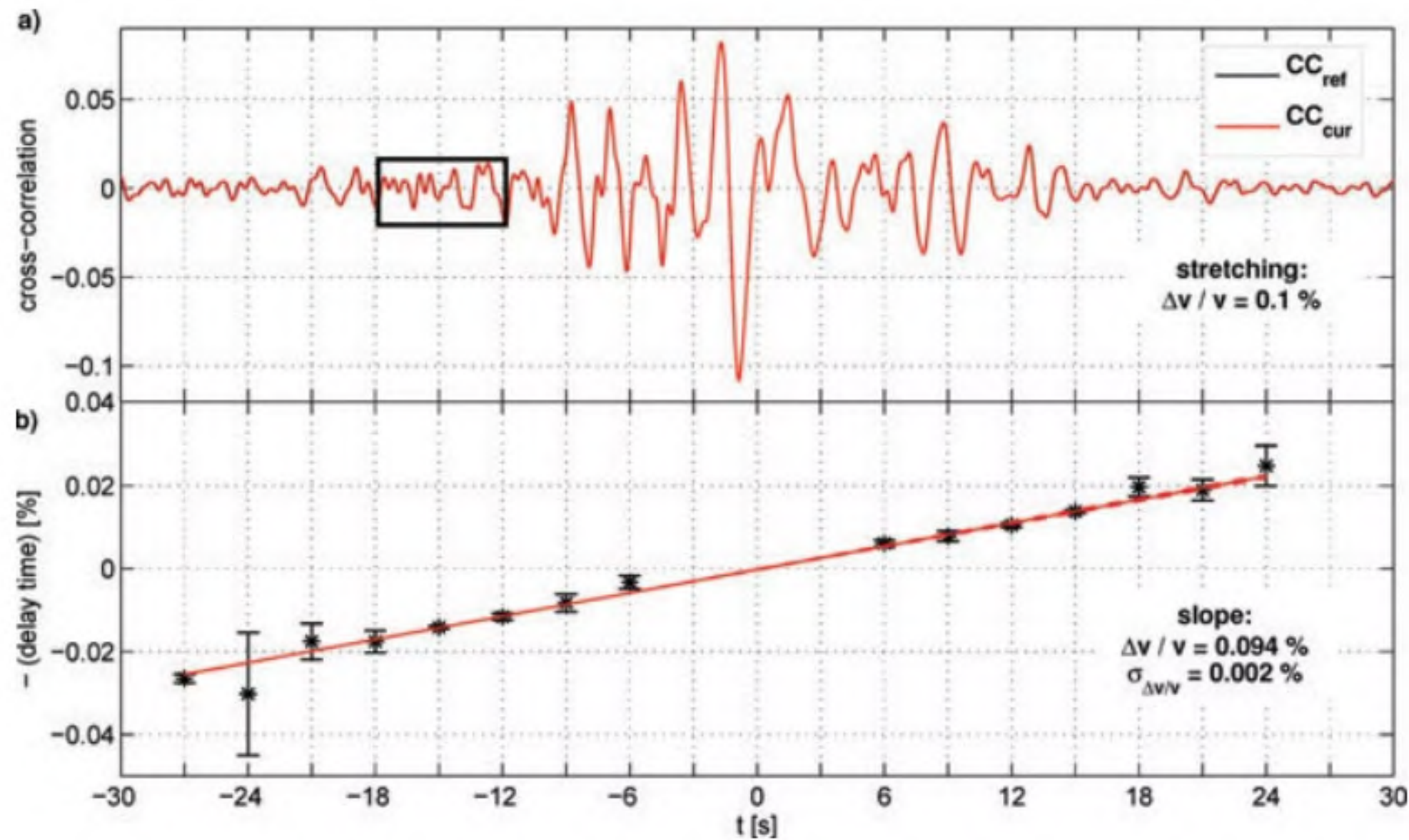
Monitoring with noise – dv/v

Detecting velocity changes over time



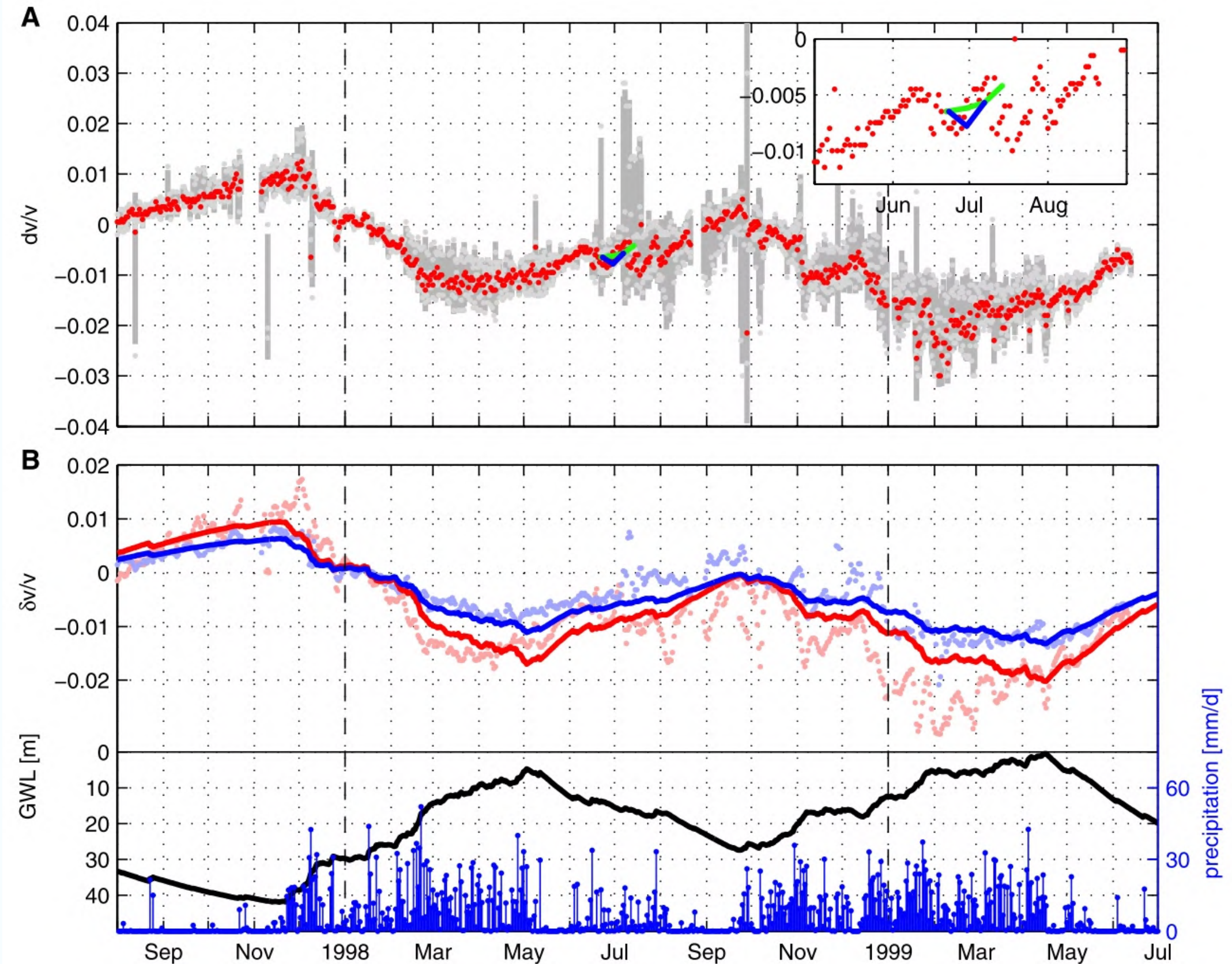
Monitoring with noise – dv/v

Detecting velocity changes over time



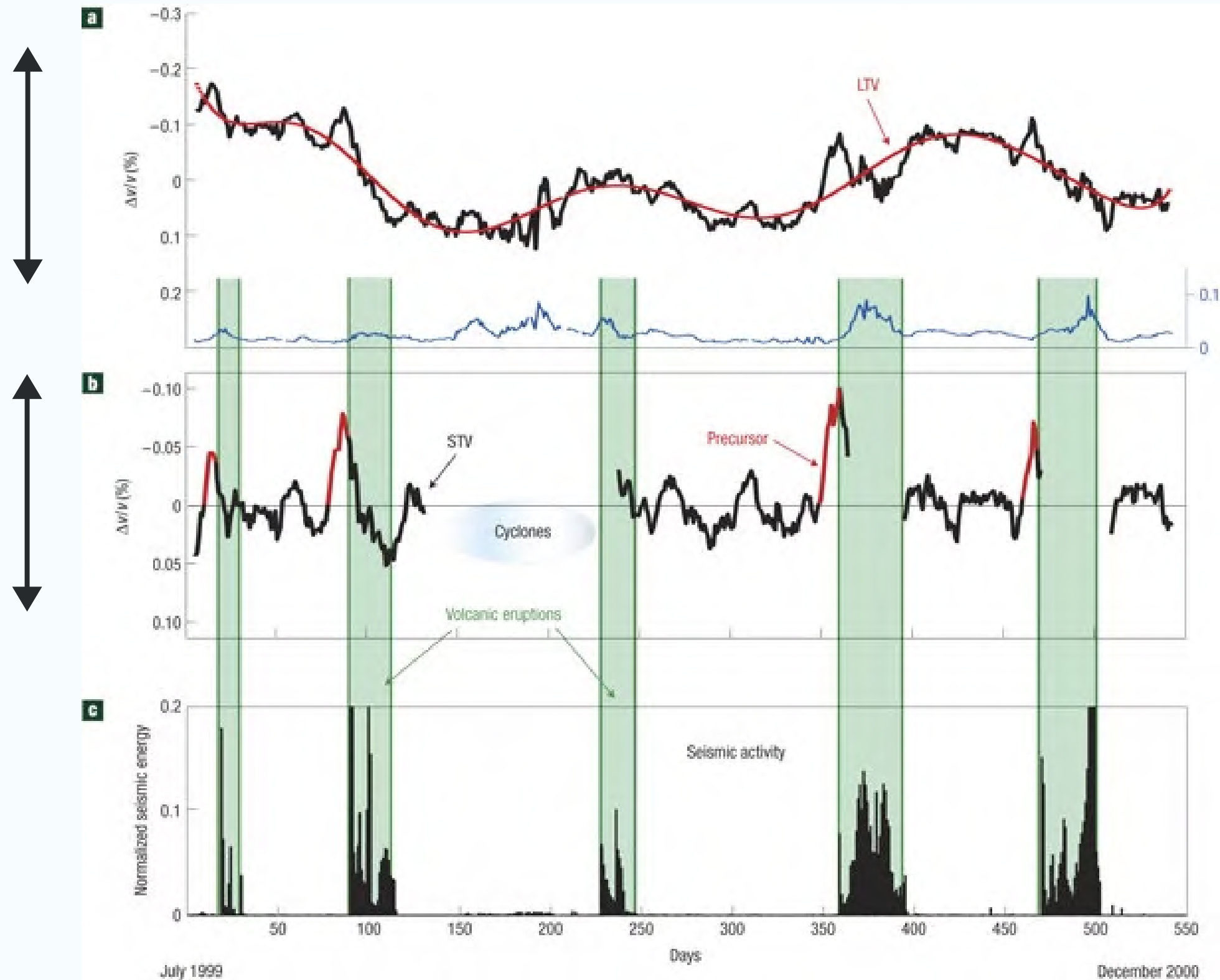
Monitoring with noise – dv/v

Detecting velocity changes over time



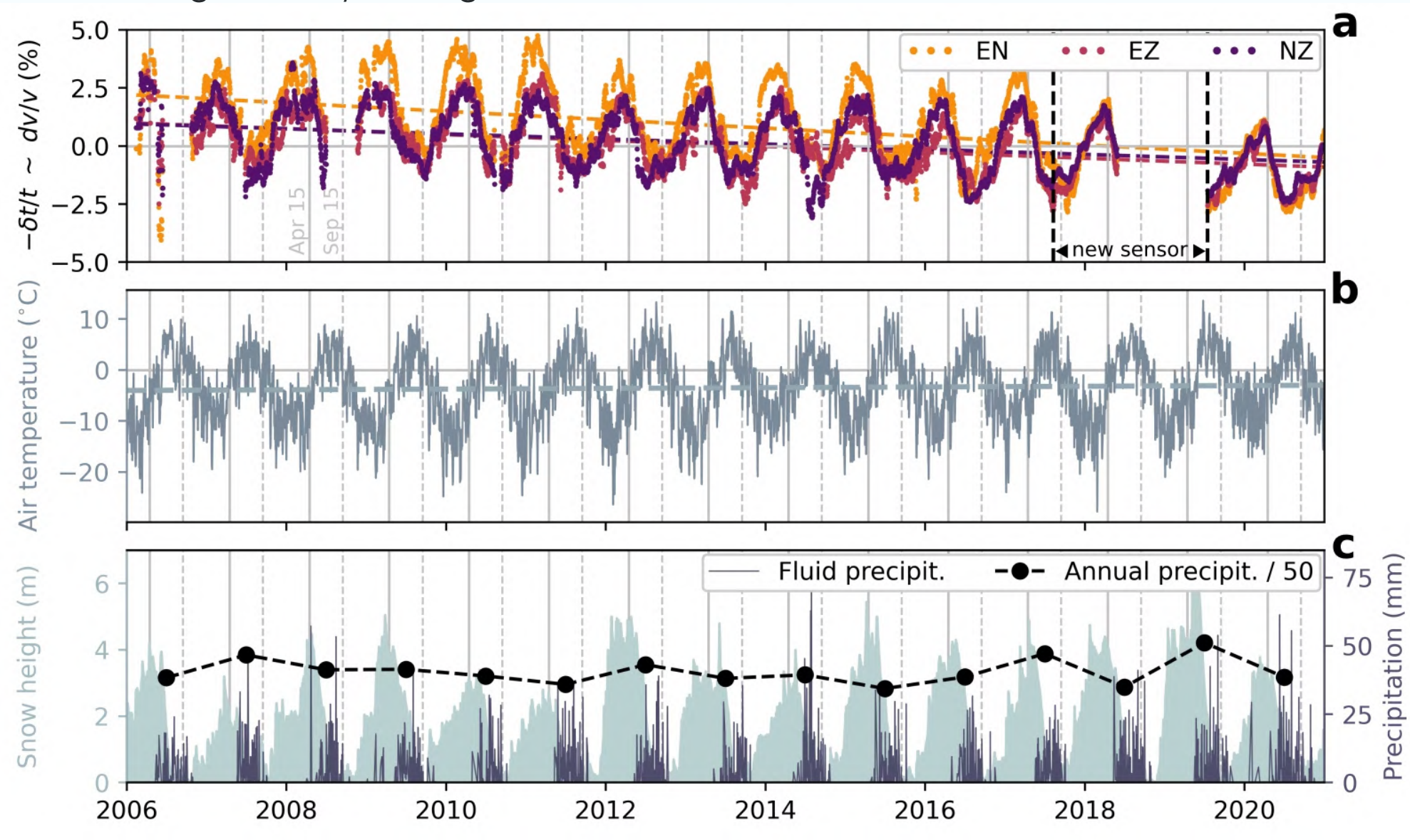
Monitoring with noise – dv/v

Detecting velocity changes over time

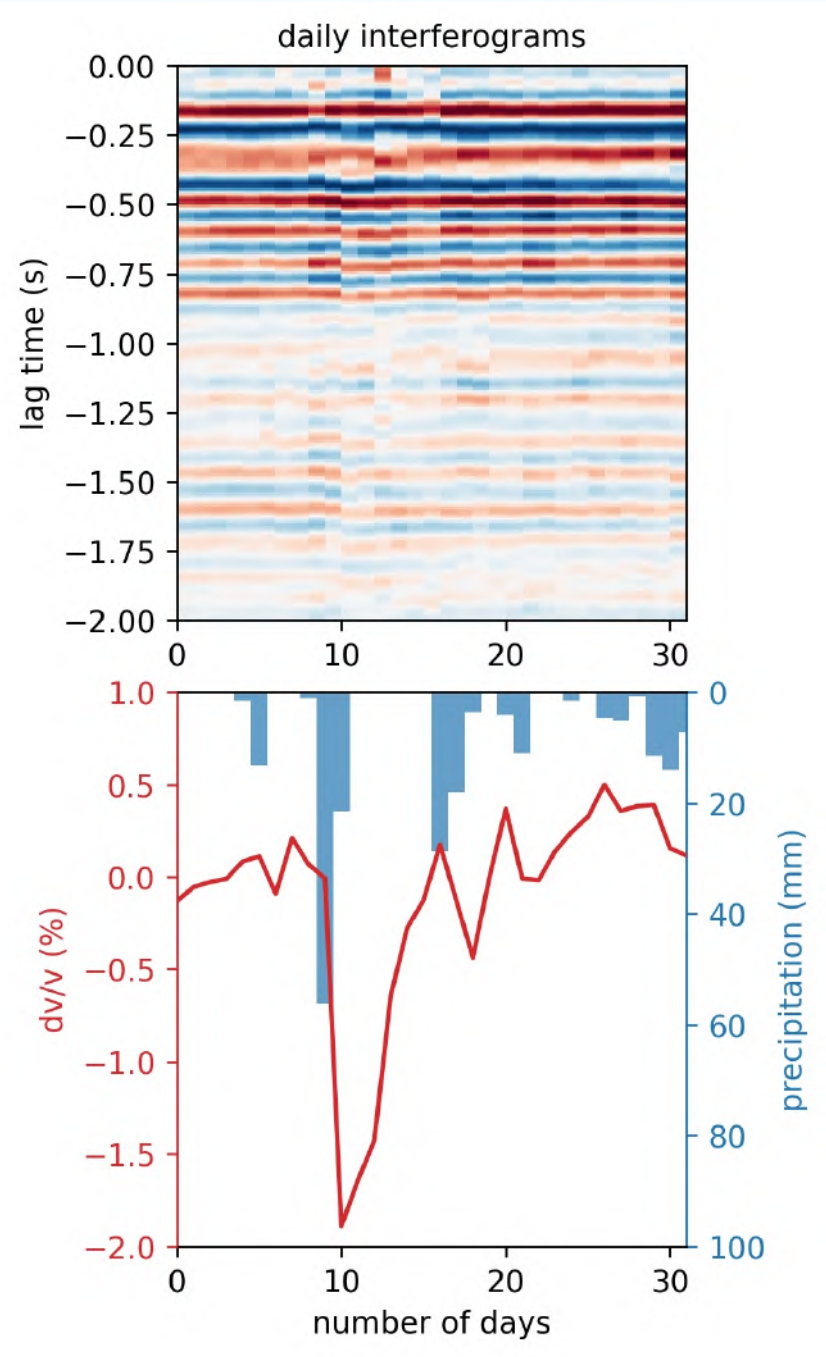


Monitoring with noise – dv/v

Detecting velocity changes over time



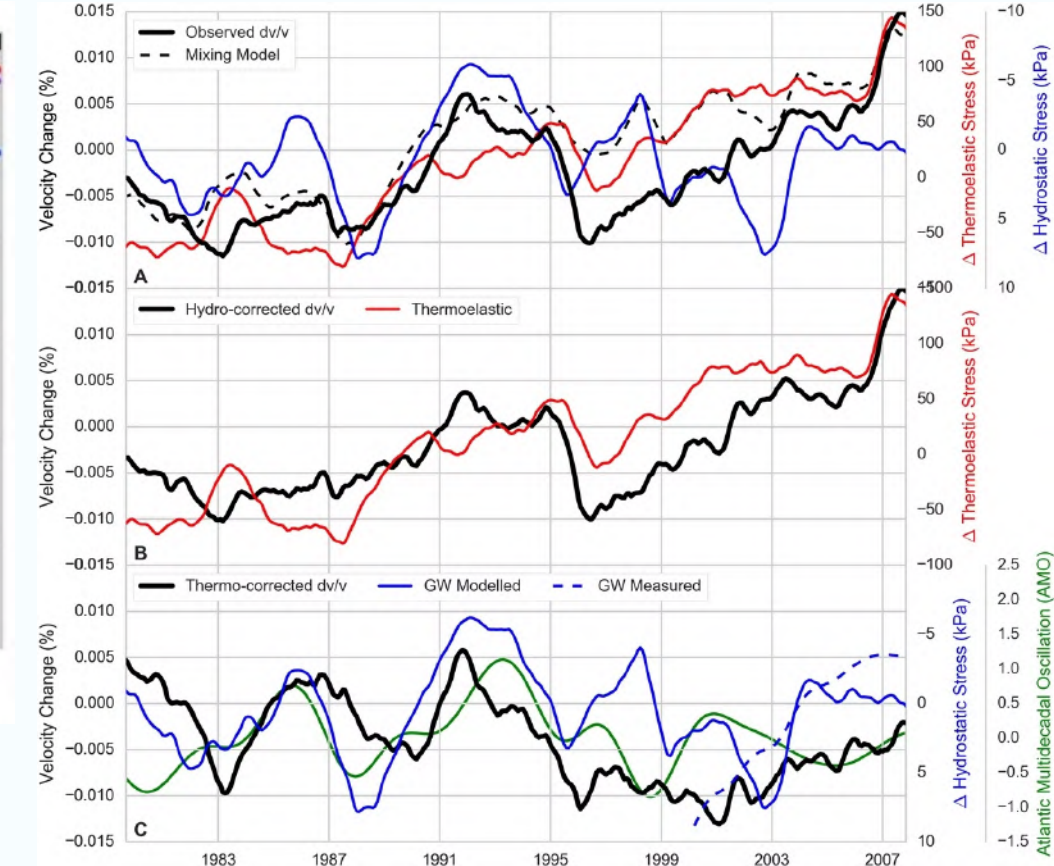
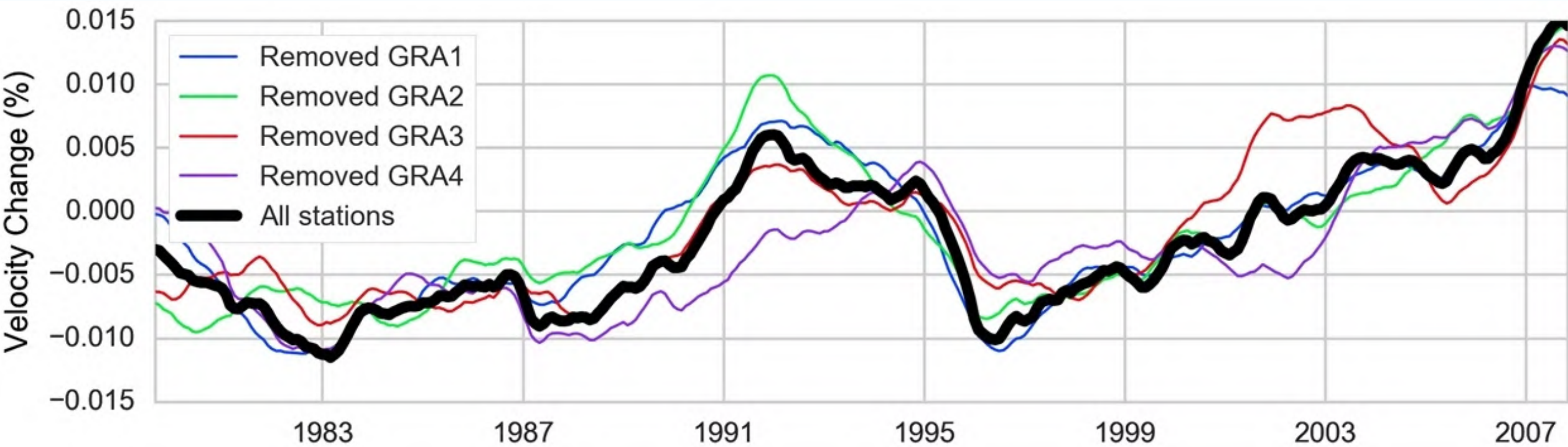
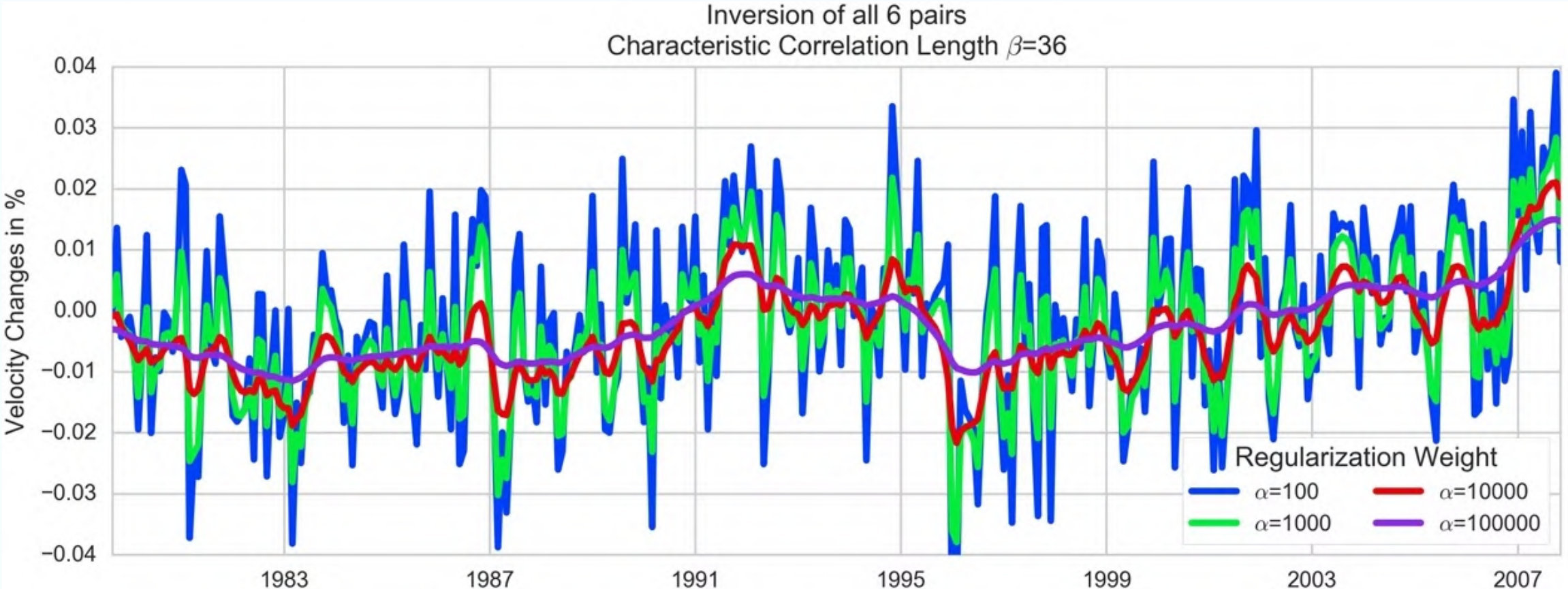
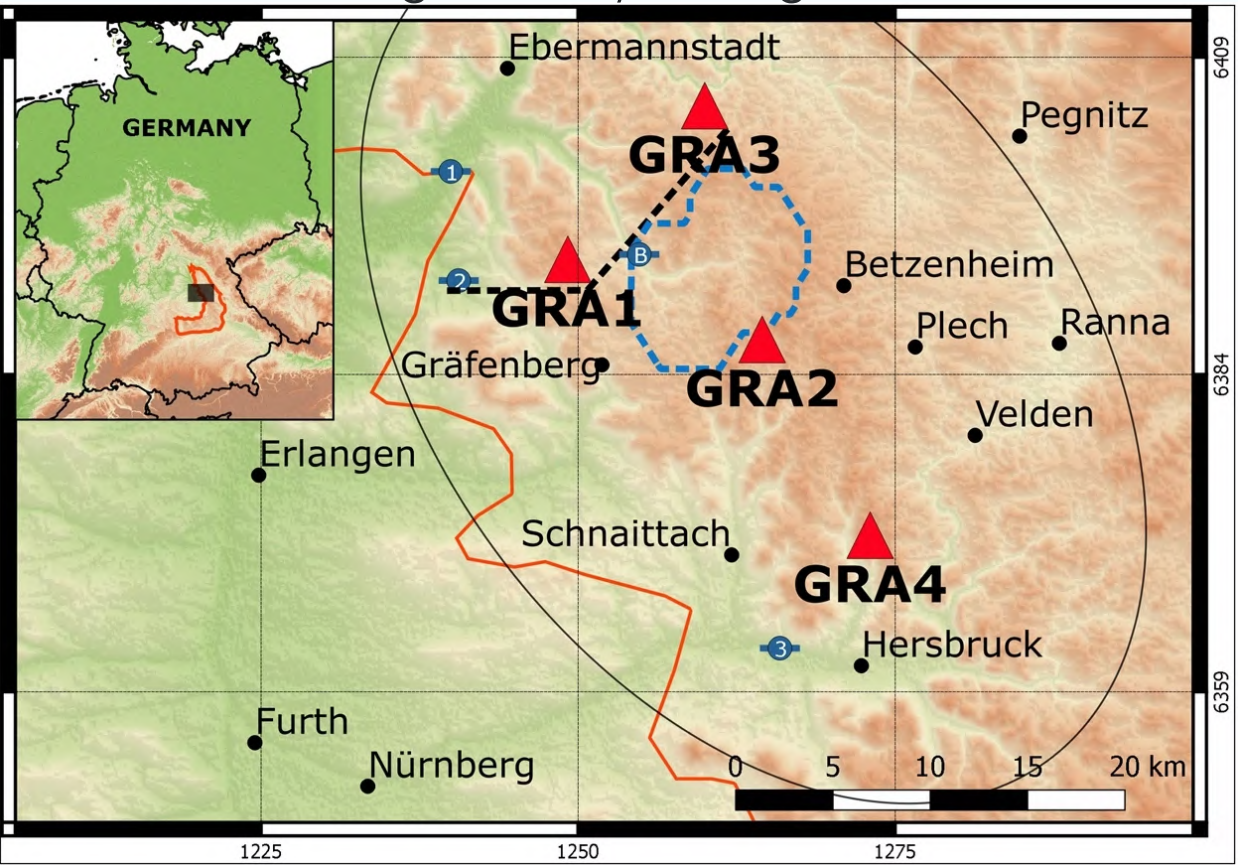
seasonal and long-term changes due to freeze-thaw and permafrost degradation: relevant for glacierized volcanoes



velocity change due to precipitation (Lindner et al, in preparation)

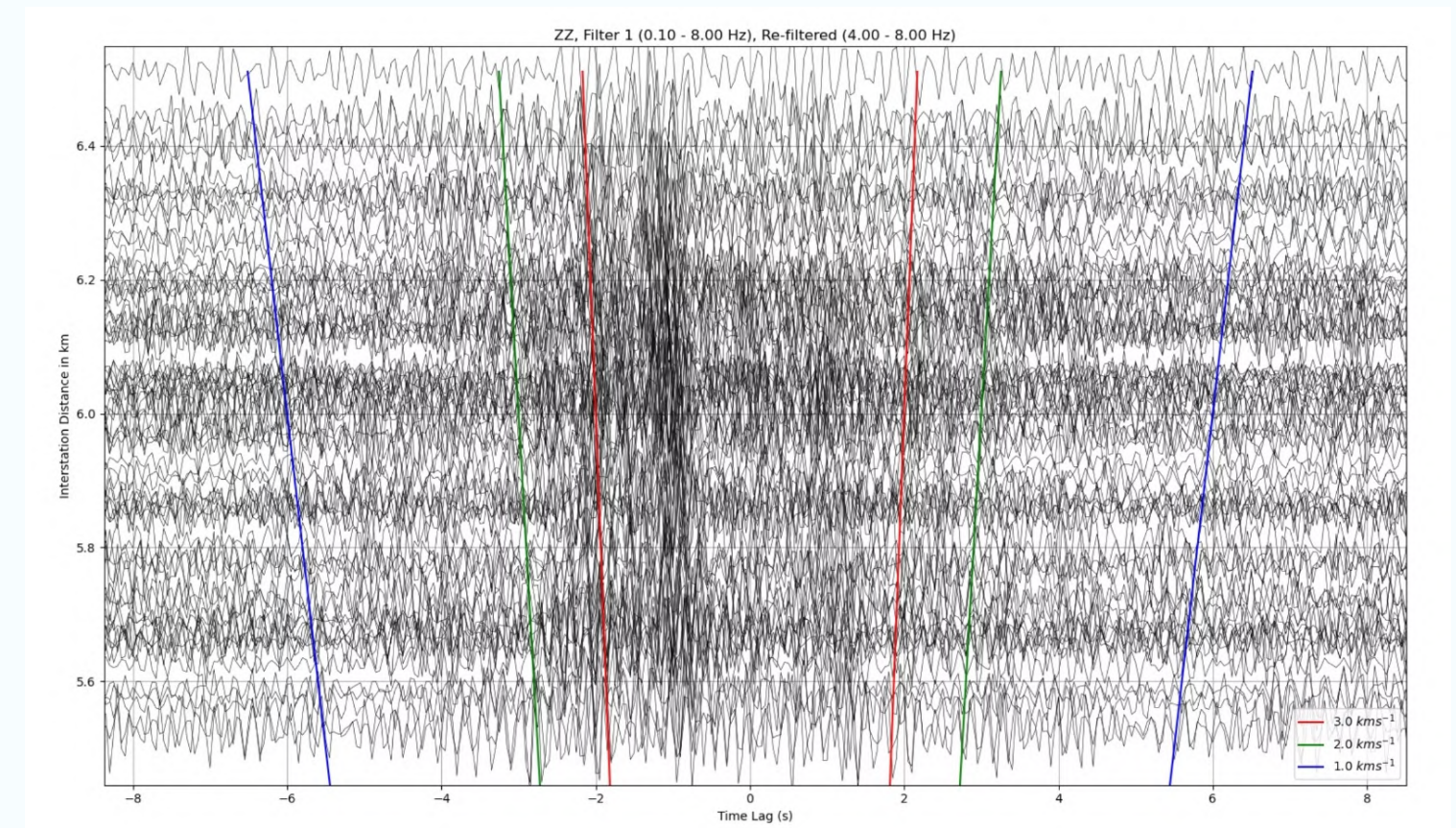
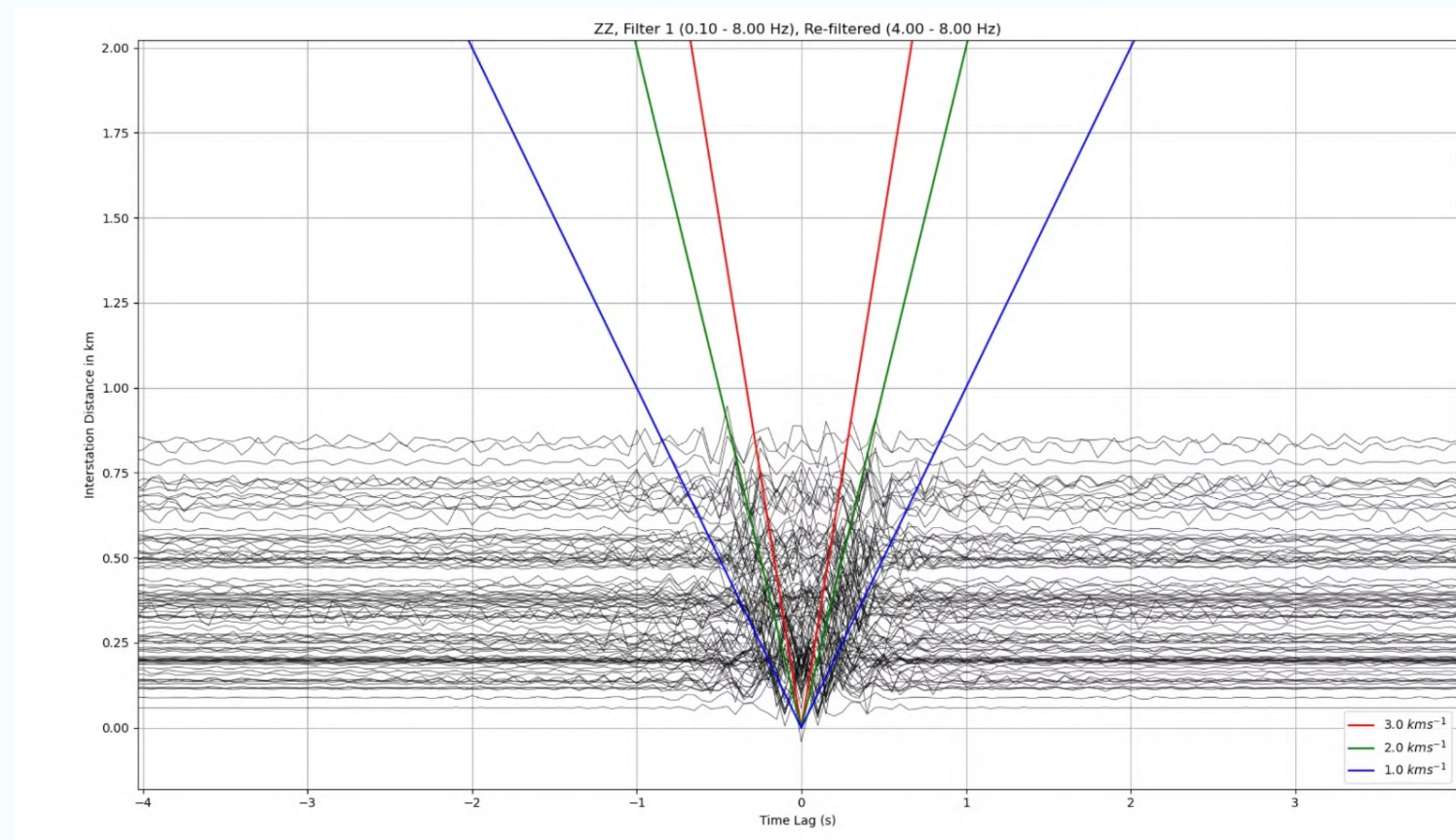
Monitoring with noise – dv/v

Detecting velocity changes over time



TIME - NOTEBOOK 30: MSNoise & CCFs

Using noise to extract propagation



Wrap up

- **The Earth is never silent:** Ground motion spans 15 orders of magnitude in amplitude and covers all frequencies — "noise" and "signal" are defined by your scientific question, not by physics
- **Measuring ground motion:** A seismometer exploits inertia; force-feedback extends its bandwidth; every instrument has a self-noise floor that limits what it can hear — choose your sensor for your target band
- **Instrument response:** Everything recorded is a distorted version of ground motion — understanding and removing $H(f)$ is not optional, it is the foundation of quantitative seismology
- **Seismic noise:** The noise spectrum is structured, predictable, and physically meaningful — microseisms, cultural noise, and tidal signals each tell a story about the Earth and human activity
- **Noise as a tool:** From HVSR site characterisation to ambient noise tomography and continuous dv/v monitoring, seismic noise has become one of the most productive data sources in modern geophysics

THE END

Noisy, and we "know" it since forever (...)

Bulletin Mensuel

DE LA

STATION GÉOPHYSIQUE D'UCCLE

(STATION ERNEST SOLVAY)

LATITUDE : 50° 47' 53" N.

LONGITUDE : E. DE GREENWICH 4° 21' 44"

JANVIER A MARS 1903

BRUXELLES

Imprimerie Veuve FERDINAND LAROIEN

25, RUE DES MINIMES

1904



Observations générales. — Le temps employé est celui de Greenwich, compté de minuit à minuit. Les pendules sont orientes : Le pendule M (éridien) dans le méridien; le pendule E (st), à 120° vers l'est; le pendule W (west), à 120° degrés vers l'ouest. T = tremors ou première onde sismique; P, onde principale ou seconde; F = fin (toujours approchée); ca = circa; da = de... à; A. S. = agitation microsismique; Tr. = macrosismes non sensibles; Tr₁ = macrosismes sensibles; M. S. = mouvements sismiques; C. = constante; Cf. = constante et forte; C^t = commencement. Les lectures sont faites à ± 8 secondes près.

Nos	DATES.	M. S.	T			P			F			A. S.	
			W h. m. s.	M h. m. s.	E h. m. s.	W h. m. s.	M h. m. s.	E h. m. s.	W h. m. s.	M h. m. s.	E h. m. s.	C ^t h. m. s.	F h. m. s.
1	1	A. S.	»	»	»	»	»	»	»	»	»	C	»
2	2	Tr.	?	?	?	?	12,56, 4	12,56, 4	?	?	?	C	»
3	3	A. S.	»	»	»	»	»	»	»	»	»	C	»
4	4	Tr.	?	5,17,34	5,17,34	5,25,32	5,25,32	5,25,32	?	?	?	Cf	»
5	5	Tr.	?	1, 0,35	0,35	?	1,11,31	1,11,31	?	?	?	C	»
6	5	Tr.	22,22,22	22,22,22	22,22,22	22,39,50	22,39,50	22,39,50	?	?	?	C	»
7	6	Tr.	?	?	?	?	?	17, 8,58	?	?	?	C	»
8	7	A. S.	»	»	»	»	»	»	»	»	»	C	»
9	8	A. S.	»	»	»	»	»	»	»	»	»	Cf	»
10	9	A. S.	»	»	»	»	»	»	»	»	»	Cf	»

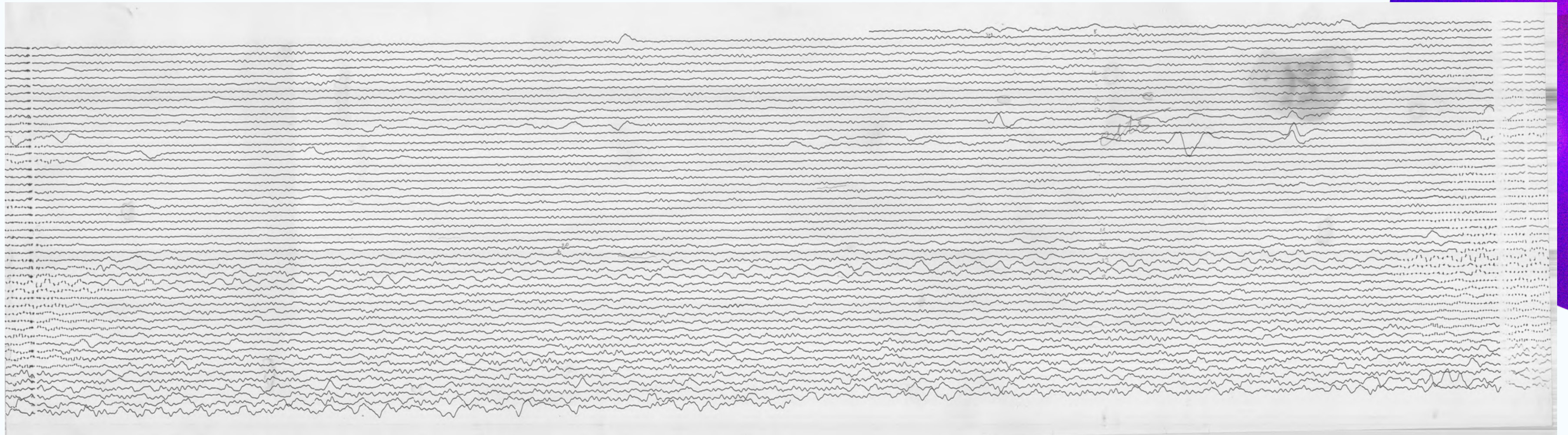
JANVIER 1903

L'agitation sismique, constante pour ainsi dire et souvent très forte que les dépressions barométriques amènent à Uccle pendant les mois d'hiver, rend la constatation générale des phénomènes dus aux tremblements de terre lointains, assez difficile. C'est une remarque qui nous paraît de plus en plus établie et dont il y aura lieu de tenir compte peut-être dans l'établissement d'une station future.

The seismic agitation, constant so to speak and often very strong that barometric depressions bring to Uccle during the winter months, makes the general observation of phenomena due to distant earthquakes quite difficult. This is a remark which seems more and more established to us and which it will perhaps be necessary to take into account in the establishment of a future station.

E. LAGRANGE, D. S.
Professeur à l'École militaire.

Noisy, and we "know" it since forever (...)

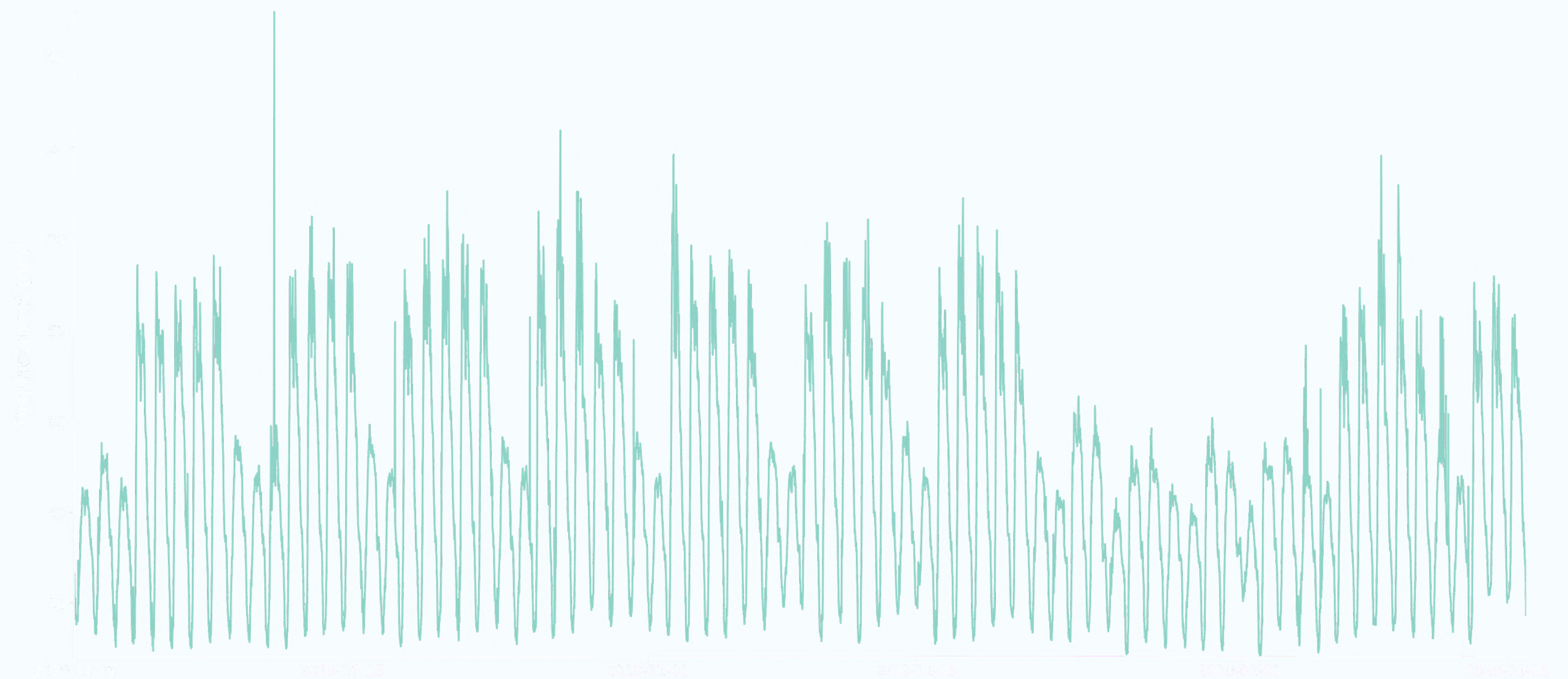


EW g
7.25
30-1-53



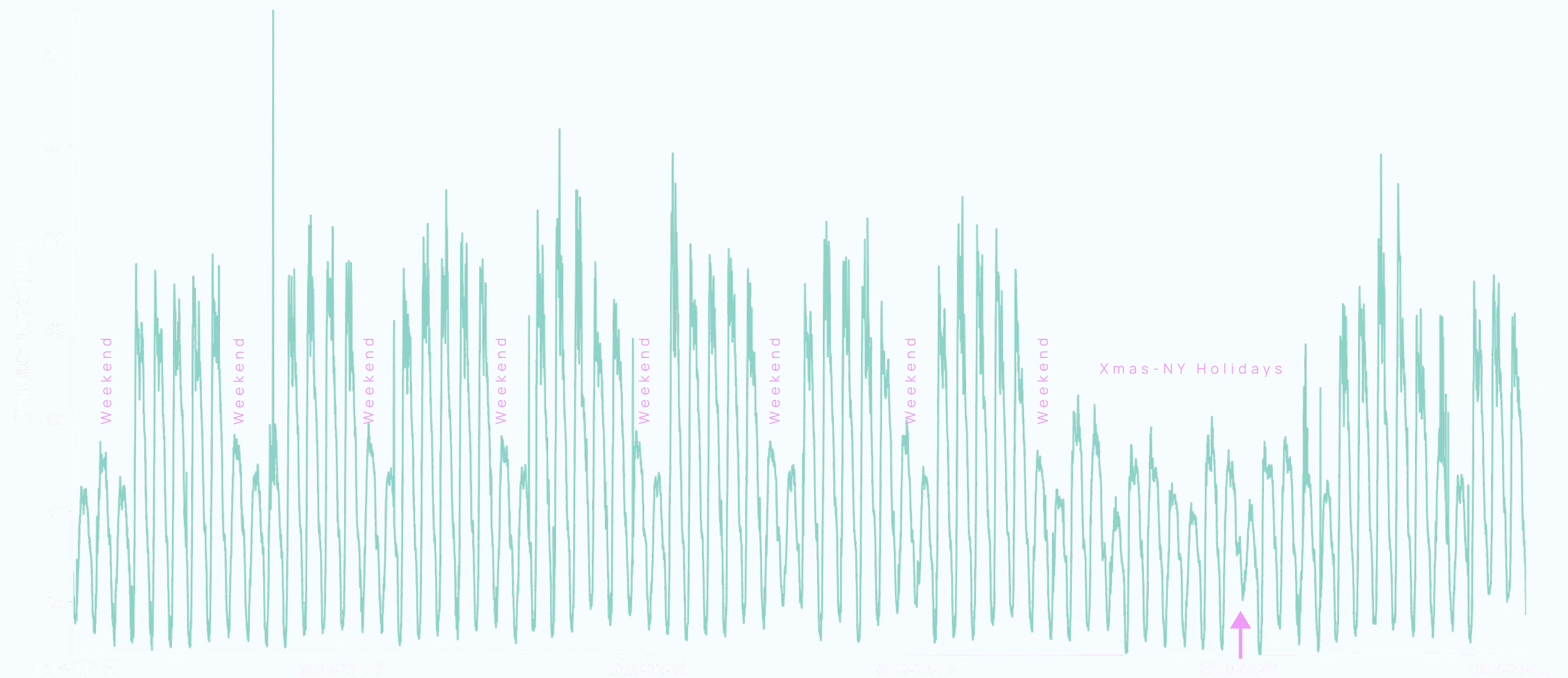
High Frequency Noise & Humans

How do we know we are guilty?



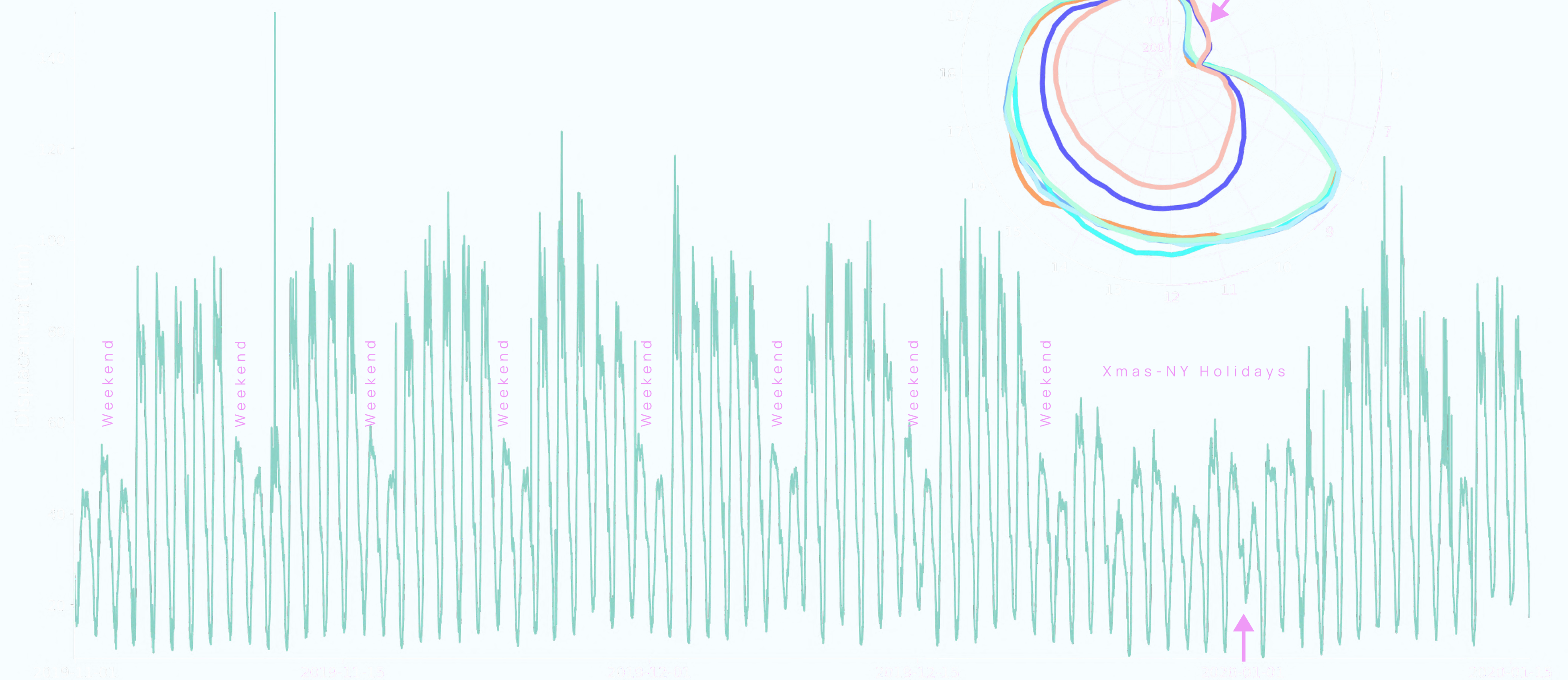
High Frequency Noise & Humans

How do we know we are guilty?



High Frequency Noise

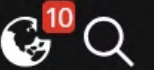
How do we know we are guilty?



Who cares about city noise?



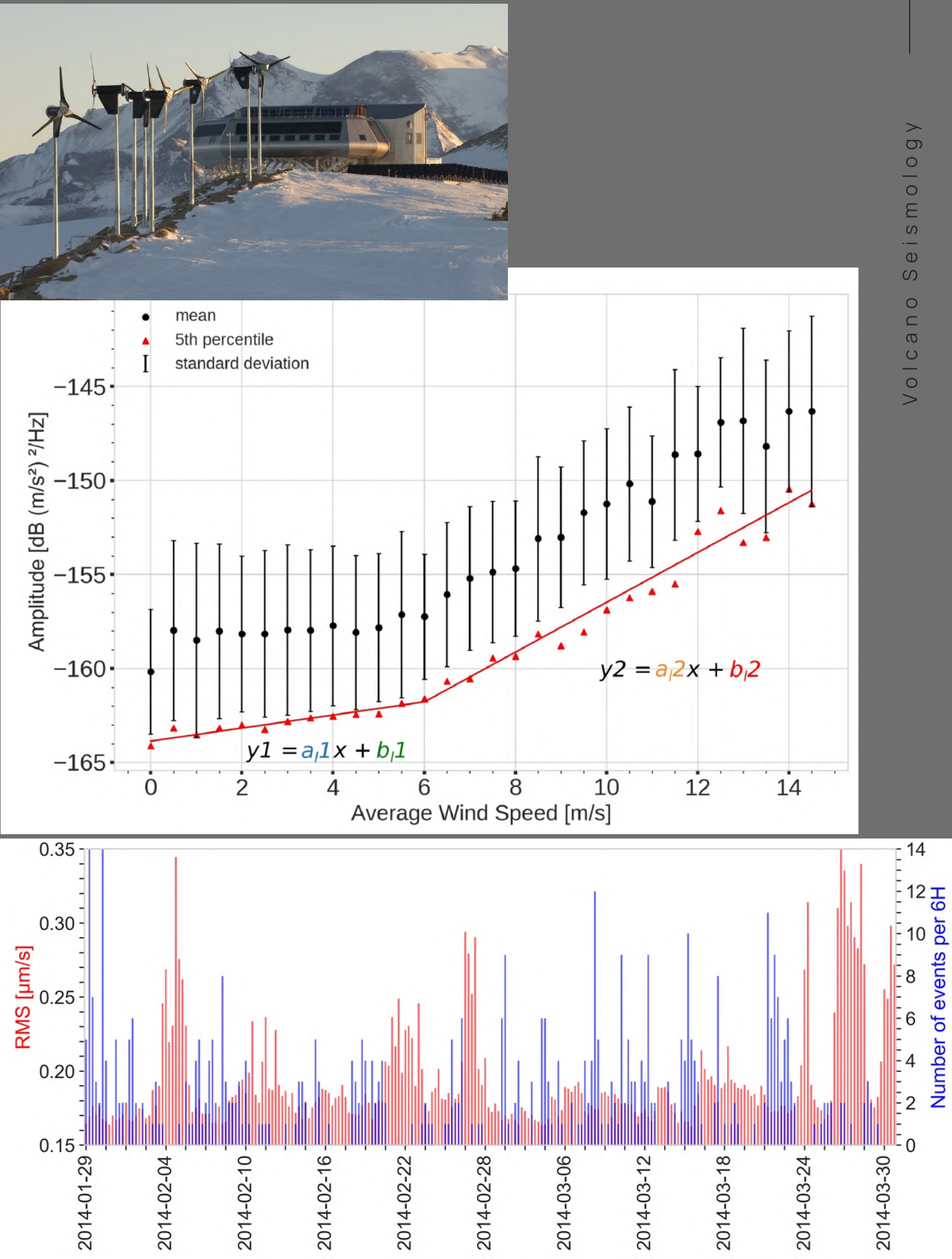
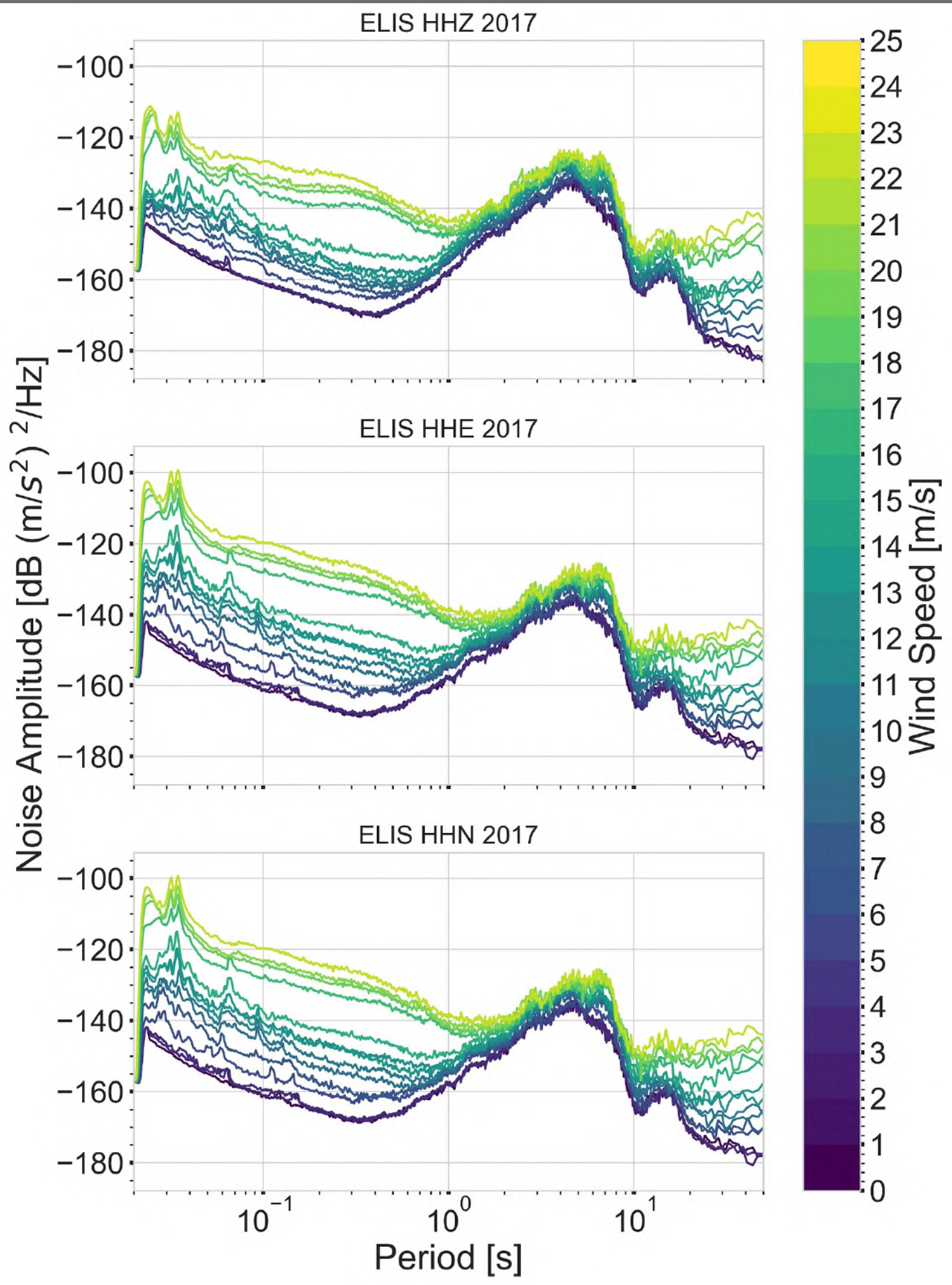
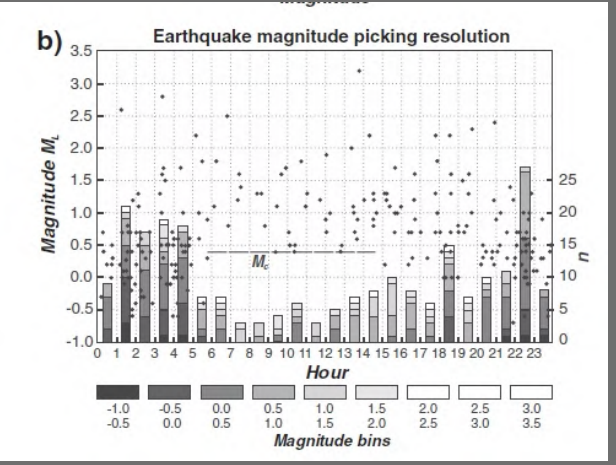
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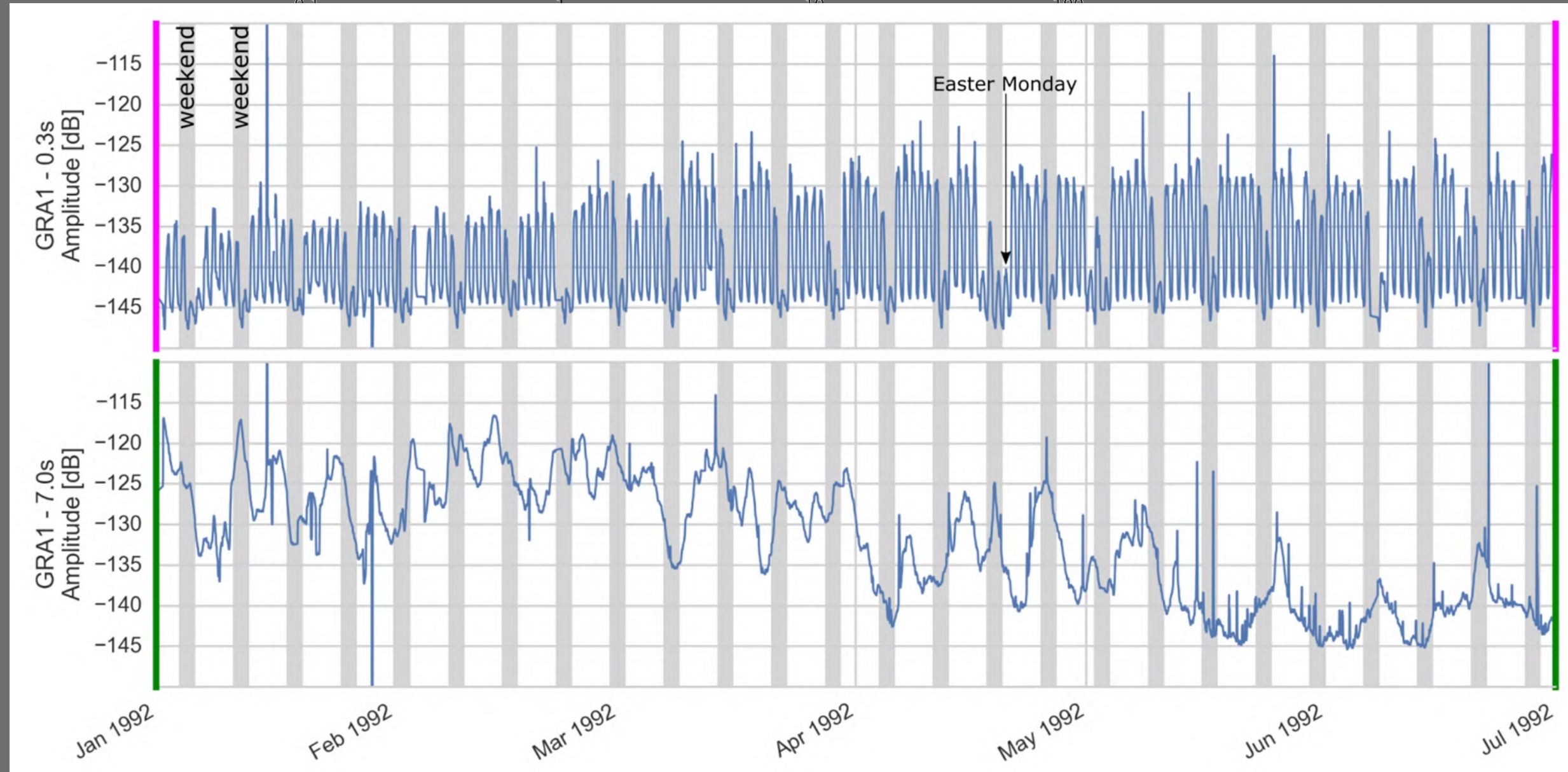
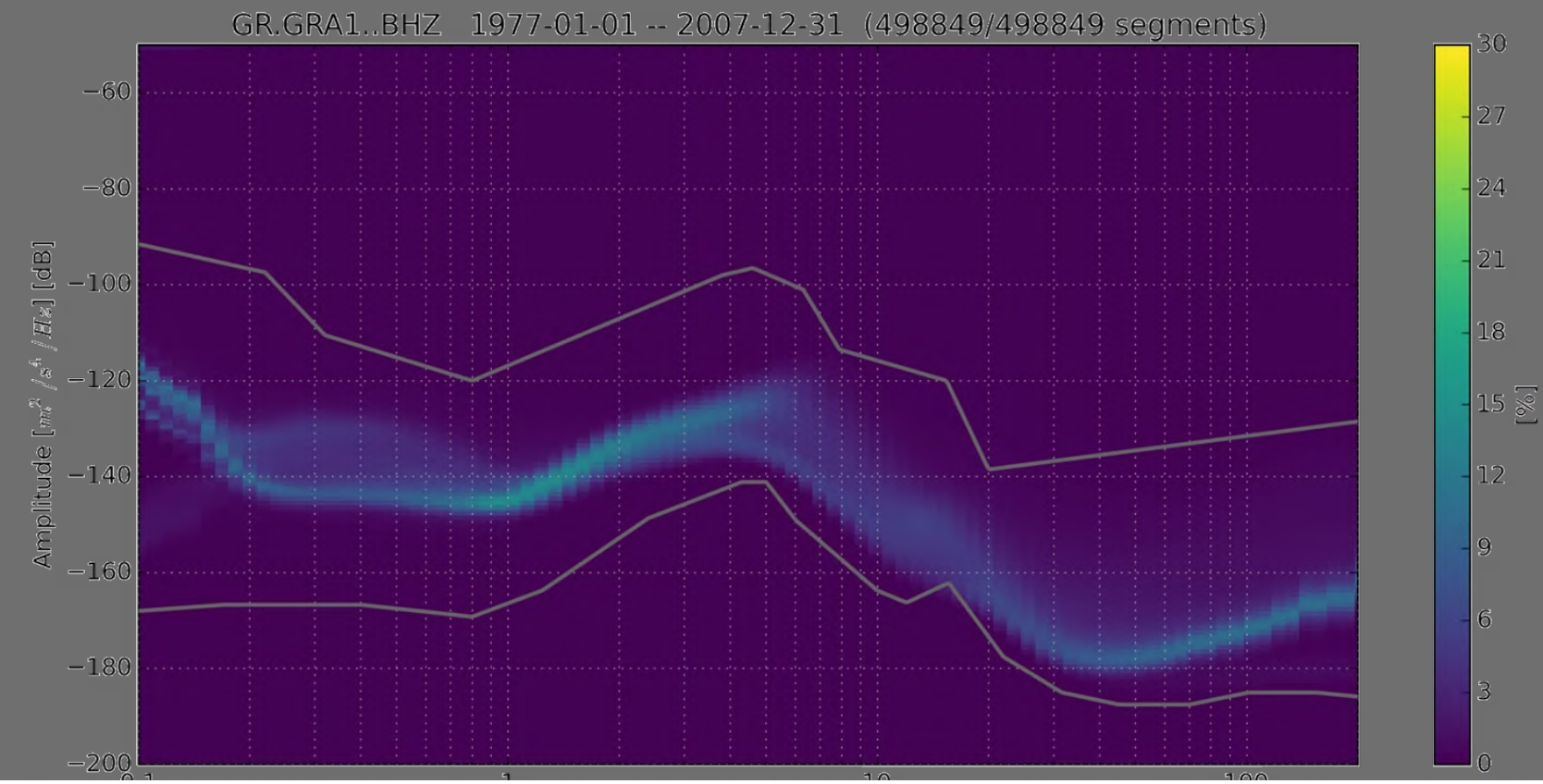
COV12 in La Antigua Guatemala

Get to know the Land
of volcanoes

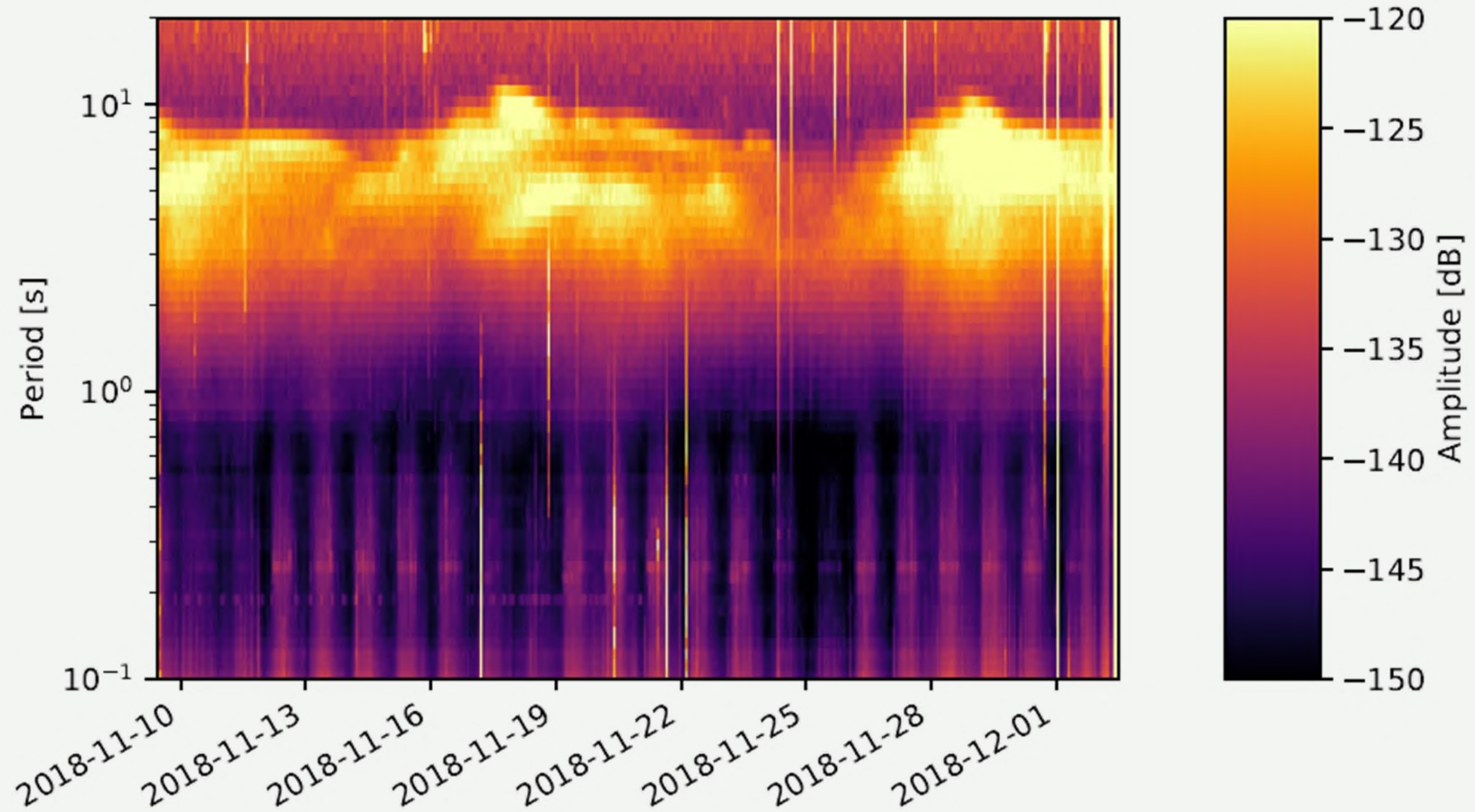
Noise ?
so what?



vs Time?



vs Time?





15th Munich Earth Science School
ObsPy goes Downtown: Seismology in Cities



Part 2

Resonance? What?

*Horizontal to Vertical Spectral Ratio
Continuous HVSR*

Open the 10. Continuous HVSR Geopsy notebook

24 February 2025



Geopsy.exe Instruction



For Windows:

```
geopsy_exe = 'YOUR_PATH/geopsypack-win64-3.4.2/bin/geopsy-hv.exe'
```

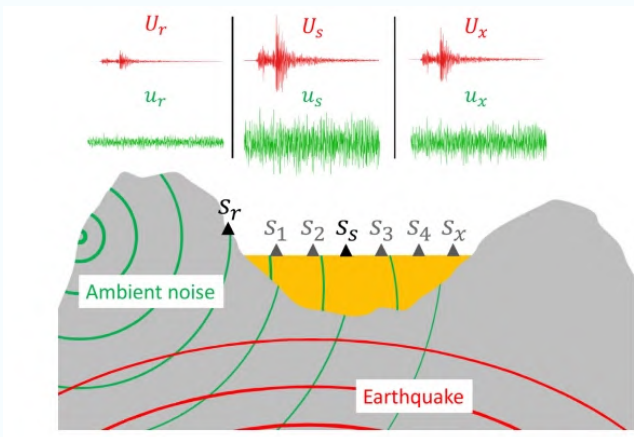
For Linux:

```
geopsy_exe = "geopsy-hv"
```

Marc Wathelet, Jean-Luc Chatelain, Cécile Cornou, Giuseppe Di Giulio, Bertrand Guillier, Matthias Ohrnberger, Alexandros Savvaidis;
Geopsy: A User-Friendly Open-Source Tool Set for Ambient Vibration Processing. *Seismological Research Letters* 2020;; 91 (3): 1878–1889.
doi: <https://doi.org/10.1785/0220190360>

H/V spectral ratio analysis

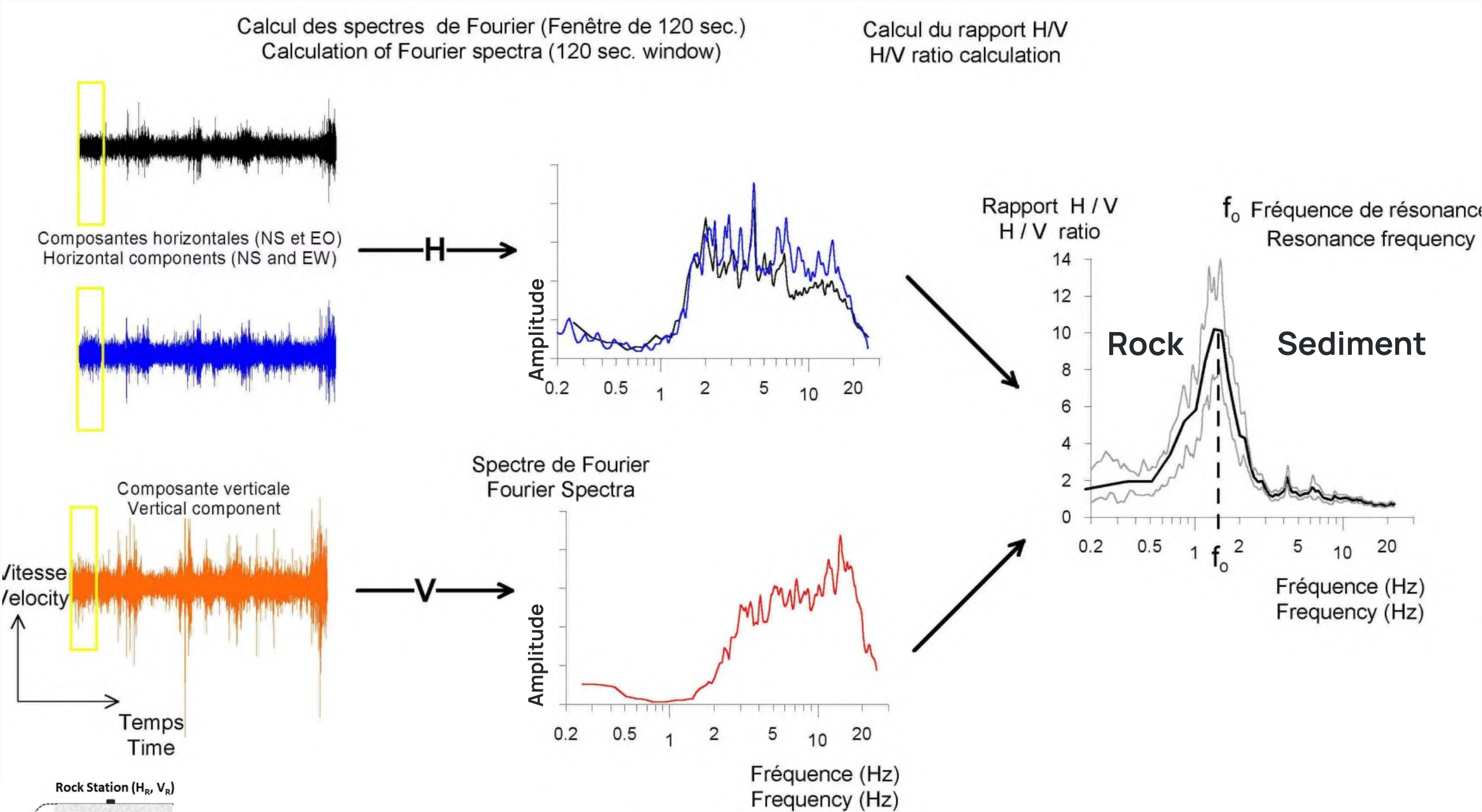
Noise field: surface waves



N-comp.

E-comp.

Z-comp.



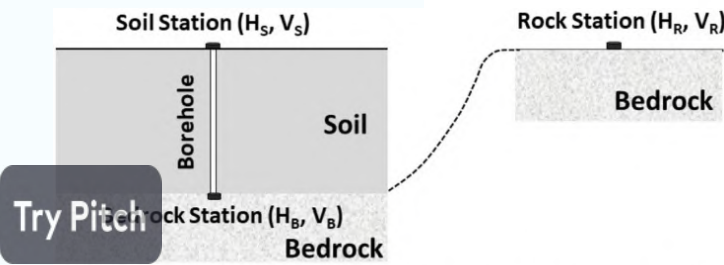
Theoretical
background
=
DEBATED

J. Seismol. (2022) 26:653–685
<https://doi.org/10.1007/s10950-021-10062-9>

ORIGINAL ARTICLE

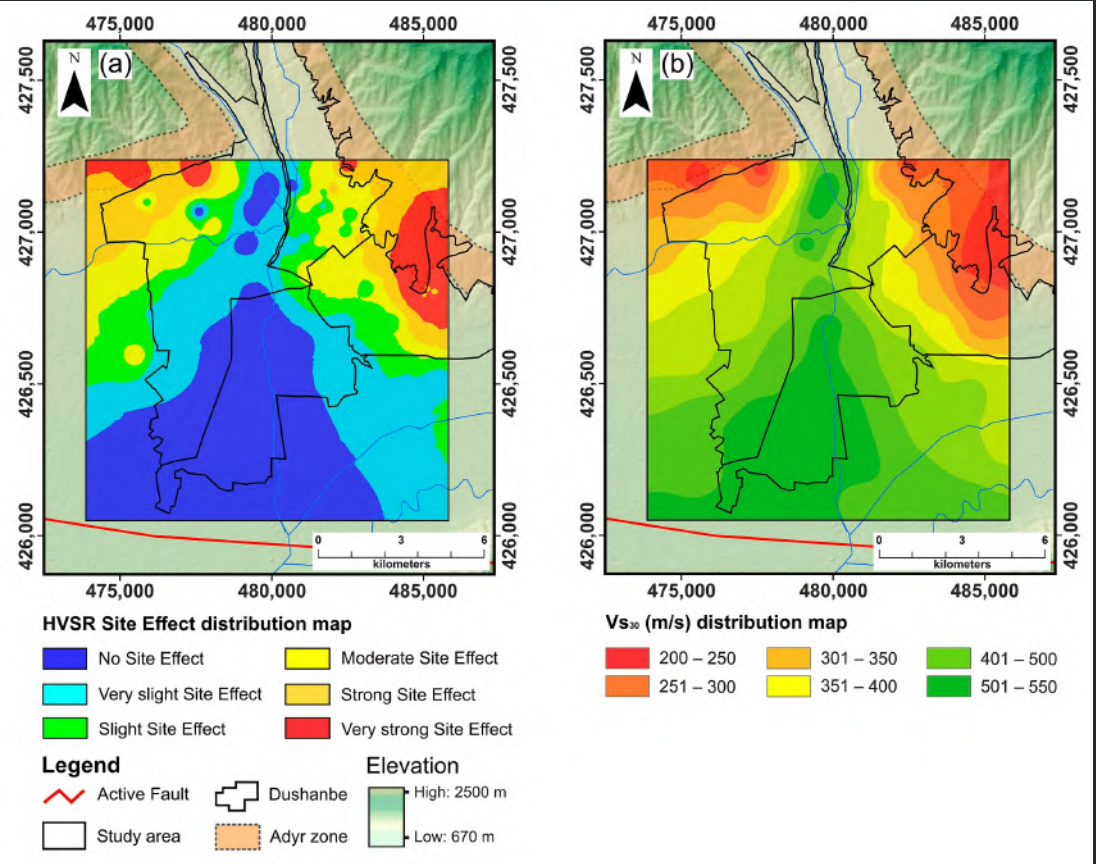
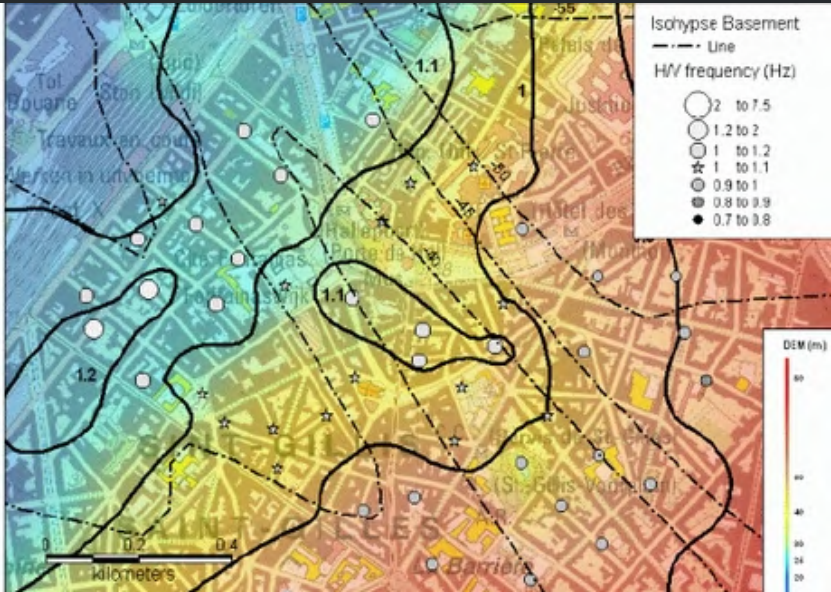
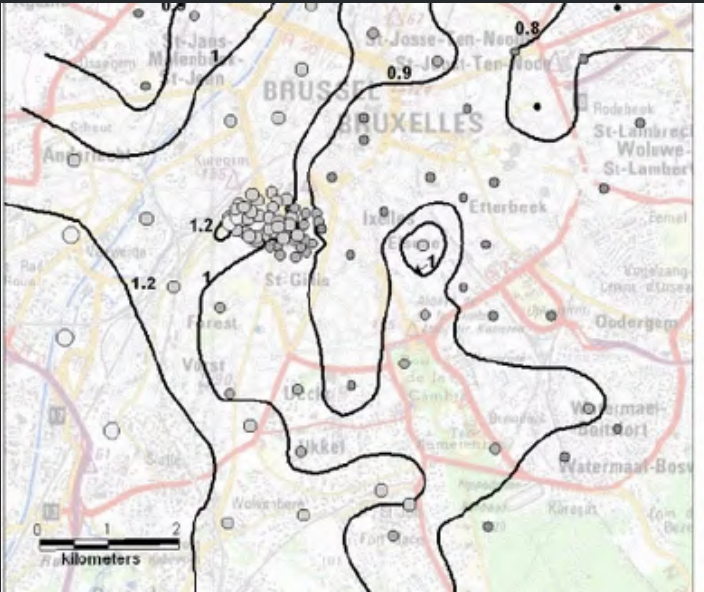
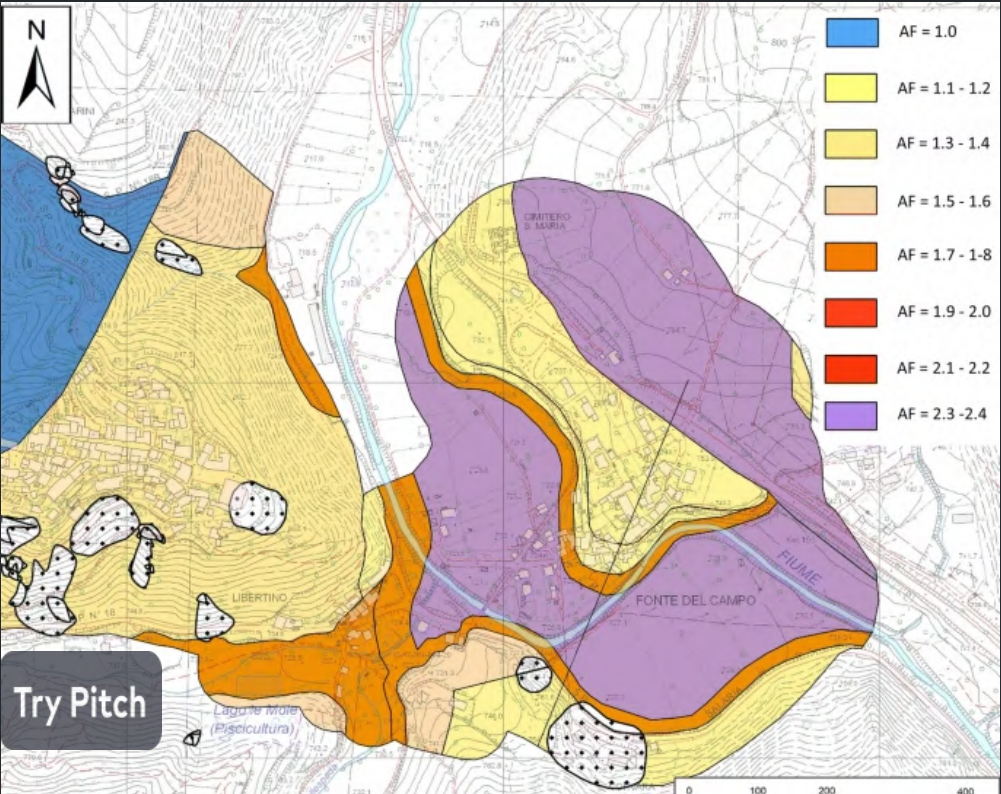
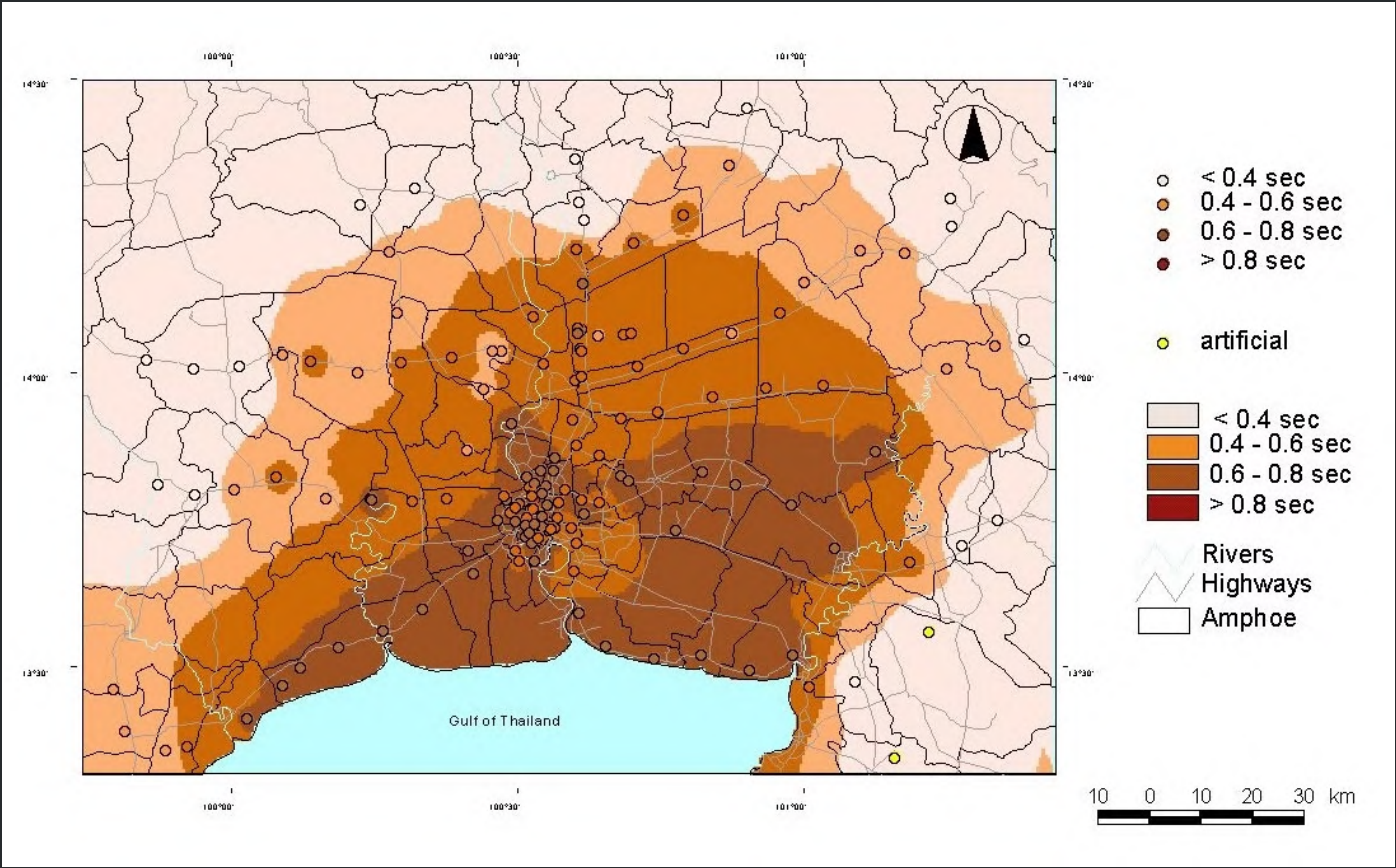
A review of the microtremor horizontal-to-vertical spectral ratio (MHVSR) method

S. Molnar · A. Sirohey · J. Assaf · P.Y. Bard · S. Castellaro · C. Cornou · B. Cox · B. Guiller · B. Hassani · H. Kawase · S. Matsushima · F. J. Sánchez-Sesma · A. Yong



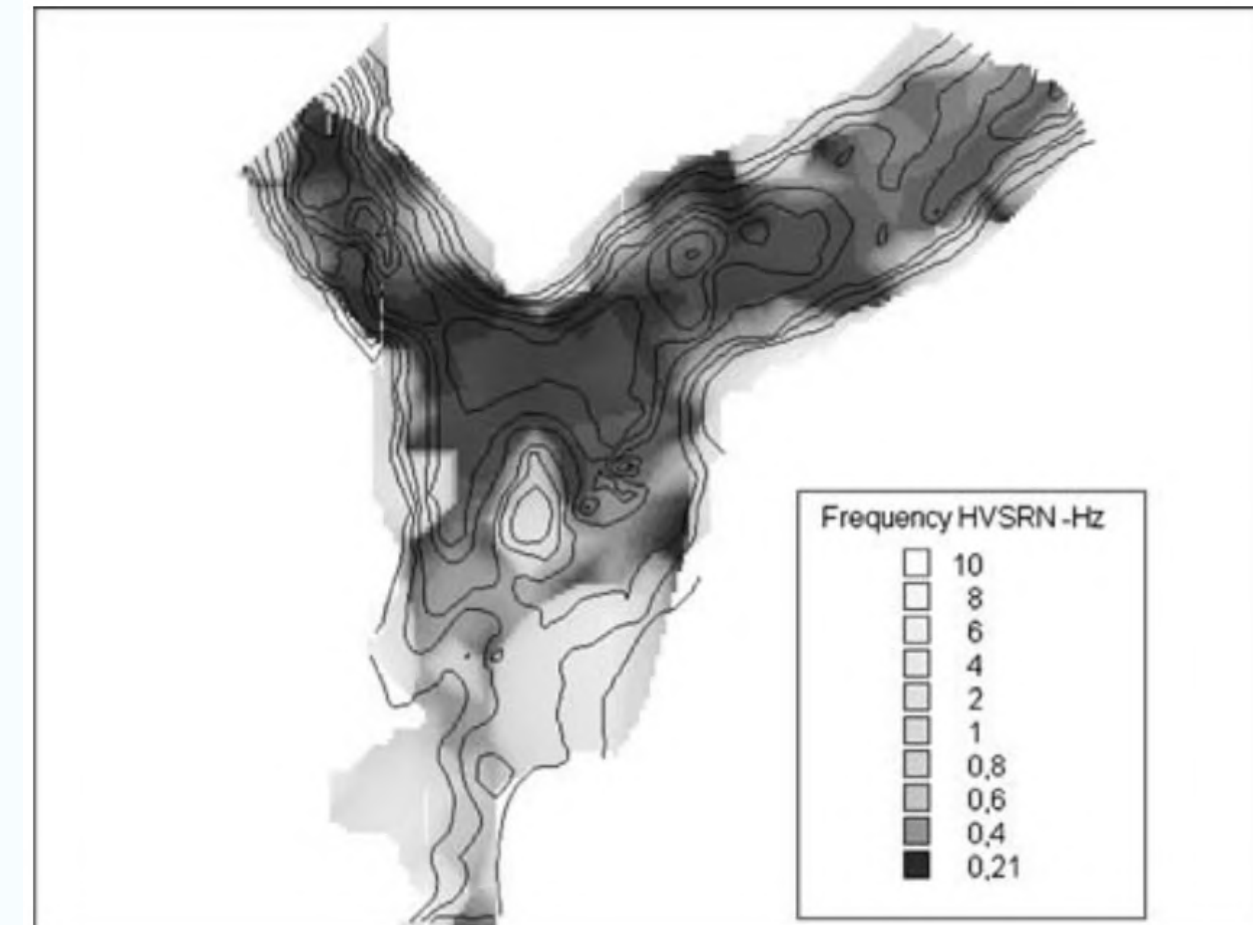
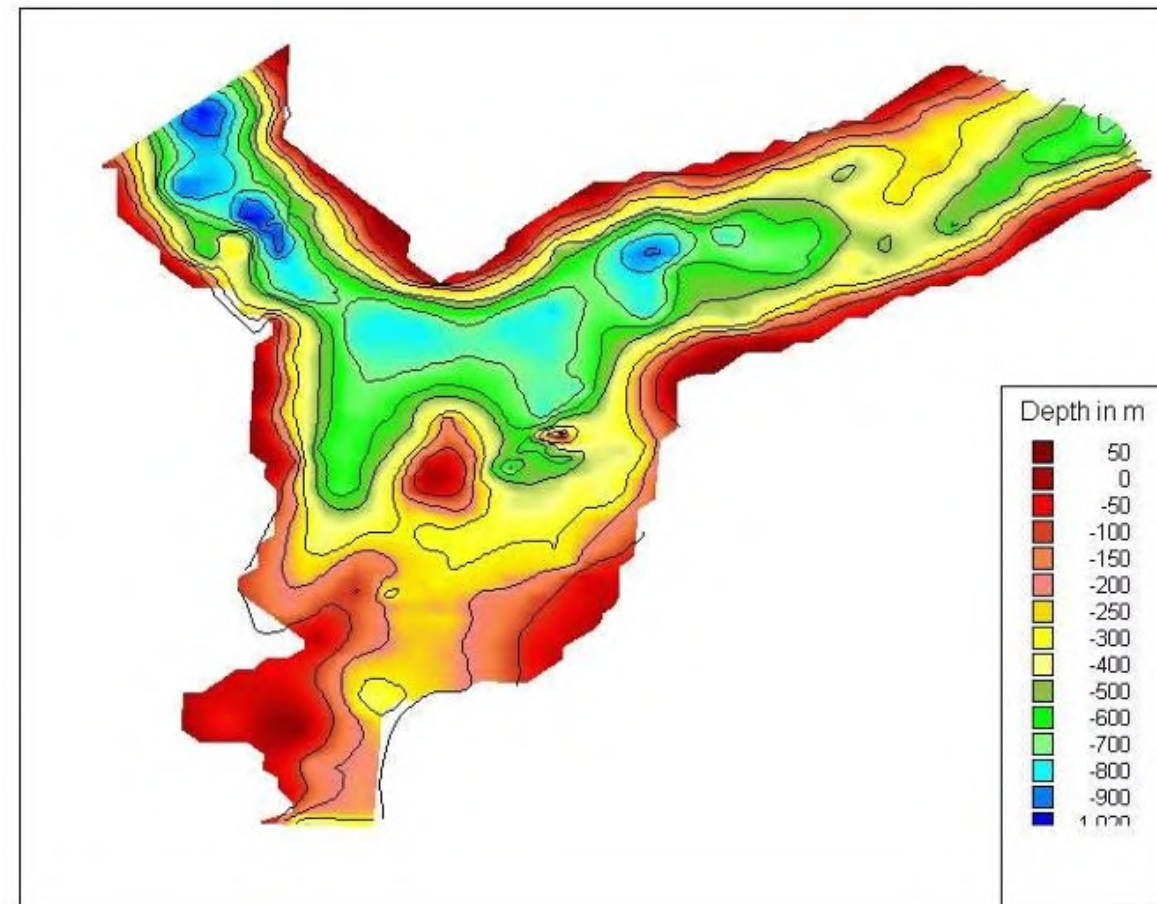
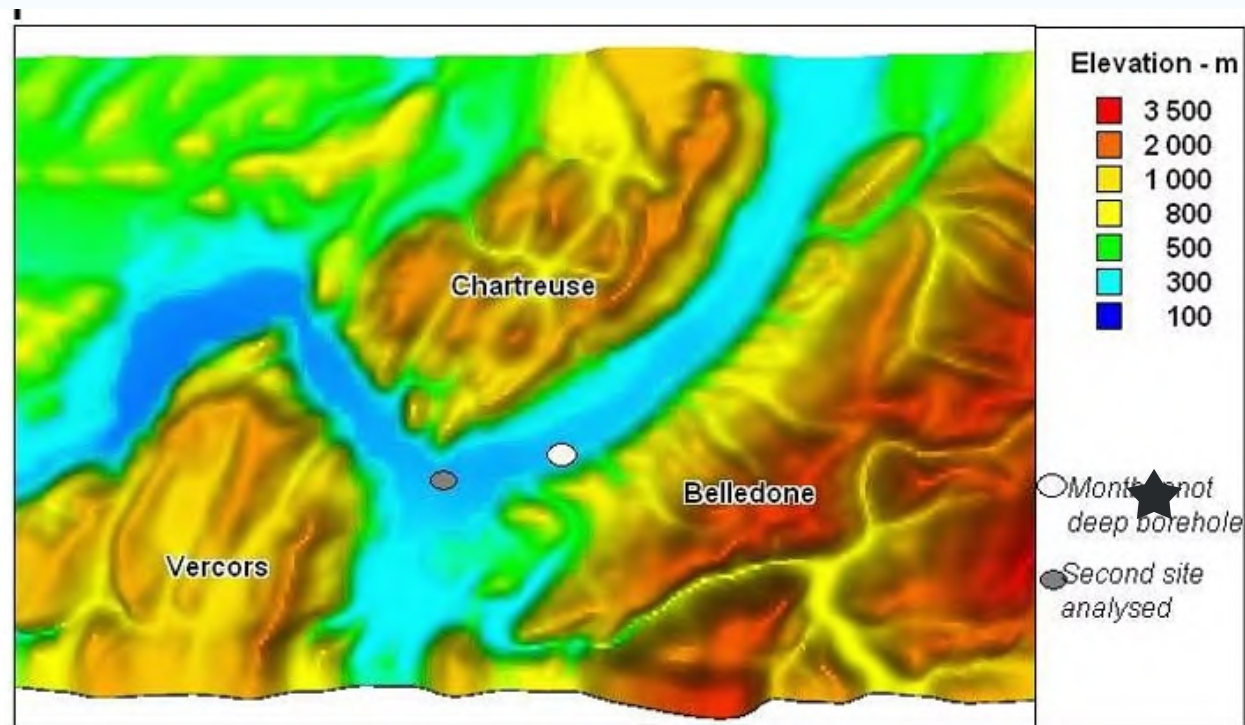
Schematic diagram of soil, reference rock, and bedrock recordings

*Resonance frequency scales
with sediment thickness !*





Grenoble basin - resonance



10.1617/s11527-024-02540-6

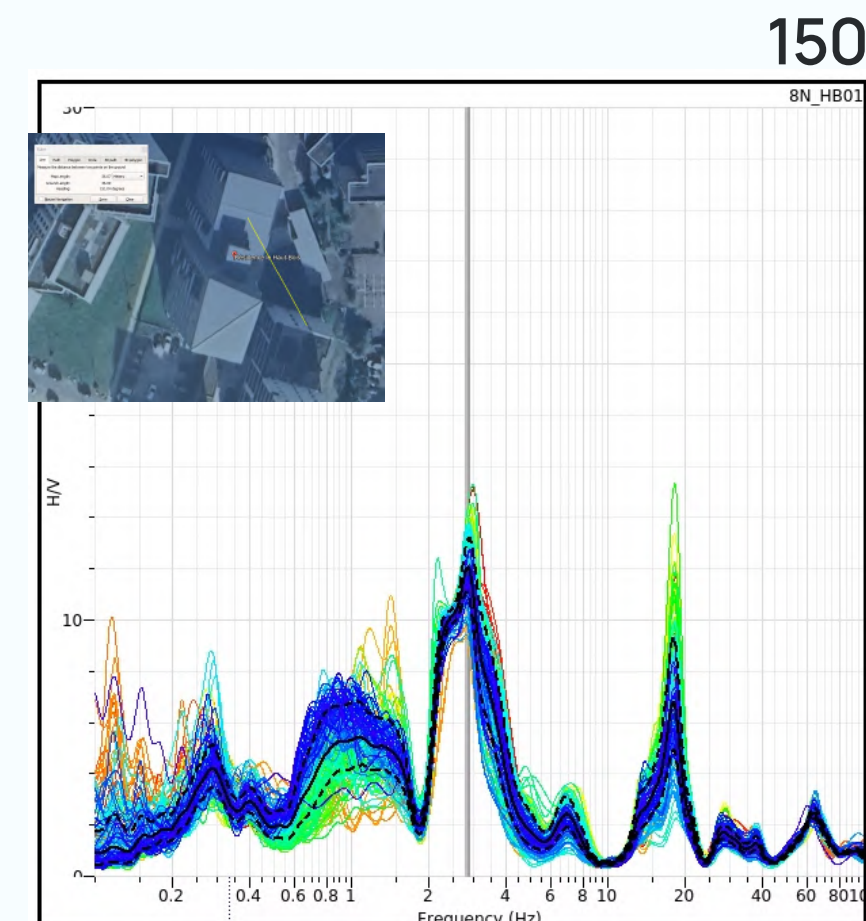
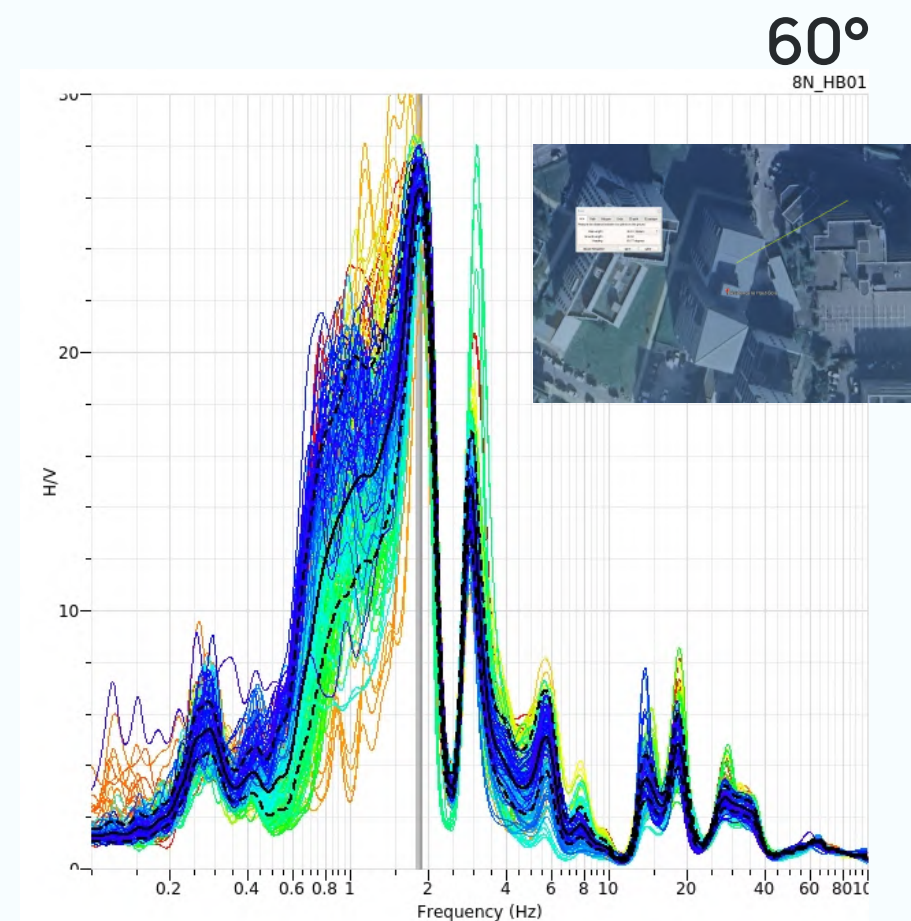
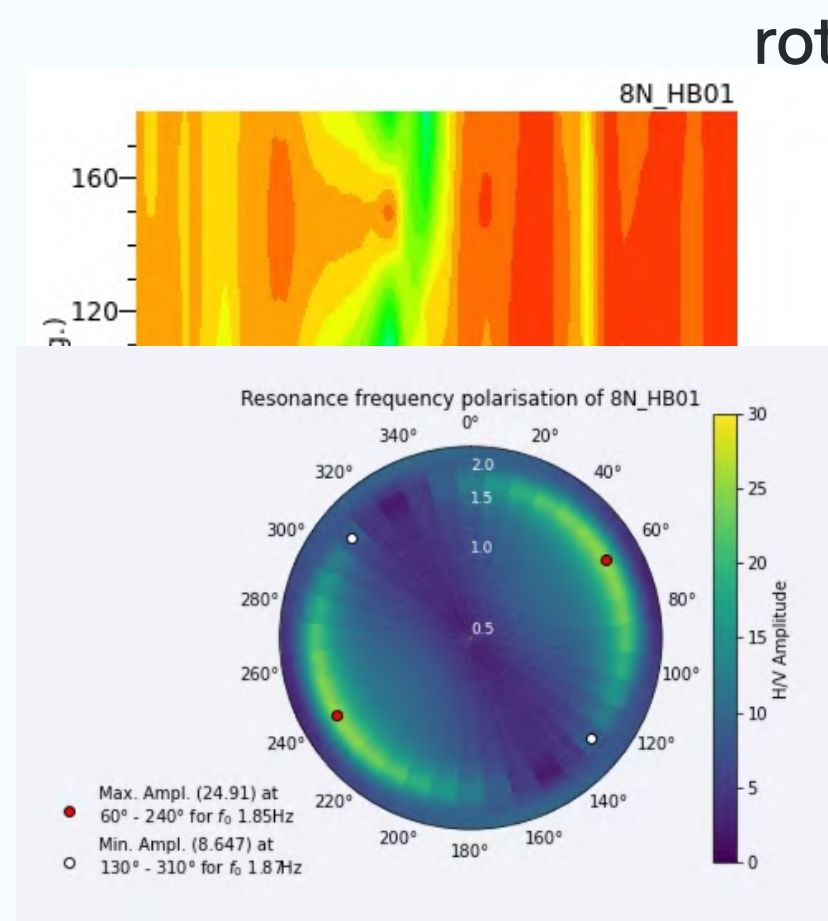
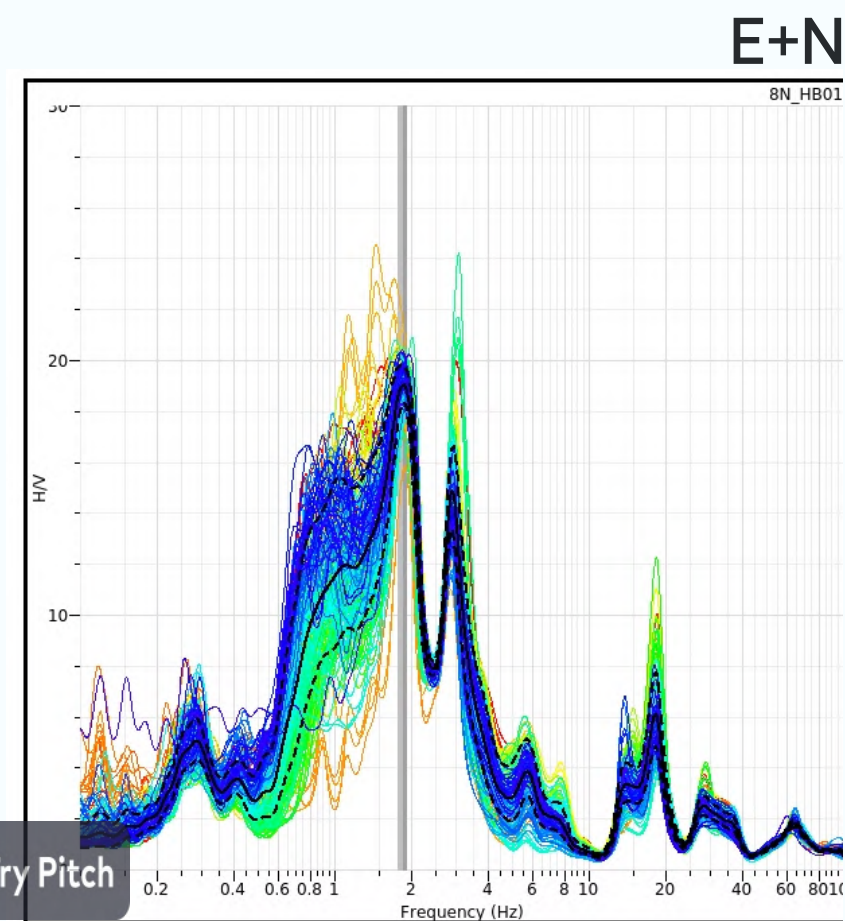
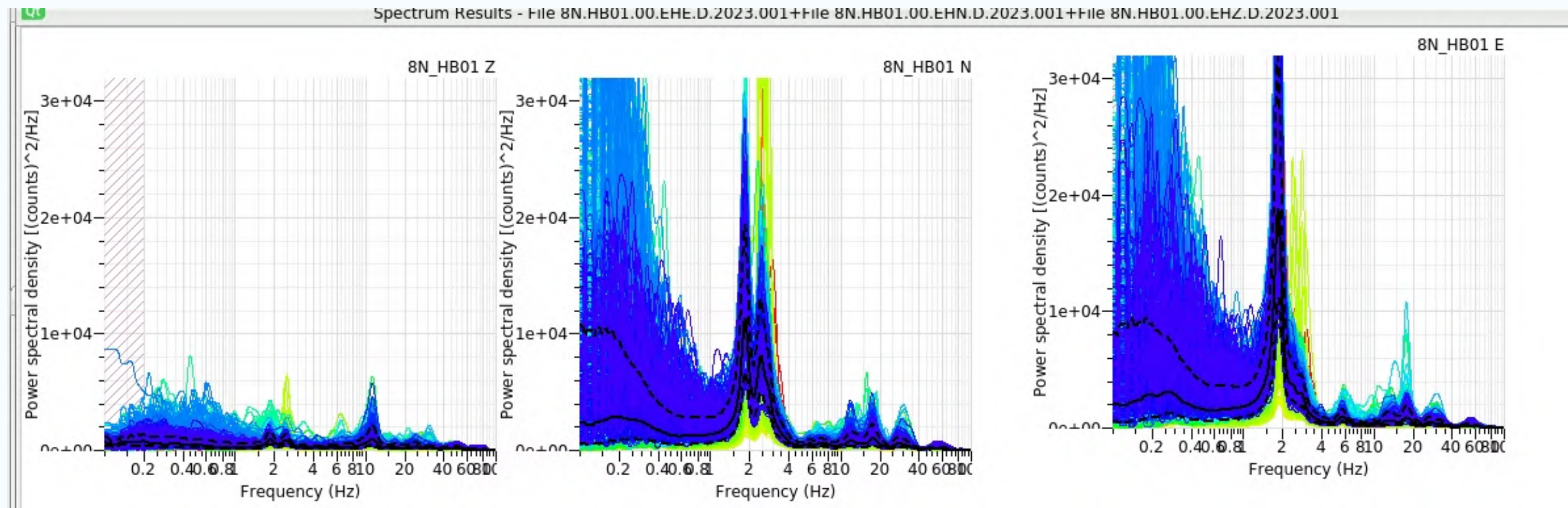
As mentioned and validated previously by Lebrun et al. (2001), the lower frequency is linked to the S-wave resonance frequency of the entire column of sediments.

24 February 2025

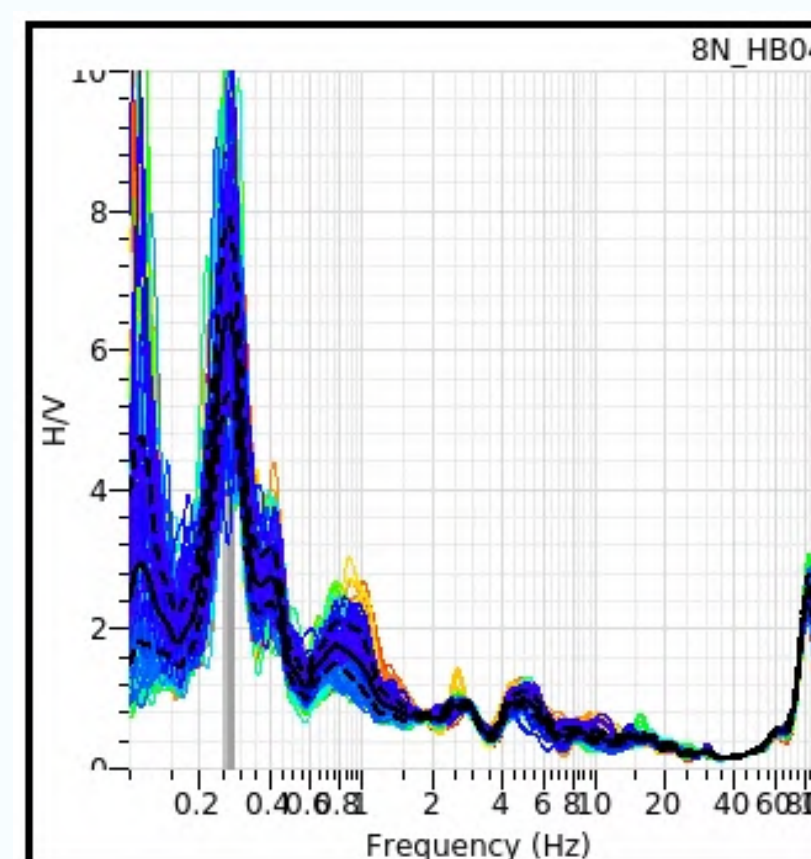
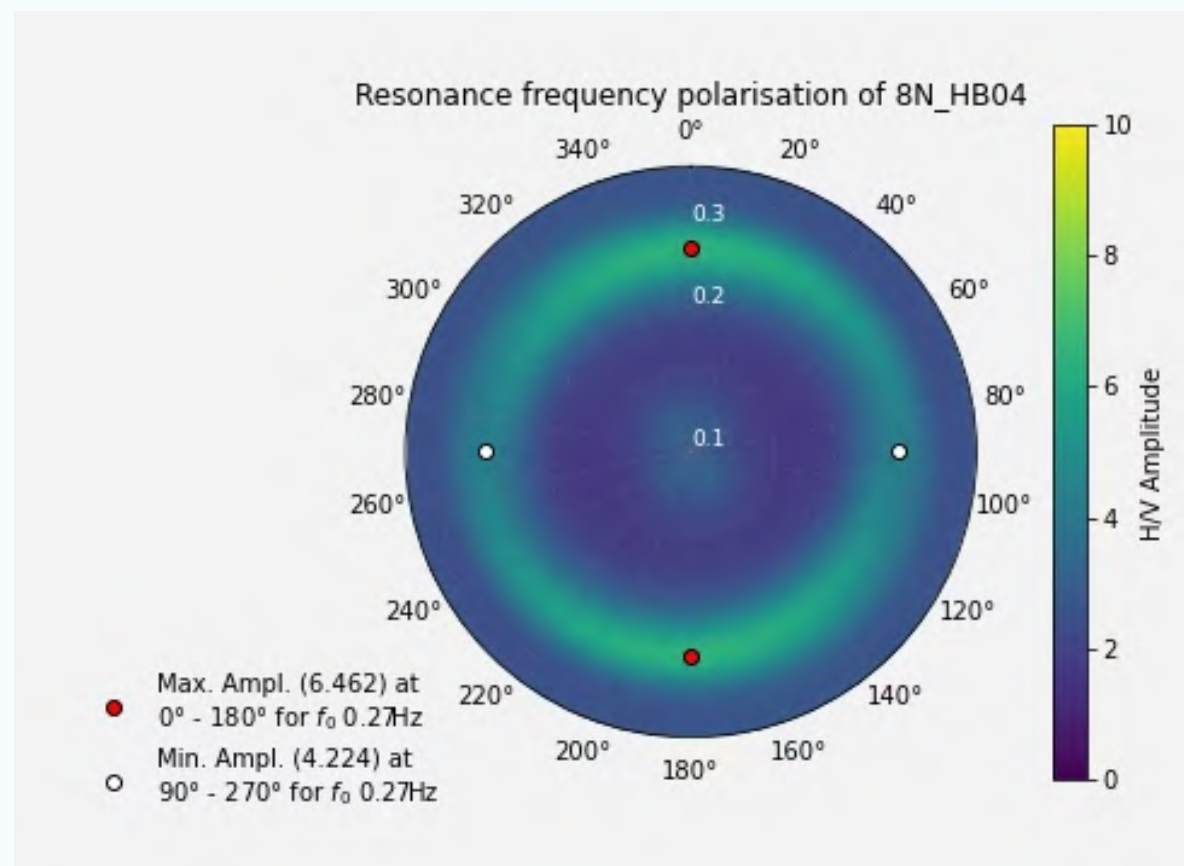
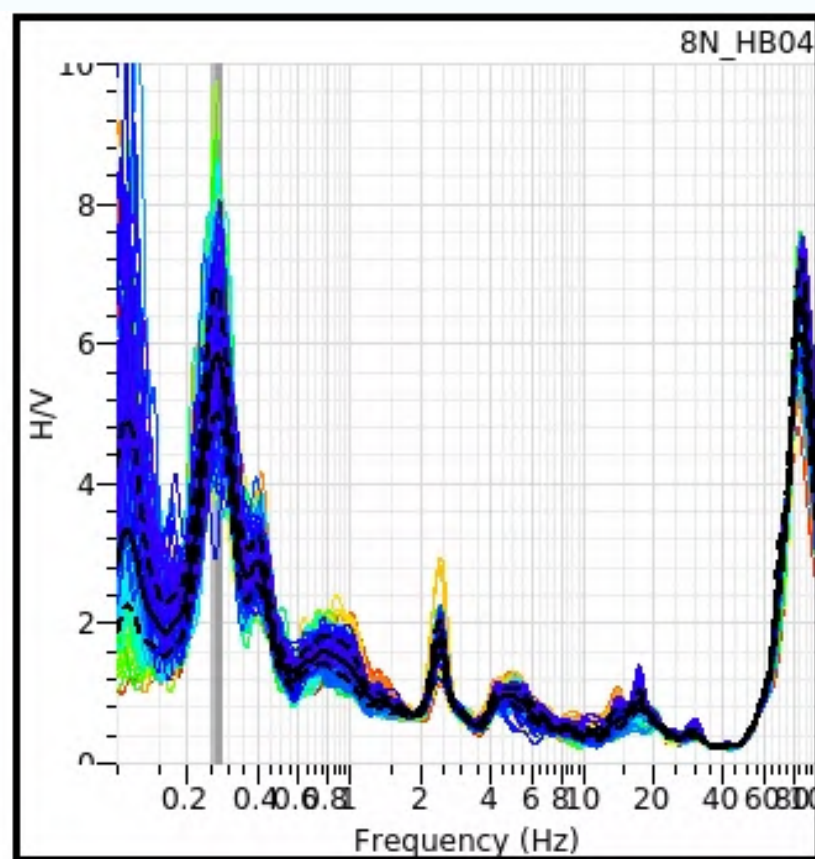
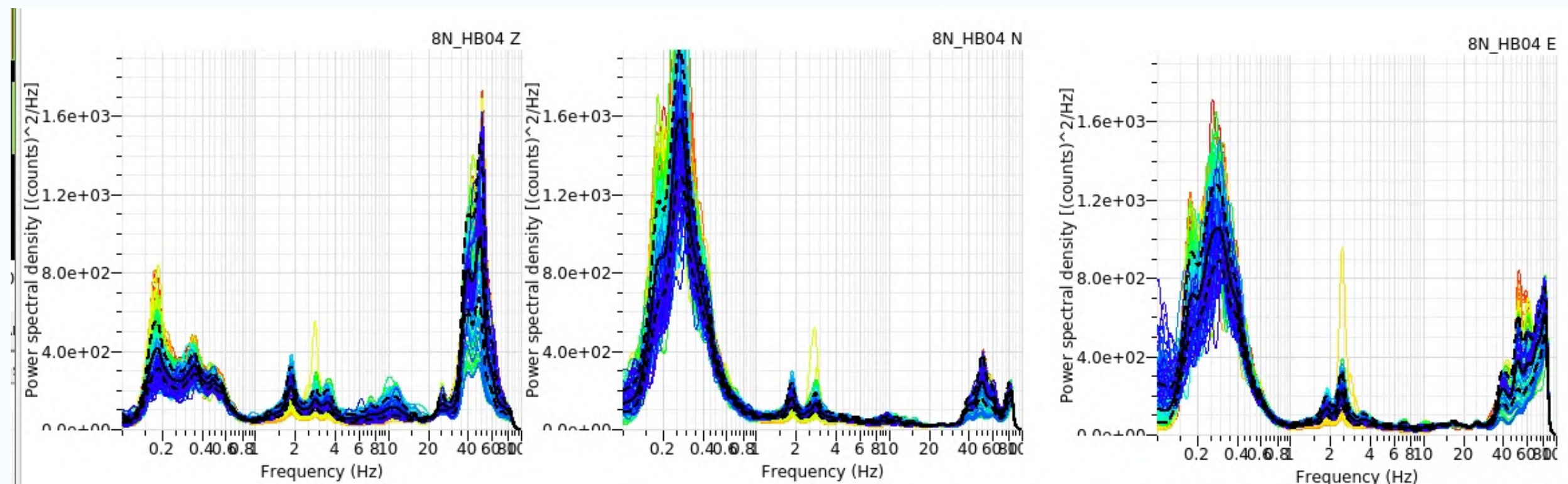


Tutorial 10 results

Continuous HVSR



HB04 – Base



0°



15th Munich Earth Science School
ObsPy goes Downtown: Seismology in Cities



PRACTICALS

Part 2: Resonance is fun

Tutorial 10:

Automatic Processing for 1 day: 30 January 2024

Explore January 2024 continuous HVSR (precomputed)

Explore time & frequency

24 February 2025



Questions

- * **Question 1: how much data is needed to know the resonance of a building: minutes? hours? days? months? **
- * **Question 2: how much data is needed to know the resonance frequency of the soil: hours? days? months? **
- * **Question 3: Can you see the resonance frequency of the soil at the rooftop in HB01, HB02, HB03 ? **
- * **Question 4: Can you see the resonance of the building at the base of the building in HB04 ? **



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